

# Sup'Biotech *projects*

**SUP**  
biotech

Volume 4 • January 2020

- ☐ IGEN : aptamers for microbial detection
- ☐ Antitumoral activity of Curcumin
- ☐ Compost revalorization
- ☐ Directed genetic evolution
- ☐ IGEN : microbial fuel cells

- ☐ 3D imaging
- ☐ Organoids & microfluidics
- ☐ Endocrine disruptors
- ☐ Antifungal activity of biocompounds



**APPLIED TRANSVERSAL  
PROJECTS PROGRAM**

**Fil Rouge  
2019-2020 edition**

## **SUP'BIOTECH PROJECT TUTORS**

**Session 2019-2020**

Agnès Saint-Pol

Jacqueline Bert

Élise Delage

Frank Yates

Vanessa Perdiz

Férid Nassor

Estelle Mogensen

Rafika Jarray

## **SUP'BIOTECH PROJECTS JOURNAL EDITORIAL COMMITTEE**



**Frank Yates, PhD**

Assistant Prof., CellTechs research team, Sup'Biotech



**Vanessa Perdiz, PhD**

Assistant Prof., CellTechs research team, Sup'Biotech



**Estelle Mogensen, PhD**

Assistant Prof., Head of the student laboratories, Sup'Biotech

*All articles published in Sup'Biotech Projects Journal are written by students, under the supervision of tutors. Articles are peer-reviewed by students before publication.*

# EDITORIAL

One more edition of the Sup'Biotech Projects journal means that one more promotion is about to leave us... It is with a little twinge in our hearts, of course, but above all with great pride that we publish this 4th edition of the Sup'Biotech Projects journal, dedicated to the "Applied Transversal Projects" ("Fil Rouge") projects of the students of the class of 2020 who have chosen the Research & Development major.

Through these "Fil Rouge" projects, the students show us, once again, how far they have come since their arrival at the school. They mobilise their knowledge and skills to identify a problem linked to current research projects and to develop a scientific approach and an experimental strategy to solve it. They find themselves confronted with the problems and difficulties encountered in a research project, and must develop their sense of curiosity, innovation and reasoning in order to solve them, together, but independently. They interpret, synthesize and communicate their results in the form of peer-reviewed posters and scientific articles. These are the articles you will find in this issue.

This year, the "Fil Rouge" programme has gained further momentum, as 11 projects were supported between January 2019 and January 2020 by Sup'Biotech's assistant professors, but also by external partners. For instance, one of these companies, DM Compost, specializes in the reduction and recovery of bio-waste in urban areas. Two multidisciplinary teams have also been formed to participate in the IGEM international competition, in partnership with other schools in the IONIS group, and with the Pasteur Institute.

And since innovation is part of Sup'Biotech's DNA, it was with pleasure and interest that the first student symposium was organized by the students themselves in November, with the support of the school and the Biocampus association. During this symposium, each group of students invited an expert working in a field close to their project to present their work and research. I would like to sincerely thank each of the speakers who agreed to take part in the exercise, as well as the members of Biocampus for their invaluable help that made this day a success.

In short, dear colleagues, dear students, it was a real pleasure to spend the "Fil Rouge 2019-2020" edition by your side !

**Vanessa Perdiz, PhD**  
Asst Prof. CellTechs Research Team,  
Sup'Biotech Fil Rouge Tutor



# CONTENTS

|                                                                                                                                                                                                                                     |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>EVALUATION OF THE POTENTIAL AGRONOMIC INTEREST OF MUNICIPAL WASTE COMPOST LEACHATE FOR ITS REVALORISATION AS BIOLOGIC FERTILIZER</b>                                                                                             | <b>Page 7</b>  |
| Farah Amara, Thomas Grossetete, Maxime Laheurte, Yasmine Othmani, Léna Philip, Oriane Saboureau                                                                                                                                     |                |
| <b>COMPARATIVE ANALYSES OF EVOLVED <i>ESCHERICHIA COLI</i> STRAINS USING AN AUTOMATED DIRECTED EVOLUTION TECHNIQUE</b>                                                                                                              | <b>Page 15</b> |
| Elodie Gittard, Marjorie Grée, Charlotte Greselle, Andréa Ruty, Séverine Soille, David Leistenschneider                                                                                                                             |                |
| <b>ENDOCRINE DISRUPTORS BIOSENSOR BASED ON COMPETITIVE INHIBITION BETWEEN DYES AND ENDOCRINE DISRUPTORS FOR LIGNINOLYTIC ENZYMES</b>                                                                                                | <b>Page 17</b> |
| Quentin Weiss, Guillaume Latour, Amandeep Kaur, Sophie Jeanjean, Julien Plisson                                                                                                                                                     |                |
| <b>ANTIFUNGAL PROPERTIES OF ESSENTIAL OILS AGAINST <i>ASPERGILLOSIS</i></b>                                                                                                                                                         | <b>Page 25</b> |
| Dauphine Boutte, Julie Calmet, Adrien Dardinier, Julie Donzier, Mathilde Ducoudray, Fanny Marceau, Charlotte Visconti                                                                                                               |                |
| <b>CURCUMIN/PIPERINE AS A POTENTIAL ALTERNATIVE TO CHEMOTHERAPEUTIC AGENTS ON LUNG CANCER CELLS</b>                                                                                                                                 | <b>Page 35</b> |
| Carolina Araiza, Amira Bellalah, Marine Chamoreau, Yasmine Daïk, Alexandra Lozano                                                                                                                                                   |                |
| <b>CURCUMIN AND CROCIN AS NATURAL SUBSTITUTES FOR CURRENT CHEMOTHERAPEUTIC AGENTS</b>                                                                                                                                               | <b>Page 41</b> |
| Antoine Retho, Léa Lisiecki, Sofia Hervais, Minerva Esparza, Tais Romero                                                                                                                                                            |                |
| <b>ADAPTING A MICROFLUIDIC DEVICE FOR APPLICATIONS IN 3D CELL CULTURE</b>                                                                                                                                                           | <b>Page 55</b> |
| Pierre-Emmanuel Corbet, Mary-Amélie Masson, Pauline Mégalli, Thibaud Metais, Julie Six                                                                                                                                              |                |
| <b>MICROWELL STRATEGY FOR HIGH THROUGHPUT UNIFORM MICROTISSUE GENERATION</b>                                                                                                                                                        | <b>Page 61</b> |
| Maxime Mauviel, Elodie Rosa, Younes Bakhalek, Ulysse Godet                                                                                                                                                                          |                |
| <b>LIGHT SHEET FLUORESCENCE MICROSCOPY: A QUALITATIVE COMPARISON OF FOUR DIFFERENT CLEARING METHODS IN PLANTS</b>                                                                                                                   | <b>Page 69</b> |
| Julien Ballbé Y Sabaté, Anne Fauvel, Agathe Claude, Sarah Pierre, Angéline Mole, Yanaëlle Rousseau                                                                                                                                  |                |
| <b>IGEM - CINERGY : DEGRADATION OF CIGARETTE BUTTS TO PRODUCE ELECTRICITY THANKS TO A MICROBIAL FUEL CELL</b>                                                                                                                       | <b>Page 77</b> |
| Ambre Leleu, Lucas Bourmancé, Mélissa Chagnaud, Caroline Suant, Chloé Doizelet, Eva Zanditenas, Julie Breiner, Marina Firpion, Mathilde Barbier, Sarah Duquesne                                                                     |                |
| <b>IGEM – DIANE : AN APTAMER-BASED DEVICE FOR RAPID DETECTION OF PATHOGENIC BACTERIA IN BIOLOGICAL FLUIDS</b>                                                                                                                       | <b>Page 87</b> |
| Léa Durix, Tara Fournier, Elise Cambon, Chloe Charendoff , Mathieu De Jode, Clara Delamare, Quentin Mosagna, Marie Sabatou, Vincent Saverat, Ronan Soudy, Laurence Vigne, Maria-Vittoria Mazzuoli, Alexandre Zidat, Deshmukh Gopaul |                |
| <b>FIRST FIL ROUGE STUDENT SYMPOSIUM : NOV, 13<sup>th</sup>, 2019</b>                                                                                                                                                               | <b>Page 95</b> |



# Evaluation of the potential agronomic interest of municipal waste compost leachate for its revalorisation as biologic fertilizer

Farah Amara, Thomas Grossetete, Maxime Laheurte, Yasmine Othmani, Léna Philip\*, Oriane Saboureau  
Sup'Biotech, Villejuif, France

\*Corresponding author: lena.philip@supbiotech.fr

**Abstract** – Fertilization is a key element to support the increasing food crop production. However, the massive use of synthetic fertilizers causes economic and environmental issues worldwide: mineral fertilizers are made from non-renewable resources and are responsible for the loss of soil microbiome and quality. Thus, there is a growing interest in bio-fertilizers. Among these products, compost is already widely used for its fertilizing properties. To this day, liquid leachates formed during the composting process have been treated like wastewater. However, since compost is known for its interesting composition in terms of plant nutrients, compost leachates are likely to share this parameter and thus represent a valuable source of nutrients. This article presents the study of the physical-chemical properties of compost leachates as well as its effect on plant growth. We explored the physical-chemical properties of compost leachate for a comparison with French regulation. We also assessed the phytotoxicity of compost leachates at two stages of plant development (germination and early growth). The results pave the way for a revalorisation of compost leachates as a liquid organic fertilizer.

**Index Terms** – Compost leachates, agronomy, fertilizer, revalorisation.

## INTRODUCTION

The massive growth of population that followed industrial revolution lead to the need of an increase of food production (Bockman *et al.*, 1997). Indeed, the total cultivated land worldwide increased by 466% from 1700 to 1980 (Matson *et al.*, 1997). This tremendous intensification of food production (due to both increased cultivated land and yield) was and is still supported by the massive use of fertilizers (in addition to high-yielding crop varieties, irrigation and pesticides). Fertilizers provide a supply of mineral nutrients needed for plant growth, mainly nitrogen (N), phosphorus (P) and potassium (K). However, the use of synthetic fertilizers presents several issues that are highly discussed today. First, fertilizers will not support food production forever since these products are obtained from non-renewable resources such as gas, coal, oil or electricity via the Haber-Bosch process (Bockman *et al.*, 1997). Moreover, intensive use of

fertilizer as long with methods used for intensive agriculture cause environmental issues at local (increased erosion, lower soil fertility and loss of biodiversity), regional (pollution of ground water and eutrophication of rivers and lakes) and global (impacts on atmospheric constituents and climate) scales. Indeed, the use of fertilizers is not optimized and does not follow nature processes. The application of fertilizers is not synchronized to plant demand (Matson *et al.*, 1997), and provide immediately available nutrients for plants, whereas in nature, organic compounds undergo microbial fermentation before being used by plants. It is known that only 40 to 60% of the applied nitrogen is used by plants (Matson *et al.*, 1997). The excess is lost either by gas emitted into the atmosphere or by leaching in solution from the soil to surface or groundwater. Not only the use of fertilizer has a highly negative impact on environment, but ironically it also presents negative feedback on agricultural production. For bearing such issues, there is a current development of biofertilizers, which represent a more sustainable, economic and environmentally friendly solution, that can be obtained from mass production (Bhattacharjee *et al.*, 2014). Biological fertilization is the use of natural inputs such as fertilizers, decaying remains of organic matters, excess crops, domestic sewage, animal manure and microorganisms such as fungi and bacteria (Bhattacharjee *et al.*, 2014). In addition to be a source of nutrients for promoting plant growth, microorganisms contained in such fertilizers may improve fixation of nutrients, produce growth stimulants for plants, improve soil stability, provide biological control, biodegrade substances, recycle nutrients, promote mycorrhiza symbiosis and also develop bioremediation processes in soil (Fediala *et al.*, 2014). Among the biofertilizers, compost is a well-known product in terms of contents and physical-chemical properties. Composting is the controlled decomposition of organic matter such as leaves, twigs, wood, waste, lawn clipping, food scraps and other organic matter (Chatterjee *et al.*, 2013). It is the oldest method used for stabilizing natural waste and biological fertilization of the soil, that provides a stable, chemically and biologically rich product with micro and macro nutrients (Fediala *et al.*, 2014) equivalent to humus. The use of compost for soil fertilization presents several advantages. First, composting is a mean for waste reduction, and the interest for such method is rising (Chatterjee *et al.*, 2013). It also benefits nutrients

management, since it recycles nutrients: what comes from the soil returns to the soil. Finally, the composting process is a mean for pathogen control: the high temperature reached during the process (from 45 to 70°C) kills bacterial pathogens, phytopathogens, pathogenic viruses and weed seed (Chatterjee *et al.*, 2013). The composting process lead to two products: compost, currently used as fertilizer, and liquid leachate. So far, these liquid products are considered as waste and treated as wastewater. However, since the richness of compost in nutrients is well recognized, and since the use of compost is efficient for fertilization, we may think that compost leachate also contains interesting agronomic properties. This article presents a study of the chemical properties of compost leachates as well as their phytotoxic characteristics to investigate the potential of compost leachate as fertilizers.

## MATERIALS AND METHODS

### I. Collection and recovery of leachates

#### I.1 Compost bin installation

In order to reproduce composting process at a laboratory scale and to evaluate potential differences between different compost compositions, a system of boxes was set up in the garden of the association Biocampus within the campus (Fig. 1). Boxes with a capacity of 27 L. were perforated several times at the bottom. A 14 L box was then fixed under each 27 L box. The 27 L boxes were used to contain the compost and the 14 L boxes were used to harvest the leachates during compost fermentation. Boxes were then filled with wet matter coming from DM compost company, and dry matter or miscanthus coming from Biocampus association. Boxes were filled with the following composition (Table 1: the box called "control" composition corresponds to two thirds of fresh organic matter for one third of dry matter, which is the usual composition for composting process).



FIGURE 1. PICTURE AND SCHEME OF THE BOXES SYSTEM FOR COMPOST AND LEACHATE PRODUCTION

Boxes were put under a tarpaulin for being covered from rainwater. The goal is to recover pure leachates non previously diluted by rain or outside water infiltration.

TABLE 1. COMPOSITION OF COMPOST BOXES

| BOX 1            | BOX 2          | BOX 3                                        | BOX 4            | BOX 5                                  |
|------------------|----------------|----------------------------------------------|------------------|----------------------------------------|
| 1/3 DRY MATTER   | 1 DRY MATTER   | BOX 1 COMPOSITION + MOISTENING WITH LEACHATE | 1/3 MISCANTHUS   | STACKING OF 3 BOXES WITH COMPOSITION 1 |
| 2/3 FRESH MATTER | 1 FRESH MATTER |                                              | 2/3 FRESH MATTER |                                        |

#### I.2 Leachates harvesting

Leachates were harvested at the following periods: February, April, October and December 2019. The leachates were collected in 50mL tubes with filtering paper (the use of coffee filter was chosen because of good suitability - indeed, the porous of cellulose filter allows to filter solid part of leachates such as insect without retaining molecular components). The volume of each solution was recorded. For each sample the colour and the viscosity were assessed visually. The overage of leachates was stored at 4°C for optimal stability in time (Romero *et al.*, 2013).

### II. Determination of physical-chemical parameters of agronomic interest

#### II.1 Determination of pH and the presence of N, P, K

The pH leachates were measured using a classic pH-meter on raw leachates. The presence of essential nutrients (N, P, K) was determined by colorimetric tests from the soil test kit HI3895 provided by Hanna Instruments. Leachates were diluted a hundred times prior to the tests. Reagent powder was added for each test in 2mL of the previously diluted leachates. Results were read according to the standard ranges provided by the test kit.

#### II.2 Quantitative determination of physical-chemical parameters of agronomic interest and safety parameters

Quantitative data on the composition of leachates have to be determined, in line with the regulations in force (NF U 44-051), to know if the leachate could be used commercially. As the necessary equipment for all these analyses was not available at Sup'Biotech, this test was subcontracted to the analytical laboratory WESSLING. In addition to nutrients composition, safety parameters were assessed (organic trace contaminants: PAH, PCB and heavy metals).

For this purpose, specific sampling vials were sent by the laboratory. A quantity of 2L of leachate had to be collected in order to fill all the vials. In view of the small quantities recovered in our experimentation tanks, leachates were diluted by two. The obtained results were calculated again from raw leachates.



### III. Microbiology assays

#### III.1 Assessment of the presence of regulatory bacteria in leachates

The presence of two regulatory bacteria, *Salmonella* and *Listeria monocytogenes* was performed on each leachate samples. A total of ten samples were obtained from the different boxes containing the compost leachate. The first five samples were pure (not diluted) and the other 5 samples were diluted (1:10 dilution).

In order to detect the presence of *Salmonella*, Hektoen Enteric Agar medium (Sigma Aldrich: 51490 Hektoen Enteric Agar) was prepared by suspending 76.67g in one liter of distilled water, adding 10g/L of agar and letting it soak 15 minutes. The mix was heated up by constant stirring until boiling. It was left to cool down to 55-60°C and poured into sterile Petri dishes (the medium is very thermolabile and thus overheating should be avoided).

For the detection of *Listeria monocytogenes*, Listeria mono differential Agar (77408 Listeria mono Differential Agar, Base) were prepared by suspending 36g in 460ml distilled water. The medium was boiled with continuous stirring to dissolve it completely. After boiling, the mix was sterilized by autoclaving at 121°C for 15 minutes, then it was left to cool down to 45-50°C. Once the medium was solidified 100µL of leachate (pure or previously diluted) were added and spread over all the surface of the Petri dish (Vaidya et al., 2018). After the inoculation, the Petri dishes were incubated at 35°C for two days. Then observations of the different Petri dishes were performed in order to conclude on the presence or the absence of the pathogen.

#### III.2 Identification and quantification of bacterial strains

Due to security and technical limits, quantification and the exact identification of bacterial strains could not be performed at Sup'Biotech laboratories. Further microbiological analyses of compost leachates was thus subcontracted to WESSLING.

### IV. Phytotoxicity assays

#### IV.1 Assessment of the effect of leachate on seed germination

Several protocols for phytotoxicity on germination assay were tested, in order to select the best one. Water agar gel was prepared with a concentration of 1%. The medium was autoclaved, then poured into petri dishes with 2mL of leachate. Commercial liquid fertilizer was used as a reference, and sterile distilled water was used as a positive control. Two seeds of bean were placed in each Petri dish. Each petri dish was made in duplicate and kept at room temperature. The number of germinated seeds and rootlets were assessed after 7 days.

#### IV.2 Assessment of the effect of leachate on plant growth

Lamb's lettuce seeds were first planted in seedling plates and then transplanted into small growing pots. Six growing pots were set up in duplicates, corresponding to the five leachates and the control. Five drops of each leachate were added on the corresponding growing pots every three days, as well as water. Control pots were only treated with water. Due to the lack of materials for controlled plant growth and vegetal experiments, two parameters could not be controlled (temperature and humidity). Plants were exposed to artificial light during eight-hour cycles.

## RESULTS

### I. Macroscopic observation of compost leachates

During macroscopic observations, all leachates appear to be similar, they were all dark and opaque, closed to black. During several harvesting, the quantity of leachate produced by the fourth box (2/3 fresh matter for 1/3 of miscanthus), was lower than the other boxes. As the miscanthus is known to have an important absorptive capacity, this low amount of leachate harvested was expected.

### II. Assessment of plant-essential nutrient contents

#### II.1. Validation of the presence of N, P, K on compost leachates

The pH analysis showed that all the solutions have a similar basic pH between 8,3 and 9,3. First colorimetric results confirmed the presence of the macronutrients N, P and K. Indeed, every sample, even diluted 100 times, showed at least a non-negligible amount of N, P and K. Those mineral salts are needed for plant growth, those first results prove the potential of leachates for being a fertilizer. Other quantitative experiments must be done in the future such as the analysis of the exact mineral composition of leachates, the follow of the variation of this composition depending on the input materials and the time of compost maturation.

#### II.2. Quantification of plant nutrients in compost leachates

Results of the analysis performed by WESSLING allowed the precise quantification of plant major and secondary nutrients in leachates. Concerning the three main nutrients, there are all found in the leachates but with important differences between their concentration. Indeed, phosphorus has been found in small quantities (82 mg/L) which represent only 3% of the quantity of total NPK. Nitrogen represent 11% of NPK with a concentration of 612, 36 mg/L. Finally, Potassium has been found in large amount (5400 mg/L) which represent 86% of the quantity of NPK. Thus, leachates could be considered as a high source of potassium and medium source of nitrogen fertilizer. Moreover, secondary nutrients have also been found in satisfying quantity.

Magnesium, necessary for chlorophyll production and calcium, important for cell wall stability, were found respectively at 130 mg/L and 220 mg/L. Finally, organic carbon has been found at 3200 mg/L. In order to characterize a fertilizer, the ratio C/N has been calculated to give information about nitrogen availability. Ration C/N is 6, which indicates a good degradation and nitrogen release for plants growth.

### III. Assessment of safety parameters in compost leachates

Safety parameters include heavy metals (*i.e.* mercury (Hg), arsenic (As), lead (Pb), cadmium (Cd), chrome (Cr), copper (Cu), nickel (Ni), selenium (Se), and Zinc (Zn)), and organic compounds such as fluoranthene, benzo(b)fluoranthene and benzo(a)pyrene.

Results of the quantification of heavy metals are summarized in Table 2. Results are given in milligram per kilograms of dry matter.

| Heavy metal | Regulatory threshold (mg/kg) | Experimental value (mg/kg) |
|-------------|------------------------------|----------------------------|
| Hg          | 2                            | < 0,01                     |
| As          | 18                           | < 0,06                     |
| Pb          | 180                          | 0,06                       |
| Cd          | 3                            | < 0,04                     |
| Cr          | 120                          | 0,14                       |
| Cu          | 300                          | 1,18                       |
| Ni          | 60                           | 0,28                       |
| Se          | 12                           | < 0,06                     |
| Zn          | 100                          | 6,30                       |

TABLE 2. QUANTIFICATION OF HEAVY METALS IN COMPOST LEACHATES.

Content in heavy metals is significantly lower than the thresholds fixed by regulation (NF U 44-051) for each metal assessed. Thus, heavy metals are not a limitation for the use of compost leachates as liquid fertilizers.

Table 3 show the results of quantification of organic compounds in compost leachates, as long as the values

| Organic compound     | Regulatory threshold (mg/kg) | Experimental value (mg/kg) |
|----------------------|------------------------------|----------------------------|
| Fluoranthene         | 4                            | < 1                        |
| Benzo(b)fluoranthene | 2,5                          | < 1                        |
| Benzo(a)pyrene       | 1,5                          | < 1                        |

TABLE 3. QUANTIFICATION OF ORGANIC COMPOUNDS IN COMPOST LEACHATES.

As for heavy metals, the content in organic compounds is lower in compost leachates to thresholds fixed by French regulation (NF U 44-051). Organic compounds do not represent a limitation for the use of compost leachates

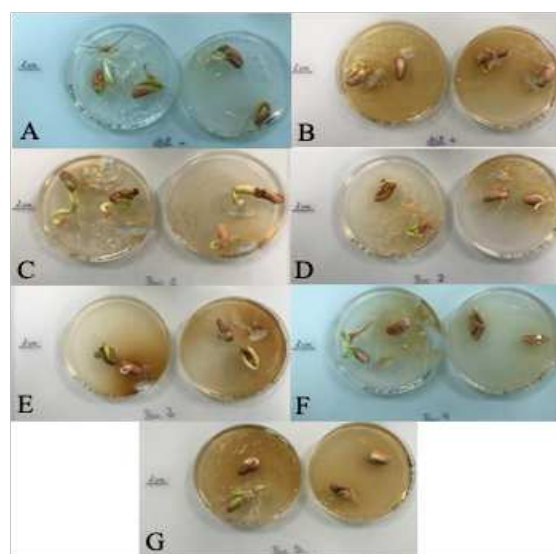


FIGURE 2. GERMINATED BEAN SEEDS ON AGAR GEL 1%. A. POSITIVE CONTROL. B. REFERENCE. C. BOX 1. D. BOX 2. E. BOX 3. F. BOX 4. G. BOX 5.

### IV. Phytotoxicity of compost leachates

#### IV.1 Compost leachates are not toxic for seed germination

Results of the phytotoxicity on germination showed an important number of tested seeds germinated (Table 4, Figure 2). Table 4 presents the results obtained with bean seeds. Experiment was made in duplicate, with two seeds each. The first column presents the number of germinated seeds in each duplicate. The second column shows the number of rootlets per seed in each replicate ("number of rootlets on seed 1 – number of rootlets on seed 2). Water was used as the positive control, and a commercial fertilizer was also tested as a

reference. All the results shown in the table 2 look similar. Because of that, it cannot be concluded that the leachate is a good fertilizer, but we can say that the studied leachates does not present toxicity on bean seeds (Figure 2).

#### IV.2 Compost leachates does not appear to be toxic to the growing plant

Unfortunately, these results cannot be considered as totally scientifically valid because the parameters were too variable during the experimentation. Nevertheless, they seem encouraging regarding the non-toxicity of compost leachates for plant growth. Indeed, during two tests, five growing pots containing lamb's lettuce and watered frequently with raw leachates, only one plant did not survive (data not shown). In contrary, both controls exposed only to water showed difficulty to grow and died after 1 week. The experience has been made in poor nutrition conditions for plants, which could explain the death of control plants compared to lettuce that receive additional nutrients from compost leachates.

TABLE 4. PHYTOTOXICITY OF COMPOST LEACHATES ON SEED GERMINATION.

| Condition        | Number of germinated seeds / number of total seeds |       | Number of rootlets per seed |        |
|------------------|----------------------------------------------------|-------|-----------------------------|--------|
|                  |                                                    |       |                             |        |
| Reference        | 1 / 2                                              | 2 / 2 | 1 - 0                       | 10 - 1 |
| Positive control | 2 / 2                                              | 2 / 2 | 8 - 10                      | 5 - 1  |
| Box n°1          | 2 / 2                                              | 2 / 2 | 6 - 9                       | 1 - 6  |
| Box n°2          | 2 / 2                                              | 2 / 2 | 1 - 0                       | 6 - 1  |
| Box n°3          | 2 / 2                                              | 2 / 2 | 1 - 9                       | 1 - 1  |
| Box n°4          | 1 / 2                                              | 1 / 2 | 9 - 0                       | 0 - 0  |
| Box n°5          | 2 / 2                                              | 2 / 2 | 8 - 0                       | 13 - 1 |

#### V. Identification of the two regulatory bacteria in compost leachates

##### V.1. Observation on Petri dishes inoculation

For the identification of both regulatory bacteria, controls were performed with water and the respective medium alone. Each condition was made in duplicate. A control with LB medium was also made to confirm the good conditions of operation in terms of sterility.

##### V.1.i. *Listeria monocytogenes*

Leachates from each box were inoculated in *Listeria*-specific medium. This medium is not specific to the strain *Listeria monocytogenes*. However, the presence of blue to green colonies demonstrates the presence of *Listeria*.

Figures 3A and 3B show the presence of blue to green colonies on the Petri dishes inoculated with the leachates, both diluted and non-diluted. This result showed the presence of *Listeria* in leachates from all composition. Still, regulation applies for the precise *Listeria monocytogenes* strain. Thus, precise identification of the strain of *Listeria* was performed by the laboratory WESSLING.

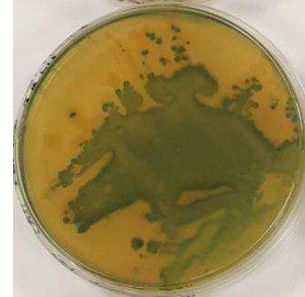


FIGURE 3. IDENTIFICATION OF *LISTERIA* IN COMPOST LEACHATES.

##### V.1.ii. *Salmonella*

Leachates from each boxes were inoculated on Hektoen medium. This medium is specific to *Salmonella*. The presence of the target-bacteria is revealed by the appearance of green colonies with a black centre on the medium.

The identification of *Salmonella* is easier than for *Listeria monocytogenes* because the green colonies with a black centre are specific to this precise pathogen strain. Figure 4 shows the presence of green colonies with a black centre (circled colony), though the colours are not well visible on this figure.



FIGURE 4. IDENTIFICATION OF *SALMONELLA* ON HEKTOEN MEDIUM.

## V.2. Precise identification and quantification of regulatory bacteria

Further microbiology assays were performed by the laboratory WESSLING in order to determine precisely the presence or absence of the two regulatory bacteria. These tests showed the absence of both *Salmonella* and *Listeria monocytogenes*. This result is encouraging for the study. Indeed, absence of these two strains in fertilizing products is required for their commercialization.

## DISCUSSION AND PERSPECTIVES

Compost leachates, up to now considered as scrap and treated like waste-water, seem eligible to be used as an organic fertilizer. Indeed, leachates appear very likely to contain interesting nutrients just like compost, that could be useful in fertilizing processes. This would allow to reduce the use of synthetic fertilizers, that are widely controversial. Moreover, it would be a way of revalorisation of an actual waste. This article investigates the physical-chemical properties of compost leachates at laboratory scale. The composition in nutrients, the safety parameters, the microbiological composition as well as the phytotoxicity of leachates were investigated. The goal was to evaluate the agronomic interest of this product and to ensure that this product is in line with French regulation.

The nutrient composition has been analysed by WESSLING laboratory and results are encouraging. The presence in large amount of potassium and the presence of Nitrogen indicates that leachates could be used as "NK" fertilizer with an approximate 2-8 ratio. C/N ratio has been calculated and is 6. Meaning that the Nitrogen will be available for the plants with the use of such liquid potential fertilizer. Also, presence of magnesium and calcium gives an other advantages to leachates because of their interest in plant growth. In terms of nutrient composition, compost leachates could so be a real good source to improve plants growth. Still, pathogenicity, presence of pollutants and phytotoxicity are discussed below.

Both *Salmonella* and *Listeria monocytogene* strains are absent from the compost leachates used for the study. The French regulation sets a zero-tolerance for both of these bacteria in fertilizing products. However, there might be variability of entrants on composting platforms. Thus, absence of these bacteria will have to be assessed on each new batch of compost leachate. In the case of presence of *Salmonella* and *Listeria monocytogene* in some of the leachates, the simplest method of treatment would be the heat such as an autoclaving process at 120°C during 10 min to kill all bacteria and their spores. This treatment has to be well studied because it can also leads to the destruction of an important quantity of nutrients.

Concerning safety parameters, heavy metals, CTO and HAP were all found in quantities lower than the french regulations

on fertilizer. There are so no limitation due to their composition to use compost leachates as fertilizer.

There was no apparent phytotoxicity of leachates, on both germination and growth steps. Further studies must be performed in order to have a better understanding of the effect of leachates on plant growth, to determine the optimal quantities and frequencies of application of leachate as a liquid fertilizer.

The perspectives of this study are mainly focused on the industrialization step of compost leachates. It would be interesting to reproduce each of those tests with leachates coming from the industrial platform in order to compare this present result with high scale compost production. This work would include the control of leachates composition variations depending on the times and the kind of inputs. Those results could confirm or infirm the stability of the leachates composition. In the second case, it is important to emphasize the fact that the composition of leachates would have to be controlled at the end of industrial processing and before the selling. It would have a non-negligible cost for the Company so it is a fact to take in account for further studies.

Concerning phytotoxicity analysis, to go further, it would be interesting to complete the same experiment that was made for this paper by testing an important amount of different seeds and by controlling luminosity and temperature and humidity. It would also be interesting to measure the roots and rootlets size to compare them with a control. It would allow us to determine the potential as a fertilizer as we already know that there is no detectable toxicity.

After all those studies it would be necessary to work on the packaging and the dilution needed to use leachates for personal consumption. Indeed, the optimal dosage in time must be fixed based on a precise analysis on plant growth.

Finally, the leachates could also be interesting as hydroponic culture inputs (Jarecki, 2007; Churilova, 2019). Indeed, for outdoor agriculture, use of compost is advantageous for it reinforcement of soil structure.

## ACKNOWLEDGEMENTS

This research was supported by the company DM compost. The company supports the installation and the maintenance of compost installations. We especially thank Mr SANDIER who provided technical expertise on the compost production that greatly assisted our researches, and ensured the follow-up of this project.

We also want to thank Mrs MOGENSEN and Mrs CHEVALLIER BOUTELL for their lab assistance, their scientific expertise and for their comment that greatly improve the manuscript.

We also worked in collaboration with the project group Elixiviat. We thank the team for helping us in regulatory studies.

## REFERENCES

R. Bhattacharjee and U. Dey, "Biofertilizer , a way towards organic agriculture : A review," vol. 8, no. 24, pp. 2332–2342, 2014.

Brown K, Ghoshdastidar AJ, Hanmore J, Frazee J, Tong AZ. Membrane bioreactor technology : A novel approach to the treatment of compost leachate. *WASTE Manag.* 2013.

O. C. Bøckman, "Fertilizers and biological nitrogen fixation as sources of plant nutrients: Perspectives for future agriculture," pp. 11–14, 1997

N. Chatterjee, M. Flury, C. Hinman, and C. G. Cogger, "Chemical and Physical Characteristics of Compost Leachates," 2013.

E. Churilova, D. Midmore. "Vermiliquer (Vermicompost Leachate) as a Complete Liquid Fertilizer for Hydroponically-Grown Pak Choi (*Brassica chinensis* L.) in the Tropics." *Horticulturae* 5:1, p. 26, 2019

W. Fediala, A. E. Mosa, L. S. Paszt, and N. A. A. El-megeed, "The Role of Bio-Fertilization in Improving Fruits Productivity — A Review," no. November, pp. 1057–1064, 2014.

M. K. Jarecki, C. Chong & R. P. Voroney "Evaluation of Compost Leachates for Plant Growth in Hydroponic Culture", *Journal of Plant Nutrition*, 28:4, pp. 651-667, 2005

P. A. Matson, "Agricultural Intensification and Ecosystem Properties," vol. 504, no. 1997, 2012.

C. Romero, P. Ramos, C. Costa, and M. C. Márquez, "Raw and digested municipal waste compost leachate as potential fertilizer: comparison with a commercial fertilizer," *J. Clean. Prod.*, vol. 59, pp. 73–78, 2013.

D. Trujillo, Font X, Antoni S. Use of Fenton reaction for the treatment of leachate from composting of different wastes. 2006.



# Comparative analyses of evolved *Escherichia coli* strains using an automated directed evolution technique

Elodie Gittard<sup>1</sup>, Marjorie Grée<sup>1</sup>, Charlotte Greselle<sup>1</sup>, Andréa Ruty<sup>1</sup>, Séverine Soille<sup>1</sup>, David Leistenschneider<sup>1\*</sup>

<sup>1</sup>Sup'Biotech Biotechnology engineering school, Villejuif, France

\*Corresponding author: [david.leistenschneider@supbiotech.fr](mailto:david.leistenschneider@supbiotech.fr)

**Article not published for  
confidentiality reasons**





# Endocrine Disruptors Biosensor based on Competitive inhibition between dyes and endocrine Disruptors for Ligninolytic Enzymes

Weiss Quentin\*, Latour Guillaume, Kaur Amandeep, Jeanjean Sophie, Plisson Julien

With the Participation of: Bertagne Baptiste, Certeaux Nicolas, Viart Nicolas

Sup'Biotech, Villejuif, France

\*Corresponding author: quentin.weiss@supbiotech.fr

**Abstract** - Endocrine Disruptors (ED) are natural or synthetic chemicals that are released in the environment and suspected to alter the functions of the endocrine system and, therefore, causing deleterious health effects to any organism, its offspring and populations or subpopulations of organisms through the disruption of the endocrine systems. Due to their extensive use during the last centuries, high amounts are found in environment and disturb the different ecosystems. Therefore, it is important to assess their presence in an efficient way. However, even if the current methods are precise, they are also expensive and time consuming, thereby, there is a need to develop a sensitive, fast and easy to use test to detect endocrine disruptors in environment. Our study aims to develop a new type of test based on enzymatic competition between EDs and dyes. Indeed, some enzymes particularly ligninolytic enzymes, produced by fungi as *Pleurotus ostreatus* have the ability to degrade aromatic compounds such as dyes and endocrine disruptors. In this context, our study aims to evaluate the potential competitive inhibition of EDs on dye degradation which are molecules easily detectable. We have shown that the competitive inhibition of ligninolytic enzymes might allow the conception of an Endocrine disruptors-dyes competitive system that could detect endocrine disruptors at environmental occurrences levels.

**Index Terms** - Biosensor, Dyes, Endocrine Disruptors, Ligninolytic enzymes, *Pleurotus ostreatus*

## INTRODUCTION

Endocrine Disruptors are defined as a group of chemicals, either natural (Oestradiol) or synthetic (Bisphenol A), that are released in the environment and suspected to alter the functions of the endocrine system and, therefore, causing deleterious health effects in an intact organism, its offspring and populations or subpopulations of organisms (US EPA., 1997). The number of molecules described as Endocrine Disruptors (EDs), their presence in environment and the understanding of their effect has been constantly increasing during last decades (Schug et al., 2016). Here we focus on Oestradiol and Bisphenol A (BPA) to have models as both natural and synthetic endocrine disruptors that are prevalent in environment. BPA is an industrial organic compound containing two phenol functional groups (Figure 1). BPA is classified as “moderately toxic” and “toxic” to aquatic wildlife within a range from 1.0 to 10 mg/L by the European Commission and the United States Environmental Protection Agency (Puig., 2012). However, studies of BPA effects on wildlife indicate that it might be toxic at concentrations below 12 µg/L (Puig., 2012). Oestradiol (Figure 1) is a well-known mammalian female hormone derived from cholesterol responsible for the apparition of secondary sex characteristics and is the most potent of oestrogens. Oestradiol is generally released in environment through wastewater and animal excrement. Concentration in natural

water (rivers, coast bays, lakes) were found to range from 0.8 to 5 ng/L upstream away from sewage treatment plant or agricultural lands and can reach more than 70 ng/L near or downstream sewage treatment plant (Dorabawila and Gupta., 2005).

Due to the presence in environment of compounds with deleterious impact such as growth disorders, sex differentiation disorders or increase of thyroid cancer risks on both human and wildlife (Street et al, 2018), the development of innovative detection systems is necessary to assess the level of prevalence of EDs in water, soil or food. However, current detection methods are generally expensive and use complex approaches such as cell proliferative assay or flow cytometry assay. In fact, the most sensitive methods currently available for the detection of (anti)estrogenic activity are cell-based assays such as ER-CALUX, MELN, T47D-KBLuc and the Yeast Estrogen Screen (YES) assays. Although these methods have been successfully developed to overcome animal testing and reach limit of detection of pg-ng/L, they have the disadvantages to require 24 h for test completion and specific laboratories equipment and personnel (Rita La Spina., 2018).

Hence, we propose to develop low-cost, fast and easy to use test based on an EDs-dyes competitive system for ligninolytic enzymes produced by *Pleurotus ostreatus*. *Pleurotus ostreatus* is a Basidiomycota that belongs to the genus *Pleurotus*, also known as oyster mushrooms. This species can be cultivated for production of food, feed,

enzymes or medicinal compounds. They can also be used for waste degradation or detoxification (Gregori et al., 2007). They are also well-known for their ability to degrade lignin through ligninolytic enzymes (Leonowicz., 1999). Indeed, these enzymes including laccase, manganese peroxidase, and lignin peroxidase are used by *P.ostreatus* in order to degrade lignin, a complex structure of diverse phenylpropane randomly linked (Chen., 2014). Thus, since most of the EDs contain aromatic structure such as lignin, their degradation by ligninolytic enzymes has already been observed (Puig., 2012). Simultaneously, number of dyes, such as Remazol Brilliant Blue R (RBBR) and Brilliant Green (BG) have been described to be sensitive and degradable by diverse ligninolytic enzymes produced by more than a hundred fungi and leading to their discoloration (Kátia et al., 2005). These fungi include as example *Marasmius cladophyllus* (Sing et al., 2017) and *Pleurotus ostreatus* (Kátia and Dácio., 2006). This sensitivity and degradability might be explained by their molecular structures containing at least one aromatic function (Figure 1). The discoloration resulting from the degradation may be explained by the important and well-known role of aromatic structures in the colors of chemicals (Sklar., 1937). Therefore, the ligninolytic enzymes are supposed to degrade both EDs and dyes leading to a competition of these compounds for the enzymes resulting in the slowdown of the degradation of the colorant meaning a slower discoloration.

In this study, we have compared two approaches. On one hand, we have evaluated the competition between EDs and dyes for the enzyme during *Pleurotus ostreatus* growth on agar medium. On the other hand, the competitiveness between EDs and dyes has been tested after enzyme extraction. We have been able to show the competition between dyes and EDs and evaluate the sensitivity of this biosensor and start experiments to optimize this system.

## MATERIAL AND METHODS

### 1 DYE DISCOLORATION DURING FUNGAL GROWTH ON AGAR PLATE

#### 1.1 Culture media preparation

PDB (Potato Dextrose Broth) and PDA (Potato Dextrose Agar) have been prepared as previously described (NC, 2018). Their preparation is extremely similar since the only difference lies in the presence of Agar in PDA medium. These media have been prepared from raw materials being potatoes and glucose. 200 g of unpeeled potatoes have been sliced and displayed in 1 L of boiling distilled water for 30 min. The mixture was then filtered through a cheesecloth, the retentate was discarded and the flow-through was kept. For PDB, the flow-through was mixed with 20 g of dextrose and the volume was completed to reach 1 L with distilled water. In the case of PDA, an extra 15 g Agar was added, and the mix was boiled in order to correctly dissolve the

Agar. The media were finally autoclaved 15 min at 121 °C and stored in 500 mL bottles at 4 °C.

#### 1.2 Dye preparation (RBBR and BG)

Different concentrations of dyes in water have been prepared. Mother solutions of Remazol Blue Brilliant R (RBBR) (SIGMA, R8001-25G) have been prepared from dehydrated form at 159.6 µM (100 mg/L) used for the competitive enzymatic assay and 319.2 µM (200 mg/L) used for the competition during *Pleurotus ostreatus* growth. Mother solutions of Brilliant Green (BG) (SIGMA, B6756-100G) have been prepared at 26.3 µM (12.5 mg/L) used for the competitive enzymatic assay and 103.9 µM (50 mg/L) used for the competition during *Pleurotus ostreatus* growth. Solutions were finally autoclaved at 121 °C for 15 min and store at 4 °C.

#### 1.3 Medium preparation for competitive tests

Two different culture media have been compared, PDA and PDB supplemented with Yeast extract (PDY) to evaluate the efficacy of dye discoloration versus *Pleurotus ostreatus* growth. The two dyes (RBBR and BG) were used at 319.2 µM (200 mg/L) for Remazol Blue and 103.9 µM (50 mg/L) for Brilliant Green and mixed with the medium containing 5 % of 30 mmol/L acetate buffer pH 5.0.

#### 1.4 Assays in agar medium

Fragment of mycelium were taken from an established culture of *Pleurotus ostreatus* via p1000 micropipette tip used to plug mycelium in order to have a reproducible amount of mycelium (-5 mm diameter) during the sampling. Fragments of mycelium were then displayed on PDA medium in 90 mm petri dishes. A particular care has been taken to place the fragment in order to maximize the surface contact between the mycelium and the medium. The plates were incubated 6 days at 28 °C.

### 2 ENZYMATIC ASSAY AFTER LIGNINOLYTIC ENZYMES EXTRACTION

#### 2.1 Fungal culture on whole cereal grain

The enzymes used to produce the biosensor are secreted by the fungus *Pleurotus ostreatus* which has been provided by the Agri-food Engineering laboratory of Sup'Biotech. A plug of 5 mm of fungal mycelium has been inoculated in 25 ml of PDB and cultivated for 6 days at 28 °C under shaking condition (120 rpm / 0.5 g). A fungal solid-state fermentation was performed on whole cereal grain of wheat. For this, the grains were steam cooked for 50 min and then sterilized at 121 °C for 15 min. Distilled water was added in order to reach 65 % moisture content. 25 g of whole cereal grain were inoculated with the fungi cultivated in PDB and then incubated at 30 °C for 10 days in a wide neck Erlenmeyer.

## 2.2 Enzyme extraction

As the ligninolytic enzymes are secreted, they can be easily extracted from the fungal culture. The solid-state fermentation was washed three times with 50 mL extraction buffer (30 mM sodium phosphate buffer, pH 5.0). The extraction buffer recovered was then concentrated 3 times by ultrafiltration (Amin Cone Ultra quatre Centrifugal filter, UFC 910) in centrifuge at 5000 g, 4°C for 15 min. The concentrated ligninolytic enzymes were stored at -20 °C.

## 2.3 Endocrine disruptor preparation

Solutions have been prepared in 50 mL falcons. Respectively 25 mg and 100 mg of Oestradiol and Bisphenol A were dissolved in 20 mL, firstly in 10 mL of ethanol and then, after complete solubilization, 10 mL of distilled water have been added in order to obtain respective concentration of 3.67 mmol/L (1 g/L) and 17.52 mmol/L (4 g/L).

## 2.4 Enzymatic assay

The effect of the endocrine disruptors on the dye degradation was tested in 96-deep well plates. In each well, 15 µL of enzyme was added to fix its quantity, 5 µL of dye, 2 to 20 µL of mother solutions of endocrine disruptors and completed by the buffer until reaching 200 µL. The plates were incubated during different period of times (3 h, 4 h, overnight) at 30 °C. The competitive inhibition of ED on dye degradation was evaluated through a comparison of the degradation of dye in absence of ED. Moreover, controls allowing to evaluate the impact of ED on dye absorbance (in absence of enzymes) and controls allowing to evaluate the impact of enzyme on absorbance measurement have been performed in order to adjust the calculation of dye degradation. The measures of absorbance of the dyes were performed for RBBR at 535 nm to detect the formation of a pink color, 625 nm for the blue wavelength and for the Brilliant Green at 632 nm.

## 2.5 Membrane filtration /concentration of BPA

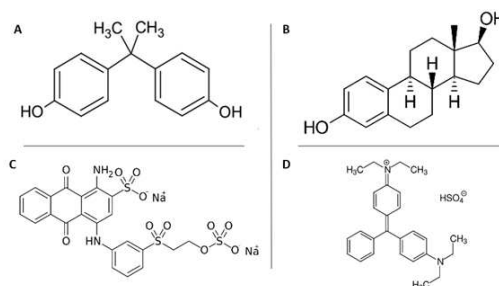
Due to the limit of detection of our test, the development of a protocol allowing the increase of this threshold of detection was considered. Therefore, we aimed to develop a process of concentration of the endocrine disruptors. We have so decided to concentrate the ED solution using binding on specific membranes.

As syringe filters with nylon membrane (0,22µm pore, 13mm Ø: SFNY-122-100) show higher recoveries and lower relative standard deviation than other membranes (Li et al., 2015), the EDs were fixed on a nylon membrane by syringe's injections. Preliminary study has been performed to determine the loading limits of the filter. Several tests of quantity and volume have been performed and the measurement shows that 34 µg no matter the volume injected of BPA is the saturation limit. For assays, 1 ml of

the diluted ED solution was loaded on the filter and elute with 0,5 ml of Acetonitrile. To be able to control exactly the volume and concentration of BPA for further tests, the acetonitrile has been evaporating overnight, and the ED were resuspended in 50 µl of Acetonitrile, leading to a 20-fold concentration.

## RESULTS

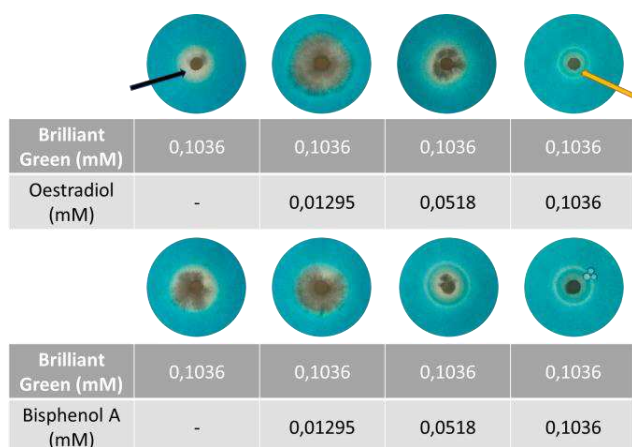
### 1 DYE DISCOLORATION DURING *PLEUROTUS OSTREATUS* GROWTH



**Figure 1:** Structures of BPA (A), Oestradiol (B), RBBR (C) and Brilliant Green (D)

In a first experiment, the discoloration of dyes (BG and RBBR) during *Pleurotus ostreatus* growth has been assessed, as it has been previously described (Kátia and Dácio, 2006) in liquid and agar culture medium. The dye discoloration on liquid media was not conclusive as the dye discoloration assessment lasts 4 weeks for complete discoloration of the medium and the cultures grown in presence of Brilliant Green did not survive probably due to toxic effects of the dye combined with the limits of *Pleurotus ostreatus* growth on liquid culture conditions (data not shown). For the approach on agar culture, a discoloration has been observed for both dyes during the growth of *Pleurotus ostreatus* and the medium was completely discolored when the fungus had colonized the whole plate after 3 weeks of culture (data not shown). Then, the discoloration of the two dyes in absence or in presence of EDs (Bisphenol A and Oestradiol) has been tested at equimolar concentration (103,6 µM), 1/2 ratio (EDs = 51,8 µM) and 1/8 ratio (EDs = 12,95 µM).

The Figure 2 presents the discoloration of Brilliant Green which is observed through the formation of a clear ring around the mycelium plug (black arrow). When increasing the concentration of endocrine disruptors, either BPA or Oestradiol, the discoloration of the dye (mostly in terms of intensity) is partly inhibited (orange arrow). These results suggest that the presence of both endocrine disruptors decreases *Pleurotus ostreatus* ability to discolor the plate and so to degrade the Brilliant green. It also seems that the growth of *Pleurotus ostreatus* is affected by both EDs and dyes.



**Figure 2:** Dye discoloration after 2 weeks of *Pleurotus ostreatus* growth in the presence of Brilliant Green and with different EDs concentrations.

The same observations and conclusions were made with experiments performed in presence of Remazol Blue dye (data not shown). Although those results are encouraging for the use of this approach to assess the presence of endocrine disruptors, its potential seems limited due to the time required to evaluate the inhibition of the dye discoloration and the difficulty to assess accurately the presence of EDs (discoloration evaluated visually without quantitative measurement).

## 2 INHIBITION OF DYE DISCOLORATION IN PRESENCE OF EDs AFTER LIGNINOLYTIC ENZYME EXTRACTION

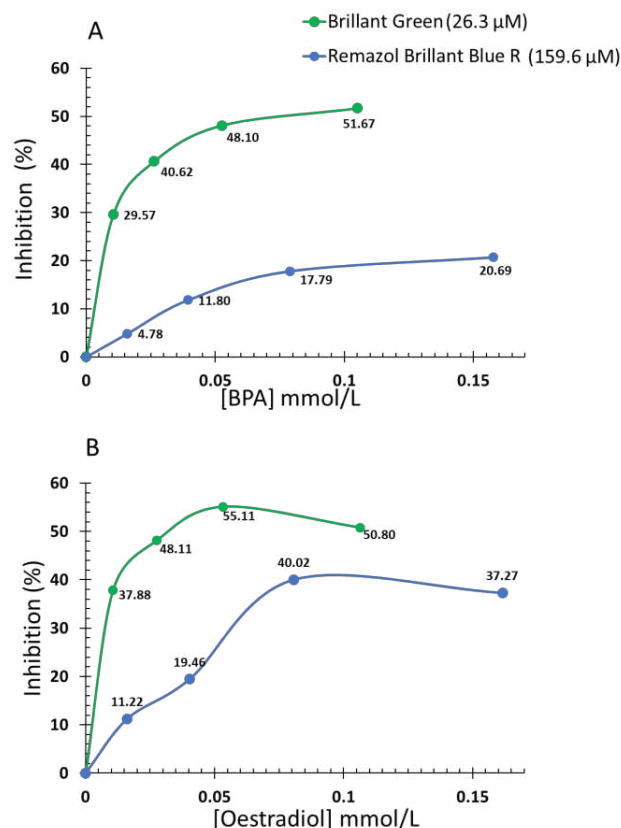
A second approach has then been carried out through the evaluation of BG and RBBR discoloration after extraction of ligninolytic enzymes from *Pleurotus ostreatus* grown on wheat grain. We have then established that this discoloration, measured by spectrophotometry, reached around 90% after overnight incubation (data not shown).

Then, we have evaluated the inhibition of this discoloration in presence of EDs (Figure 3).

To determine the percentage of enzymatic inhibition, we assessed the ratio between the absorbance of the dye (either RBBR or BG) in the presence of EDs at a particular concentration and its absorbance without EDs. Indeed, as explained in the introduction, ligninolytic enzymes degrade both EDs and dyes leading to a competition of these compounds for the enzymes resulting in the slowdown of the degradation of the colorant meaning a slower discoloration and a higher absorbance of the dye in presence of EDs. An inhibition of RBBR and BG has been observed in presence of either BPA (Figure 3, panel A) or either Oestradiol (Figure 3, panel B).

This inhibition was more efficient when Brilliant Green was used as dye and reached 40 % when EDs and dye were at equimolar concentration. This inhibition was also present when Remazol Blue was used as a dye with 20 % and 40 %

of inhibition observed at equimolar concentration of Bisphenol A and Oestradiol respectively.



**Figure 3:** Inhibition of dye discoloration after 18h of incubation in presence of BPA (A) and Oestradiol (B)

## 3 FIRST LEADS TOWARDS AN OPTIMIZATION OF THE COMPETITION ASSAY

The results obtained (Figure 3) have then shown that a competitive assay between dyes and EDs might be used as an EDs detection system. However, an optimization is required to develop such model. Firstly, in term of specificity as the competitive assay give information about the presence of aromatic compounds limiting the degradation of dyes. Secondly, in term of sensitivity as the threshold of detection is much higher than the concentration of EDs found in environment. Visual observation of the formation of a specific color during the process opens a new field of investigation to evaluate the specificity of the detection system. Indeed even if the inhibition of the enzyme activity with the couple Bisphenol A/Remazol Blue was the slowest (Figure 3), we observed an apparition of a pink coloration during the enzymatic assay which does not appear with the other endocrine disruptors/dyes couple (Oestradiol/Remazol Blue) (Figure 4, compare panel A to panel B). These observations might be explored to further identify the existing link between dye discoloration and specificity of EDs.

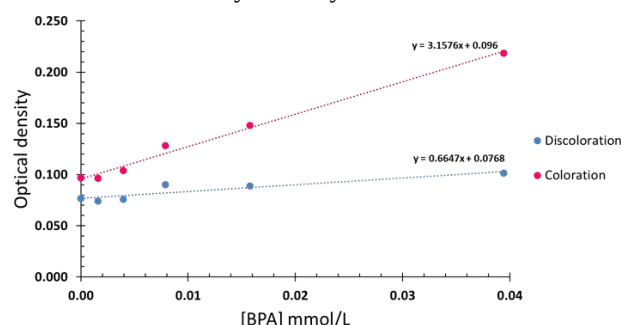
|               |     |    |    |      |   |   |   |
|---------------|-----|----|----|------|---|---|---|
| <b>A</b>      |     |    |    |      |   |   |   |
| RBBR (157 μM) | +   | +  | +  | +    | + | - | + |
| BPA (μM)      | 157 | 79 | 39 | 15.6 | - | - | - |
| Enzyme        | +   | +  | +  | +    | + | + | - |

|                 |     |    |    |    |   |   |   |
|-----------------|-----|----|----|----|---|---|---|
| <b>B</b>        |     |    |    |    |   |   |   |
| RBBR (162 μM)   | +   | +  | +  | +  | + | - | + |
| Oestradiol (μM) | 162 | 81 | 41 | 21 | - | - | - |
| Enzyme          | +   | +  | +  | +  | + | + | - |

**Figure 4:** Discoloration evolution of Remazol Blue in presence or absence of endocrine disruptors (BPA in A and Oestradiol B) of enzymatic assay after 12 h of incubation.

As mentioned above, an optimization in term of sensitivity is also required. In this context we have compared the absorbance values of the initial RBBR (625 nm) and the pink color (535 nm). As shown in Figure 5, the assessment of the apparition of the pink color seems easier to evaluate than the discoloration of Remazol Blue with the spectrophotometer, as the optical density slope is higher for the pink coloration than the initial Remazol Blue dye. As example, at 0,015 mmol/L of Bisphenol A, the enzymatic inhibition is only estimated at 5 % (Figure 3) according to the dye discoloration which can be considered negligible to assess the presence of a potential endocrine disruptor. However, for this concentration of BPA (Figure 5), the absorbance detected is increased by 60 % (from 0,08 (625 nm, blue) to 0,14 (535 nm, pink)) which can be sufficient to detect the inhibition of the enzyme activity and so the presence of an endocrine disruptor. Those preliminary results show that the use of secondary dye absorbance might increase the sensitivity of the system.



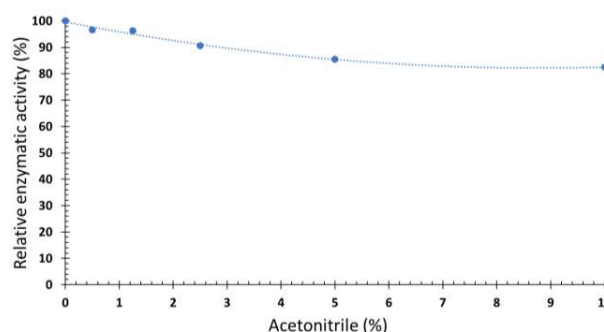
**Figure 5:** Comparison of the dye degradation and the apparition of the pink color.

#### 4 OPTIMIZATION BY CONCENTRATION OF INITIAL SAMPLE CONTAINING EDS

As generally observed in the environment, EDs are present at really low concentration (Dorabawila and Gupta., 2005). It is then necessary to increase the sensitivity of our easy-to-use system. It is possible to concentrate EDs using

binding on nylon membrane (Li et al., 2015). Extraction and concentration of endocrine disruptors have been performed on such membranes. The bond EDs have then been eluted with acetonitrile. However, such organic solvent might inhibit the activity of the enzymes present in the biosensor. In order to determine the impact of this solvent, the enzymatic assay has been carried out overnight in presence of 0.5, 1.25, 2.5, 5 and 10 % (v/v) of acetonitrile (Figure 6). Optical density was measured at 625 nm to evaluate the degradation of RBBR. It appears that acetonitrile reduces the activity of the enzyme up to 18 % in presence of 10 % acetonitrile (Figure 6). In the case of the experiment, a reduction of the activity 9 % seems to be acceptable to keep a certain sensitivity of the system and 5% of acetonitrile was selected for the next experiment.

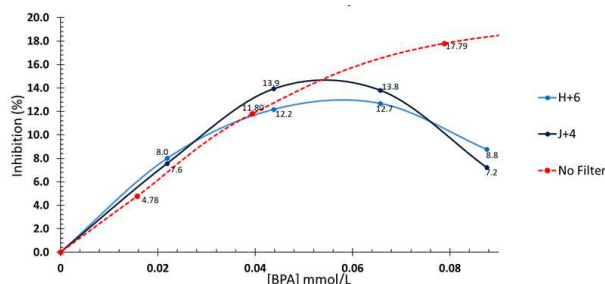
Therefore, after loading 1mL of known quantities of BPA on nylon membrane ranging from 0 to 0.175 mmol/L (0 to 40 mg/L), an elution in 50 μL of acetonitrile was performed. Further concentration steps were done by evaporation to get theoretical concentration ranging from 0 to 3.5 mol/L (0 to 800 mg/L).



**Figure 6:** Impact of acetonitrile on enzymatic activity

Finally, an enzymatic assay has been performed on the BPA recovered after filtration in order to test if the filtration and concentration steps impact the inhibition activity of the ED. We chose to test the filter for BPA for two reasons. Firstly, it is the only tested ED that show perspectives of specificity for the system and secondly it is easier to optimize the sensitivity to reach detection limits that correspond to the level of toxicity for this endocrine disruptor. By the dilution occurring in the plate, we were able to proceed the test with BPA concentration ranging from 0 to 0.09 mmol/L. As we can see (Figure 7), the inhibition of the discoloration seems similar between the two conditions (without or after filtration on nylon membrane).

However, it seems that after 0.06 mmol/L of BPA, corresponding to a filtration of 1ml at 0.131 mol/L of BPA, the inhibitory activity of BPA seems to decrease. This could be due to an incorrect binding of BPA on the filter. So, we can conclude that the filtration and concentration do not seem to affect the competition between EDs and dyes and so it seems possible to use concentration through filtration to quantitatively measure EDs in sample.



**Figure 7:** Evolution of the inhibition of BPA in three different conditions: without filtration on nylon membrane (No Filter), after 6 hours of incubation (H+6), after 4 days of incubation (J+4).

## DISCUSSION/CONCLUSION

This study aims to investigate the potential development of a new type of biosensor based on competition between EDs and dyes for ligninolytic enzymes produced by *Pleurotus ostreatus*.

Three different approaches have been compared. The first two were based on competition between EDs and dyes for the enzyme during *Pleurotus ostreatus* growth on agar and liquid medium. The last approach aimed to study competitiveness between EDs and dyes after the enzyme extraction. We firstly assess the dye discoloration due to the action of ligninolytic enzymes, all approaches tested lead to a degradation of dyes and therefore an efficient discoloration of the medium except the approach focusing on the growth of *Pleurotus ostreatus* on liquid media.

In a second time, we proceed to the assessment of the competitive inhibition of ligninolytic enzymes based on the discoloration of the dyes compared with the discoloration measured in presence of EDs. Although the enzymatic assay seems to be the most adequate approach as it allows the fastest degradation rate as well as the efficiency at lowest concentration of EDs, the sensitivity of this preliminary system was not sufficient to detect the lowest level of EDs at which they are harmful for human health and the environment. Indeed, for Bisphenol A we have a detection threshold at 24 mg/L while BPA effects on wildlife has been assessed between 1 and 10 mg/L and might be toxic at concentrations below 12 µg/L (Puig, 2012). In the case of Oestradiol, we have a detection threshold of 29 mg/L while it has been shown that Oestradiol can be toxic at low concentrations (ng/L) in several organisms such as fish, amphibians or mollusks (Deblonde et al, 2013). We then have looked to overcome this limit by proposing two complementary approaches to optimize the system.

The first approach is based on the assessment of the competitive inhibition through the apparition of a pink color resulting from the degradation of the Remazol Blue in presence of BPA. Preliminary results showed a better sensitivity of the system when the apparition of pink is detected compared to when disappearance of RBBR is detected. Indeed, detection of this color formation allows to increase the initial threshold of 24 mg/L to 2,4 mg/L meaning diminution of a factor 10 therefore increasing the

sensitivity. This approach also offers perspectives of specificity that were not primarily expected as this color appear only with the couple Remazol Blue/BPA. A screening of different couple of dyes/EDs could be interesting as particular association of dyes and EDs might lead to the formation of new color during their degradation which could allow to develop a more specific biosensor.

However, it is possible that some endocrine disruptors lead to the formation of the same color which could be a bias for the specificity of the system, that is why further tests will be required. The second approach consisted to concentrate the EDs before the enzymatic assay through nylon filters. However, this approach requires to utilize organic solvents for extraction and concentration of EDs that could have an effect on the enzymatic activity and therefore being a bias for the assessment of the endocrine disruptor presence. In this purpose, as we have used acetonitrile for the extraction, we evaluated its influence on the enzyme activity, and it has been shown that its impact is limited regarding the quantity required for the extraction. Finally, we evaluated the retention capacity of the filters and our results suggest that the concentration capacity can be promising as we increase 20-fold the concentration. Concerning the limits of this approach, high quantity of endocrine disruptors lead to the filter saturation therefore further studies are needed to assess filter capacity. Besides, for this approach, we only focus on Bisphenol A, but this method could be potentially performed with other endocrine disruptors such as Oestradiol or with different organic solvents depending on which solvent is more suitable for a particular type of ED. Finally, a combination of both approaches would allow to optimize our biosensor and overcome the current limits of detection.

## Acknowledgement

We would like to thank Dr. Agnès Saint-Pol, Valentina Glicorijevic and Dr. Jacqueline Bert, for their advices and for guiding us for the bibliography research, the redaction of protocols, the presentation of the poster, the research of the speaker for the symposium and the redaction of the article.

## BIBLIOGRAPHY

- Chen, H., "Biotechnology of lignocellulose: Theory and practice", *Biotechnology of Lignocellulose: Theory and Practice*, 2014
- Deblonde et al, "Environmental impact of medical prescriptions: assessing the risks and hazards of persistence, bioaccumulation and toxicity of pharmaceuticals", *Public Health*, 127, 2013, 312-317
- Dorabawila, N., & Gupta, G., "Endocrine disrupter—estradiol—in Chesapeake Bay tributaries", *Journal of Hazardous Materials*, 120(1-3), 2005, 67–71.

- Gregori, A., Švagelj, M., Pohleven, J., "Cultivation techniques and medicinal properties of *Pleurotus* spp" *Food Technology and Biotechnology*, 45, 2007, 236–247.
- Kátia, M., G., M. & Dácio, R., M., "Biodegradation of Remazol Brilliant Blue R by ligninolytic enzymatic complex produced by *Pleurotus ostreatus*", *Brazilian Journal of Microbiology*, 1517-8382, 2006
- Kátia, M., G., M., Dácio, R., M., Vera, L., R., B., "Ligninolytic enzymes production and Remazol Brilliant Blue R decolorization by tropical Brazilian basidiomycetes fungi", *Brazilian Journal of Microbiology*, 36, 2005, 246-252
- Leonowicz, A., Matuszewska, A., Luterek, J., Ziegenhagen, D., Wojtaś-Wasilewska, M., Cho, N. S., Rogalski, J., "Biodegradation of lignin by white rot fungi", *Fungal Genetics and Biology*, 1999
- Leskinen, P., Michelini, E., Picard, D., Karp, M. and Virta, M., "Bioluminescent yeast assays for detecting estrogenic and androgenic activity in different matrices". *Chemosphere*, 61(2), 2005, pp.259-266.
- Li, S., Chen, F., Liu, F., Liu, F., Liu, Z., Zeng, R., Tang, H., Gao, Z., and Zhou, H., "Rapid detection of Bisphenol a in water samples by high-performance liquid chromatography based on syringe filters with nylon membrane extraction", *J. Liq. Chromatogr. Relat. Technol*, 38, 2015, 1474–1478.
- NC, "Potato Dextrose Agar (PDA)- Principle, Uses, Composition, Procedure and Colony Characteristics", *microbiology info.com*, 2018
- Park, J.W., Kurosawa, S., Aizawa, H., Goda, Y., Takai, M. and Ishihara, K., "Piezoelectric immunosensor for Bisphenol A based on signal enhancing step with 2-methacryloyloxyethyl phosphorylcholine polymeric nanoparticle." *Analyst*, 131(1), 2006, pp.155-162.
- Puig, D.,R.,T., "Removal of Endocrine Disrupting Chemicals by the Ligninolytic Enzyme Versatile Peroxidase", *Thesis, Department of Chemistry, Universidade de Santiago de Compostela, Santiago*, 2012
- Rita La Spina et al, « Label-Free Biosensor Detection of Endocrine Disrupting Compounds Using Engineered Estrogen Receptors", *Biosensor*, vol 8, n°1, 2018, 1-15
- Rosol, T., J., DeLellis, R., A., Harvey, P., W., Sutcliffe., C., "Endocrine System", *Haschek and Rousseaux's Handbook of Toxicologic Pathology*, 2013, 2391–2492.
- Schug, T., T., Johnson, A., F., Birnbaum, L., S., Colborn, T., Guillette, L., J., Crews, D., P., Heindel, J., J., "Minireview: Endocrine Disruptors: Past Lessons and Future Directions", *Molecular Endocrinology*, 30(8), 2016, 833–847
- Sing, N., N., Husaini, A., Zulkharnain, A., Roslan, H., A., "Decolourisation Capabilities of Ligninolytic Enzymes Produced by *Marasmius cladophyllus* UMAS MS8 on Remazol Brilliant Blue R and Other Azo Dyes", *BioMed Research International*, 2017, 1–8
- Sklar, A., L., "Theory of Color of Organic Compounds", *The Journal of Chemical Physics*, 5(9), 1937, 669–681
- Street, M., Angelini, S., Bernasconi, S., Burgio, E., Cassio, A., Catellani, C., Amarri, S., "Current Knowledge on Endocrine Disrupting Chemicals (EDCs) from Animal Biology to Humans, from Pregnancy to Adulthood: Highlights from a National Italian Meeting." *International Journal of Molecular Sciences*, 19(6), 2018, 1647
- US-EPA, "Special report on endocrine disruption: An effects assessment and analysis"., *U.S. EPA, Washington, DC*, 1997.





# Antifungal properties of Essential Oils against Aspergillosis

Boutte Dauphine, Calmet Julie, Dardinier Adrien, Donzier Julie, Ducoudray Mathilde, Marceau Fanny,  
Visconti Charlotte

Sup'Biotech, Villejuif, France

\*Corresponding author: jacqueline.bert@supbiotech.fr

**Abstract** - Aspergillosis is a nosocomial infection caused by *Aspergillus fumigatus* and a serious threat for immunocompromised patients. This disease is currently treated with drugs having a high toxicity for human kidneys, especially in already fragilized patients. This study aimed to find an alternative to chemical drug treatments using essential oils (EO). Indeed, some specific EO are known to have antimicrobial properties. In this study, the antifungal efficacy of EO was investigated using *Aspergillus oryzae*, a close parent of *A. fumigatus*, as a model strain, with Amphotericin B and Fluconazole, drugs used for aspergillosis treatment, as positive controls. The experiments consisted in cultivating and observing the growth of the fungus under the effect of 6 different EO: Basil, Thyme Thymol, Peppermint, Lavender, Tea Tree and Oregano Compact. To do so, different methods using Sabouraud medium were implemented. Drugs and EO were tested on *Aspergillus oryzae* either in liquid or solid media. First, a screening test was performed in liquid medium to identify the EO with the highest antifungal property. Then, each EO was studied to determine whether it has a fungicide or fungistatic effect using solid Sabouraud medium. A fungicide substance is a substance that eliminate or kill parasitic fungi whereas a fungistatic substance is only able to inhibit the growth of the fungi without killing it. Finally, the potential synergic effect of EO mixtures has been studied using 1:1, 1:2, 2:1, 3:1 and 1:3 ratios. Results revealed that Oregano Compact and Thyme Thymol have the highest antifungal effect. Then, some synergic effects have been observed using Peppermint with Oregano Compact at ratio 3:1 and Tea Tree with Oregano Compact at ratio 2:1 and 3:1. To go further, the toxicity of some EO at increasing concentrations was investigated on H1299 and Hela cell lines. Results demonstrated that at a concentration of 0.1mg/mL and 0.5mg/mL, EO had little or no effect on cells phenotype whereas at concentrations above 1mg/mL, signs of cell damage were observed.

**Index Terms** - Aspergillosis, *Aspergillus fumigatus*, *Aspergillus oryzae*, Antifungal properties, Essential oils, Fungistatic, Fungicide, Nosocomial infection, Basil, Thym Thymol, Lavender, Oregano, Tea Tree, Peppermint.

## I. Introduction

*Aspergillus fumigatus* is a filamentous fungus that can cause serious diseases in humans. It is present in the external environment (air, soil, plants, water ...), but also inside houses and hospitals [1] in which it can find suitable conditions for its development (humidity, insulation and lack of ventilation). It produces millions of spores that are carried in the air and inhaled into the lungs. For a major part of the population, natural defenses are enough to eliminate it before the infection. However, immunocompromised patients can contract a disease called aspergillosis when the number of

spores is too important [1] [2]. There are four types of aspergillosis induced by *A. fumigatus*. According to Pasteur Institute, allergic bronchopulmonary aspergillosis is common in patients with asthma and cystic fibrosis. Then, aspergilloma is another disease that can be contracted by patients with a history of tuberculosis or sarcoidosis: the fungus develops in the lung cavity affected by these previous diseases [3]. *Aspergillus sinusitis* is a much more serious and invasive disease for patients with a weak immune system and requires antifungal treatment. Finally, invasive aspergillosis is the fourth form of diseases caused by *A. fumigatus*. This aspergillosis spreads to other organs that the

lungs such as heart, liver, kidneys and brain through blood circulation. According to Pasteur Institute, this disease is the second cause of death from fungi infection in hospital, therefore we chose it for our case study. Existing treatments for invasive aspergillosis are powerful chemicals, as amphotericin B [4], fluconazole [5], itraconazole, that can cause adverse toxic effects in patients [6]. Besides, according to Kontoyiannis et al. 2012 [7], *A. fumigatus* could sometimes show a resistance for amphotericin B.

The aim of this study was to find an alternative to chemical drugs for invasive aspergillosis treatment. *A. fumigatus* is a pathogenic strain classified as Category 2. *Aspergillus oryzae*, a non-pathogenic specie, was used as a model for in vitro experiments. This specie has been suggested to be phylogenetically very close to *A. fumigatus*, even enough for sexual reproduction [8]. Similarly, to other studies which demonstrated the antifungal efficacy of some natural phenolic compounds found in Essential Oils (EO) [9], this article reports a study on six EO containing linalool, carvacrol and other phenols extracted from Tea Tree, Basil, Lavender, Oregano Compact, Thyme Thymol and Peppermint. These EO have been selected for their known antifungal properties against various *Aspergillus* strains [10]: Oregano Compact and Thymol EO showed antifungal properties against *Aspergillus flavus*, *Aspergillus niger* and *Aspergillus ochraceus* [11]; Tea tree EO [12] showed antifungal properties against 14 fungal strains including *A. fumigatus* and *A. niger*; Lavender EO was efficient against *Candida albicans* [13] and *Aspergillus nidulans*; Peppermint EO is known to have antimicrobial effects [14] and needed investigations for its potential antifungal activity; several Basil EO showed antifungal properties at a high concentration (150 µL/L as minimum inhibitory concentration) against *A. fumigatus* [15]. Synergistic antimicrobial effects of combined essential oils have been suggested in recent publications [16] [17].

## II. Materials and Methods

### *Aspergillus oryzae* strain

To assess the antifungal activity of EO, *Aspergillus oryzae*, a close species of *A. fumigatus*, was taken as an experimental model. Manipulation of *A. oryzae* had the advantage to be feasible in a class 1 microbiology laboratory. *A. oryzae* suspension was kindly provided by Ms Saint-Pol, Sup'Biotech (LRPIA). The suspension concentration was determined using dilutions and Malassez cell counting. The optimal concentration of *A. oryzae* for the experiments was determined by screening in Sabouraud liquid medium and found to be 10<sup>4</sup> conidia/mL. A conidium is a spore that ensure the asexual multiplication for a fungus.

### Study of *A. oryzae* growth at different EO concentrations

Six pure EOs from PurEssentiel® were used to determine the effects of EO on the fungal growth: Tea tree, Lavender, Thyme Thymol, Basil, Oregano Compact and Peppermint. EO were diluted in Tween 0.05%. An experimental weighting of the EO revealed a weight almost equal to the one of water. Therefore, 1 µL of EO was considered to be equal to 1 mg in the following experiments. The screening test was performed in liquid Sabouraud medium, in 24-well plates, with four different concentrations of EO going from 1 mg/mL to 4 mg/mL. Sabouraud liquid medium (250mL) is composed of 2.5g of peptone, 5g of glucose and water (qsp). Amphotericin B (Ampho B) was used as a fungicide control at concentrations between 1 and 4 µg/mL. Observations were performed using the fluorescent cell imager ZOE® (Bio-Rad).

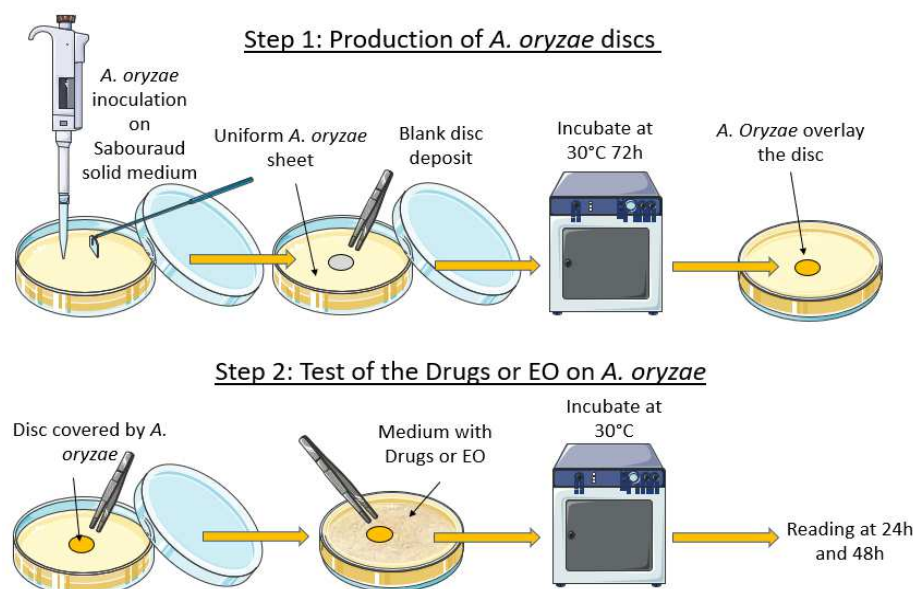
### Disc tests containing *A. oryzae*

For the discs test (**Figure 1**), solid Sabouraud media was inoculated with a liquid suspension of *A. oryzae* at 10<sup>4</sup> conidia/mL and incubated at 30°C for 72h. The solid Sabouraud medium (450mL) is composed of 4.5g of peptone, 9g of glucose, 6,75g of agar and water (qsp). Once covered with fungi, the discs were picked-up and

laid on media containing either nothing but medium, water, DMSO, Tween, Amphotericin B (fungicide), Fluconazole (fungistatic) or EO. Petri dishes were incubated at 30°C and results were observed at 24h and 48h. In order to assess if EO had a fungistatic or fungicidal effect, discs previously incubated on EO containing media were put on fresh Sabouraud media and incubated at 30°C. Fungal growth was observed again at 24 hours.

### Study of *A. oryzae* growth with EO mixtures

The study of EO mixtures was performed to demonstrate potential synergic effects between EO. Only Thyme Thymol (Th), Tea Tree (TT), Peppermint (P) and Oregano Compact (O) were tested for lack of time. EO mixtures at a final concentration of 4 mg/mL were tested with different ratio as represented in **Table 1**.



**Figure 1: Discs test with discs containing *A. oryzae*.** Step 1: Production of *A. oryzae* containing discs using *A. oryzae* suspended in liquid Sabouraud medium. *A. oryzae* was then uniformly plated using a sterile rake. Finally, a blank disc was deposited at the center and the plate was incubated at 30°C for 72 hours. Step 2: Test with the drugs or EO was performed by recovering the discs covered by *A. oryzae* and displaying them on a plate containing Sabouraud medium and drugs or EO. The discs were incubated at 30°C and readings were performed at 24 hours and 48 hours.

**Table 1: Ratio of EO mixtures.** 5 different ratios were tested for mixtures of Peppermint (P) + Oregano Compact (O); P + Tea Tree (T); O + TT; O + Thyme Thymol (Th) and TT + TH; P + Th

|              | P + O  |        | P + TT |        | O + TT |        |
|--------------|--------|--------|--------|--------|--------|--------|
| <b>1 : 1</b> | 50 µl  | 50 µl  | 50 µl  | 50 µl  | 50 µl  | 50 µl  |
| <b>1 : 2</b> | 50 µl  | 100 µl | 50 µl  | 100 µl | 50 µl  | 100 µl |
| <b>2 : 1</b> | 100 µl | 50 µl  | 100 µl | 50 µl  | 100 µl | 50 µl  |
| <b>1 : 3</b> | 50 µl  | 150 µl | 50 µl  | 150 µl | 50 µl  | 150 µl |
| <b>3 : 1</b> | 150 µl | 50 µl  | 150 µl | 50 µl  | 150 µl | 50 µl  |

|              | O + TH |        | TT + TH |        | P + TH |        |
|--------------|--------|--------|---------|--------|--------|--------|
| <b>1 : 1</b> | 50 µl  | 50 µl  | 50 µl   | 50 µl  | 50 µl  | 50 µl  |
| <b>1 : 2</b> | 50 µl  | 100 µl | 50 µl   | 100 µl | 50 µl  | 100 µl |
| <b>2 : 1</b> | 100 µl | 50 µl  | 100 µl  | 50 µl  | 100 µl | 50 µl  |
| <b>1 : 3</b> | 50 µl  | 150 µl | 50 µl   | 150 µl | 50 µl  | 150 µl |
| <b>3 : 1</b> | 150 µl | 50 µl  | 150 µl  | 50 µl  | 150 µl | 50 µl  |

### Toxicity assessment of Essential oils on two different cell lines

In order to find EO concentrations allowing a fungal growth inhibition without damaging cells, the effects of Th, O, PM, TT and Lavender (L) were investigated at different concentrations on both H1299 and Hela cell lines. The H1299 cell line was established from a lymph node metastasis. This node come from a patient who had received prior radiation therapy. The Hela cell line is a commonly used human cell line. This line was derived from cervical cancer cells. Both cell lines were prepared at confluence and provided by Mrs Perdiz (CEA). The five most efficient EOs were used for this experiment, *i.e.* Thyme Thymol, Oregano Compact, Peppermint, Tea Tree and lavender. A subculture of each human cell line was performed in 24-well plates with DMEM medium. DMEM medium was removed after 24h of culture and was substituted by DMEM containing EO at concentration levels of 0,1mg/mL, 0,5mg/mL, 1mg/mL and 2mg/mL. The phenotype of the cells was observed with a fluorescent cell imager ZOE® (Bio-Rad) for each strain after 24h and 48h of incubation at 30°C.

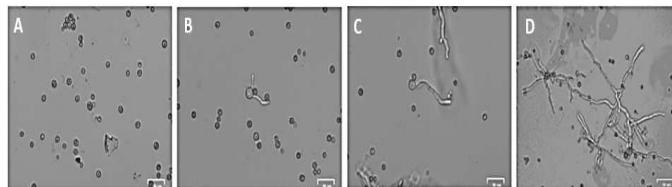
### III. Results and Discussion

The main objective of this study was to assess the antifungal activity of EO for a potential application in invasive aspergillosis treatment. Besides, a comparison with existing treatments such as Amphotericin B and Fluconazole was driven by using them as positive controls.

Experiments consisted of a screening test in liquid Sabouraud medium with different EO at several concentrations. A fungal concentration of  $10^4$  conidia/mL was used for this experiment. This concentration allowed a correct visualization of the fungal germinative tubes with a fluorescent cell imager (**Figure 2**).

### Screening on liquid Sabouraud medium demonstrated that Thyme and Oregano EOs were the most efficient at 1mg/mL and 2mg/mL

In this experiment, a normal conidial growth was observed for the positive control Amphotericin B (data not shown). As expected, no fungal growth was observed in presence of the fungicide drug, *i.e.* Amphotericin B.



**Figure 2: Growth stages of *A. Oryzae* in Sabouraud medium at 30°C during 24 hours (A: initial conidia, B: sprout of the germinative tube, C: development of the germinative tube, D: branching of the germinative tube)**

An important growth was observed in presence of the fungistatic drug, *i.e.* Fluconazole (data not shown). This was not expected, as Fluconazole is a currently used treatment against aspergillosis. Fluconazole was not very effective as a growth inhibitor control; however, Amphotericin B control was enough to interpret the results. Basil, Lavender, Tea tree and Peppermint had little or no effect on *A. oryzae* growth at a concentration of 1 mg/mL (**Table 2**). Only Oregano Compact and Thyme Thymol were effective at 1 mg/mL (**Table 2**). At a concentration of 4 mg/mL, a fungistatic effect was observed with the 6 different EOs. Indeed, Thyme Thymol and Oregano Compact EOs showed a significant growth inhibition of *A. oryzae* at 1mg/mL (**Table 2**). A progressive diminution of fungal development was observed by increasing Tea Tree, Lavender, Basil and Peppermint EO concentrations. Tea Tree showed no conidial growth from a concentration of 2 mg/mL. These results suggest that the EO with the most important antifungal activity are Thyme Thymol and Oregano Compact.

**EOs can be classified according to their efficacy as following: Thyme Thymol > Oregano Compact > Tea Tree > Peppermint > Lavender > Basil.**

The disc test was driven to determine whether EOs had a fungistatic (inhibits the growth without killing) or fungicidal (kills the fungus) effect.

The results obtained for this experiment confirmed the previous ones of the screening test: Oregano Compact and Thyme Thymol are the most efficient fungicide EOs, whereas other EOs are less efficient and only fungistatic (**Table 2**).

As shown in **Figure 3**, the fungus had well grown on medium containing water, Tween, DMSO, and only medium, meaning that the different reagents to be used did not disturb *A. oryzae* growth. The

positive controls were concilient, as Amphotericin B completely inhibited the fungal growth and Fluconazole clearly delayed it. Then, a reduced fungal growth was observed with Basil, Peppermint, Lavender, and Tea Tree EOs at a concentration of 2mg/mL. There was no fungal growth for Thyme Thymol and Oregano Compact at the same concentration. This means that Thyme Thymol and Oregano Compact EOs are able to stop the fungal growth at a concentration of 2 mg/mL, while the others only slow the growth of the fungi down. Then, for an EO concentration of 4 mg/mL, a stronger effect with Peppermint, Lavender and Tea Tree EOs was observed, while Basil EO still allowed a good fungal growth. The essential oils were then classified according to their growth inhibition effect: Thyme Thymol > Oregano Compact > Tea Tree > Peppermint > Lavender > Basil

*Table 2: Effect of EOS on conidial growth. Summary of the EOs tested and their effect on Aspergillus oryzae growth*

| EO            | Thym Thymol | Origano   | Tea Tree    | Peppermint  | Lavender    | Basil       |
|---------------|-------------|-----------|-------------|-------------|-------------|-------------|
| Effects       | Fungicide   | Fungicide | Fungistatic | Fungistatic | Fungistatic | Fungistatic |
| Concentration | 1mg/mL      | 1mg/mL    | 3mg/mL      | 3mg/mL      | 4mg/mL      | 4mg/mL      |

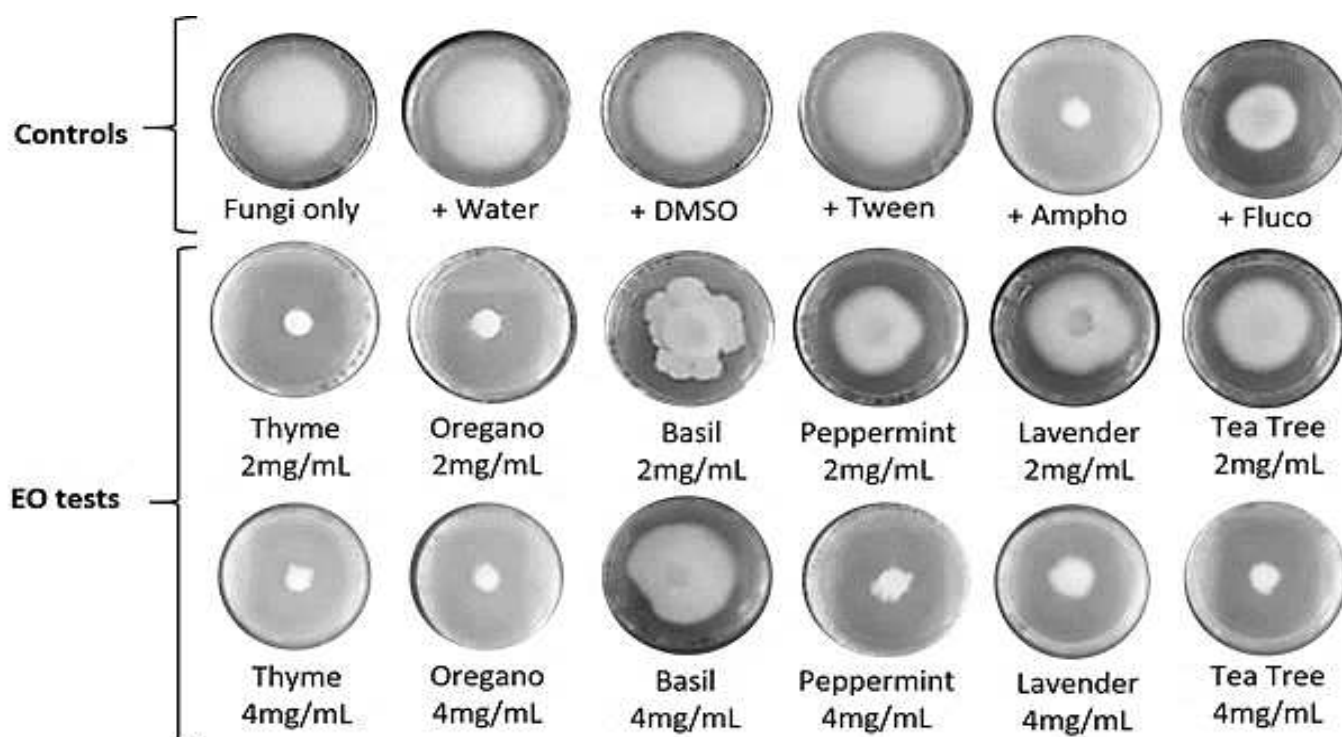


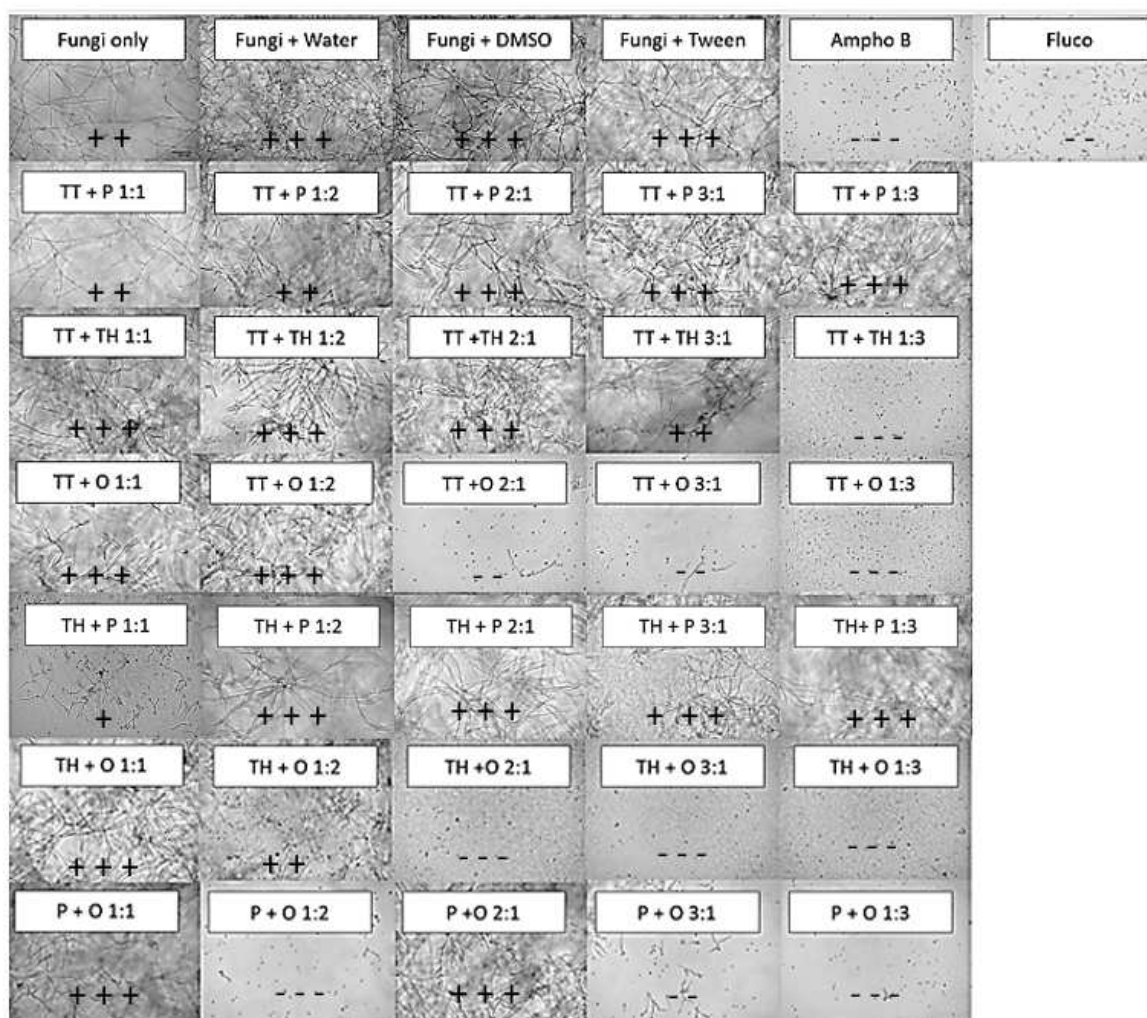
Figure 3: **Discs test with discs containing *A. oryzae*.** Essential oils of Thyme Thymol, Oregano Compact, Basil, Peppermint, Lavender and Tea tree were tested at 2 concentrations (2 mg/mL and 4 mg/mL) in Sabouraud medium. *A. oryzae* was grown 48h on discs, then discs were deposited on solid Sabouraud medium.

Additional experiments were conducted using discs soaked with EO, deposited on solid medium previously spread with *A. oryzae*. As no inhibition growth was observed even in high EO concentration, we hypothesized that EOs had evaporated in plates. Indeed, EOs are volatile and the evaporation during the 30°C incubation could be an explanation to this result.

**Peppermint with Oregano Compact at ratio 3:1 and Tea Tree with Oregano Compact at ratio 2:1 and 3:1 showed a synergic effect for fungal growth inhibition**

Since synergistic antibacterial effects of combined essential oils have already been

suggested, experiments have been performed in order to identify the mixture ratio which gives the highest growth inhibition activity possible on *A. oryzae* and potential synergic effects (**Figure 4**). EO were at final concentration of 4mg/mL. This test showed an important *A. oryzae* growth for negative controls (**Figure 4**, Fungi only, Fungi + water, Fungi + DMSO and Fungi + Tween), and a high growth inhibition by Amphotericin B and Fluconazole (**Figure 4**, Ampho B and Fluco). All ratios containing P and TT showed a fungal growth. TT + Th mixtures only showed complete growth inhibition with 1:3 ratio. TT + O mixtures showed growth inhibition for 2:1, 3:1 and 1:3 EO ratios. Th + P mixtures showed a growth inhibition only for 1:1 ratio.



**Figure 4: Screening test of essential oil mixtures effect on *A. oryzae* growth.** Positive controls used is Amphotericin B, commonly used drugs against *A. fumigatus*, and Fluconazole. The fungal growth was observed after 48h on liquid Sabouraud medium using the cell life imager ZOE®. 5 different ratios were tested, containing Peppermint (P) + Oregano Compact (O); P + Tea Tree (TT); O + TT; O + Thyme Thymol (Th); TT + Th and P + Th. The growth scale of *A. oryzae* conidia goes from no conidial growth (---) to normal conidial growth (+++).

Ratios 2:1, 3:1 and 1:3 for TH + O EO mixtures showed a maximal growth inhibition. Finally, P + O mixtures showed growth inhibition for ratios 1:2, 3:1 and 1:3. These results confirm that Thyme Thymol and Oregano Compact EOs are the most efficient against *A. oryzae* growth. Some EOs seem to have synergistic effects. Indeed, Tea Tree and Oregano Compact mixtures showed a similar complete growth inhibition for 2:1 and 3:1 ratios, but no growth inhibition for 1:2 ratio, suggesting a possible better antifungal effect for twice as much Tea Tree than Oregano Compact, compared to more

Oregano Compact than Tea Tree. Peppermint and Oregano compact mixtures also showed a complete fungal growth at ratio 1:1 and 2:1, whereas a good growth inhibition was observed at ratio 3:1. This suggested that a synergic effect could exist when there is much more Peppermint than Oregano Compact and inversely, when there is twice as much Peppermint than Oregano Compact. Surprisingly, Thyme Thymol and Oregano Compact at ratio 1:1 and 1:2 showed almost no growth inhibition. This may be explained by reactions between these two EOs, leading to almost no growth inhibition effect when there

is twice as much Oregano Compact than Thyme Thymol, and complete growth when Thyme Thymol and Oregano Compact are at the same amount. This hypothesis of reaction between EO may be confirmed by the important fungal growth with all 1:1 ratios.

### **EOs were toxic for cells at concentrations above 1mg/mL.**

Finally, experiments on human cell lines were performed in order to assess the toxicity of the EO on human cells (data not shown). Concentration of 0.1mg/mL, 0.5mg/mL, 1mg/mL and 2mg/mL of Thyme, Oregano, Tea Tree, Peppermint and Lavender EOs were tested on both Hela and H1299 cell lines at approximately 80% of confluence (data not shown). Results demonstrated that at concentrations 0.1mg/mL and 0.5mg/mL the two cell lines were in good shape. However, EO concentrations above 1mg/mL gave clear signs of cell damage. Indeed, EOs are known to be very toxic for cells (human cells) and no concentration for inhibiting the fungal growth without damaging the cells have been found. However, experiments carried on a 3D model may give different results as cell would undergo the stress differently than on a 2D model like the one carried out in this article.

### **III. Perspectives**

To go further, it would be interesting to study the different types of antifungal activities (fungicide versus fungistatic) with narrower concentration ranges. Besides, it would be interesting to study the persistency of EO inside the cells and to assess the stability of their antifungal activity over time. In addition, the same experiments should be done using *Aspergillus nidulans* then *A. fumigatus* rather than *A. oryzae* to confirm the efficacy of EOs on these strains. The toxicity experiment on cells could be repeated and the LD50 of each EO should be determined by counting dead cells stained with Trypan blue. As explained in the results, EOs have a toxic effect on human cell lines when applied directly in the medium. Therefore, experiments should be driven on cell

line with an indirect contact of the EO on a 3D model using, for instance, *organoids*. Organoids are tiny, self-organized three-dimensional tissue cultures that are derived from stem cells. Then, other EOs such as *Eucalyptus Citriodora* [18] or *Eucalyptus Globulus* should be tested. Indeed, *Eucalyptus Globulus* seems to have an antifungal property against *Aspergillus nidulans* according to the literature [19].

### **IV. Acknowledgement**

First, we would like to express our special thanks and gratitude to our project tutor, Mrs Jacqueline Bert, who gave us the opportunity to work on this research project. She invested her full efforts in guiding us through this project, which also helped us to acquire autonomy, anticipation and new skills. We appreciated the guidance given by Mrs Estelle Mogensen, who lent us her laboratory and equipment. Then, we would also like to acknowledge with much appreciation Mrs Agnès Saint Pol for providing us fungal strains and Mr Franck Yates, who suggested us to push our experiments on human cellular models. Finally, thanks to Mrs Vanessa Perdiz for providing us human cell lines from the CEA and Mr Nicolas Moniaux for having shared his knowledge on the use of animal model for experimental purpose with us.

### **V. References**

- [1] M. Kousha, R. Tadi, and A. O. Soubani, "Pulmonary aspergillosis : a clinical review," vol. 20, no. 121, pp. 156–174, 2011.
- [2] K. J. Holding *et al.*, "Aspergillosis among People Infected with Human Immunodeficiency Virus : Incidence and Survival," pp. 1253–1257, 2000.
- [3] P. D. Barnes, & K. A. Marr, "Aspergillosis: Spectrum of disease, diagnosis, and treatment." *Infectious Disease Clinics of North America*, vol. 20, no.3, pp. 545–561, vi., 2006.
- [4] D. A. Stevens, V. L. Kan, M. A. Judson, V. A. Morrison, S. Dummer, D. W. Denning, J. E. Bennett, T. J. Walsh, T.



- F. Patterson, & G. A. Pankey. "Practice Guidelines for Diseases Caused by *Aspergillus*." *Clinical Infectious Diseases*, vol. 30, no. 4, pp. 696–709, 2000.
- [5] P. Troke, R. Andrews, M. Marriott, and K. Richardson, "Efficacy of fluconazole (UK-49,858) against experimental aspergillosis and cryptococcosis in mice," *J. Antimicrob. Chemother. Oxford Acad.*, vol. 19, no. 5, pp. 663–670, 1987.
- [6] J. R. Wingard *et al.*, "Clinical Significance of Nephrotoxicity in Patients Treated with Amphotericin B for Suspected or Proven Aspergillosis," no. January 1990, pp. 1402–1407, 1999.
- [7] D. Kontoyiannis, R. Lewis, G. May, N. Osherov, and M. Rinaldi, "*Aspergillus nidulans* is frequently resistant to amphotericin B," *Mycoses*, 2002.
- [8] James E. Galagan, Calvo, S. E., Cuomo, C., Ma, L.-J., & Wortman, J. R. "Sequencing of *Aspergillus nidulans* and comparative analysis with *A. fumigatus* and *A. oryzae*", *Nature*, no. 428, pp. 1105–1115, 2005.
- [9] M. Zabka and R. Pavela, "Antifungal efficacy of some natural phenolic compounds against significant pathogenic and toxinogenic filamentous fungi," *Chemosphere*, 2013.
- [10] F. C. Silva, S. M. Chalfoun, V. M. De, D. M. S. Botelho, N. Lima, and R. Luís, "Evaluation of antifungal activity of essential oils against potentially mycotoxigenic *Aspergillus flavus* and *Aspergillus parasiticus*," vol. 22, no. 5, pp. 1002–1010, 2012.
- [11] N. Paster, M. Menashero, U. Z. I. Raviol, and B. Juven, "Antifungal Activity of Oregano and Thyme Essential Oils Applied as Fumigants Against Fungi Attacking Stored Grain e," vol. 58, no. 1, pp. 81–85, 1995.
- [12] K. A. Hammer, C. F. Carson, and T. V. Riley, "Antifungal activity of the components of *Melaleuca alternifolia* (tea tree) oil," *J. Appl. Microbiol.*, pp. 853–860, 2003.
- [13] F. D. D. Auria, M. Tecca, V. Strippoli, G. Salvatore, L. Battinelli, and G. Mazzanti, "Antifungal activity of *Lavandula angustifolia* essential oil against *Candida albicans* yeast and mycelial form," no. August, pp. 391–396, 2005.
- [14] G. İşcan, N. Kirimer, M. Kürkcüoğlu, H. Can Başer, and F. Demirci, "Antimicrobial Screening of *Mentha piperita* Essential Oils," *J. Agric. Food Chem.*, 2002.
- [15] K. Koba, P. Poutouli, C. Raynaud, J. Chaumont, and K. Sanda, "Chemical composition and antimicrobial properties of different basil essential oils chemotypes from Togo," *Bangladesh J. Pharmacol.*, vol. 4, no. 1, 2009.
- [16] S. Fahimi, H. Hajimehdipoor, H. Shabanpoor, F. Bagheri, and M. Shekarchi, "Synergic antibacterial activity of some essential oils from Lamiaceae," *J. Pharmacol.*, vol. 2, no. 3, pp. 23–29, 2015.
- [17] W. Ouedrhiri, B. Mounyr, E. H. Harki, and S. Moja, "Synergistic antimicrobial activity of two binary combinations of marjoram, lavender, and wild thyme essential oils," *Int. J. Food Prop.*, vol. 20, no. 12, pp. 3149–3158, 2017.
- [18] M. Javed and A. Shoaib, "Analysis of phytochemical constituents of *Eucalyptus citriodora* L. responsible for antifungal activity against post-harvest fungi," *Nat. Product Res.*, vol. 28, no. 18, 2011.
- [19] C. Miyamoto, J. Rocha de Sant'Anna, C. Da sylvia Franco, M. Cunico, and O. Miguel, "Genotoxic Activity of *Eucalyptus globulus* Essential Oil in *Aspergillus nidulans* Diploid Cells," vol. 54, no. 6, pp. 493–494, 2009.



# Curcumine/Piperine as a potential alternative to chemotherapeutic agents on lung cancer cells

Carolina Araiza, Amira Bellalah, Marine Chamoreau\*, Yasmine Daïk, Alexandra Lozano  
Sup'Biotech, Villejuif, France

\*Corresponding author : marine.chamoreau@gmail.com

**Abstract** - Concerns over the toxicity of chemotherapeutic agents has led to the research of natural alternatives to limit side effects on healthy cells. Curcumin has been widely studied for its anti-cancer properties but its low bioavailability remains an issue for a pharmacological approach. Piperine has been shown to increase the bioavailability of curcuminoids. The aim of this study is to investigate the potential anticancer effect of two natural compounds, curcumin and piperine, on two non-small lung cancer cell lines H1299 (p53 inactive) and A549 (p53 active) as an alternative to chemotherapies. Both cell lines were treated with 20, 40  $\mu$ M curcumin and 350, 700 nm piperine either alone or in combination and with Gemcitabine or cisplatin chemotherapeutic agents. Their antiproliferative activities were determined using a WST-1 assay after 30h treatment and a FLICA assay was conducted to better understand the mechanism underlying the cytotoxicity induced by these molecules. Our results suggest a synergic effect between both natural compounds, and a similar antiproliferative effect between the mix and chemotherapies for both cell lines, which suggests that p53 is not involved in the anti-proliferative pathway. Furthermore, caspase 3 and caspase 7 activities were detected after curcumin treatment suggesting that the cell death mechanism may be associated with the caspase-3 apoptotic pathway.

**Index Terms** – Apoptosis, Caspase Chemotherapeutic agents, Curcumin, Non-small Lung Cancer Cells, Piperine.

## INTRODUCTION

In 2018, the World Health Organization estimated there were around 9.6 million deaths caused by cancer from which 1.8 million deaths were marked as lung cancer making it the most common cancer and cancer-related death worldwide [1]. Lung cancer is known to be a highly invasive and rapidly metastasizing cancer. Primary lung carcinomas are divided into small (SCLC) or non-small cell lung cancer (NSCLC) the latter covering around 85% of total lung cancer and is known to be less aggressive than SCLC [2]. Several treatments have been developed for lung cancer such as surgery, radiation therapy, immunotherapy, and chemotherapy which is the most common treatment for NSCLC. However, this type of lung cancer is the least responsive to chemotherapy [1].

Treatments can vary depending on the stage and type of lung carcinoma [3]. Surgery is one of the best known treatment for removing lung tumors. Another option is radiation therapy, which consists in using high-powered energy from sources such as x-rays and protons to kill cancer cells. When cancer cells carry some mutations, targeted drug therapy can also be an alternative. When lung cancer is classified as a stage four cancer, immunotherapy can also be used [3].

Chemotherapeutic agents, which are given to the patient through the bloodstream, tend to be cytotoxic, meaning that they prevent the replication, growth or can induce the death of cancer cells [4]. Two often used chemotherapy drugs which efficiency has been proved are cisplatin and gemcitabine [5]. However, both chemicals induce toxic side-effects on surrounding healthy cells which leads to complications on multiple organ systems [5].

There has recently been a dynamic increase of the research for natural compounds usable in cancer treatments mainly because of chemotherapies toxic side-effects [6]. Numerous studies have addressed the anti-cancer potential of curcumin [7], and piperine, two natural compounds commonly used as food spices [8]. Curcumin is a polyphenol (diferuloylmethane) derived from the roots of *curcuma longa* (turmeric), a plant of the ginger family [9]. It is the main biologically active compound of this plant, and it has been associated with “antioxidant, anti-inflammatory, anti-cancer, antiviral and antibacterial activities” [10]. Recent studies suggest that curcumin may induce cell cycle arrest and activate apoptosis pathways in several cancer cells, including A549 lung cancer cells (p53 active), without promoting the development of side effect [7], [9],[11]. Piperine is a natural alkaloid isolated from black pepper (*Piper Nigrum*) and long peppers (*Piper longum*) [8]. Recent investigations demonstrated the chemo preventive and antioxidant activities of this compound along with several other pharmacological actions such as anti-inflammatory, anti-fungal, and hepatoprotective properties [8] [12].

Lin's study (2014) showed cytotoxic effect of piperine against A549 cells in a dose-dependent manner while showing no effect on WI38 human lung fibroblasts [8]. The anticancer effect of this molecule might be attributed to cell DNA damage and apoptosis [13].

As curcumin and piperine have each been shown to be safe and well tolerated in clinical studies [11], testing the combination of curcumin and piperine would allow us to gain some insights on their combined effects on NSLC. Moreover, the addition of piperine to curcumin may help address the low bioavailability issue of curcumin as piperine proved to increase the bioavailability of curcumin *in vivo* [14], [15].

In this study, the combination of curcumin and piperine as efficient all-natural alternative to gemcitabine and cisplatin chemotherapies was investigated.

## MATERIALS AND METHODS

**Media.** Dulbecco's Modified Eagle Medium - High Glucose (Sigma, cat. D5796) with 4500 mg/L glucose, L-glutamine and NaHCO<sub>3</sub> have been used for cell culture. DMEM was supplemented with 10% of FBS (Dutscher, cat. P30-800), 1% PenStrep (Thermofisher, cat. 15070063), 1% non-essential amino acids (Sigma, cat. M7145).

**Cells.** A549 (Non-small lung cancer epithelial cells with functional p53), and H1299 cell lines (Non-small lung cancer cells from lymph node with defective p53) were gently provided by the CellTechs research laboratory located in CEA Fontenay-Aux-Roses (France), as well as BJ primary cells (fibroblast). Cells were cultured in complete DMEM at 37°C in a incubator in an atmosphere of 95% oxygen, 5% carbon dioxide and 90 % humidity. Cells were manipulated under laminar flow hood. Cells were observed with ZOE Fluorescent Cell Imager (Bio-Rad) microscope and pictures pre- and post-treatment were taken using the same microscope.

**Stock solution preparation.** Curcumin powder (Merck, cat. C1386) was first dissolved in DMSO at a final concentration of 16 mM. Serial concentrations (160, 80 µM) of curcumin were prepared by dilutions of stock solution in sterile water. Piperine powder (Merck, cat. P49007) was first dissolved in DMSO at a final concentration of 28 mM. This stock solution was then serially diluted with sterile water to prepare different concentrations (2.8 µM, 1.4 µM). Combined solution of piperine/curcumin were made by diluting stock solutions.

Gemcitabine powder (Merck, cat. G6423) was initially dissolved in sterile water at a final concentration of 40 mM. The stock solution was serially diluted with sterile water to obtain a concentration of 80 µM. Cisplatin powder (Sigma, cat. P5493) was first dissolved in PBS at a final concentration of 10 mM and a concentration of 80 µM were also prepared via serial dilution of the stock solution using sterile water.

All solutions were vortexed and diluted in complete DMEM in order to obtain the following final solution :

|       |                                             |      |                 |
|-------|---------------------------------------------|------|-----------------|
| G20   | Gemcitabine 20 nM                           | I20  | Cisplatin 20 µM |
| C40   | Curcumin 40 µM                              | P700 | Piperine 700 nM |
| C20   | Curcumin 20 µM                              | P350 | Piperine 350 nM |
| C + P | Curcumin 20 - 40 µM + Piperine 350 - 700 nM |      |                 |

**WST-1 assay.** Cell lines were seeded onto 96-well plates at a density of 6000 cells/well in 200 µl DMEM. After 24 hours, 200 µl of the extemporally prepared solutions were added in triplicate and plates were incubated for 30h at 37°C. Negative control were performed with: PBS, DMSO or DMEM. After treatment, cells were incubated with 100 µL of 1% WST-1 solution in complete DMEM and optical density was measured using a microplate reader at 450 nm (with 620 nm as a reference) at 30 min, 1h and 1h30 post-incubation.

**FAM-FLICA Caspase Assay.** Activation of caspase 3 and 7 was detected using the FAM-FLICA staining (Bio-Rad, FAM FLICA Caspase-3/7 kit ref. ICT03). FLICA was reconstituted into DMSO to form 150X FLICA and then diluted in PBS at final concentration of 30X. A549 and H1299 cell lines were seeded as mentioned above. Staining was performed with FAM-FLICA working solution of 1X. Washing steps were performed using apoptosis washing buffer (0.1% w/v sodium azide) supplied with the Bio-Rad kit. Counter-staining was done with 0.5% v/v Hoechst solution and dead cells were stained with 0.5% v/v propidium iodide. For each staining step, incubation was performed at 37°C for 40 min, in the humidified incubator and plates were protected from light using aluminum foil. Observations and pictures were performed using ZOE Fluorescent Cell Imager (Bio-Rad).

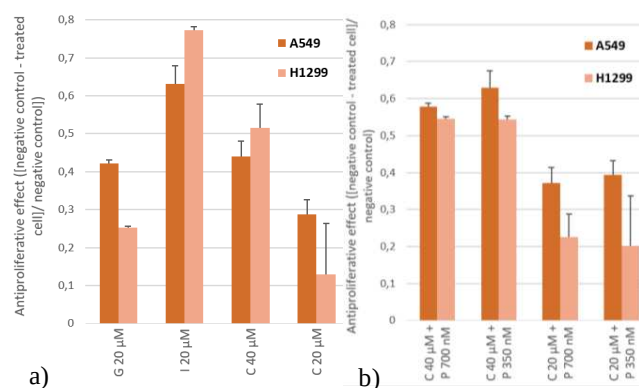
**Statistical analysis.** Results were analyzed using Excel software. The anti-proliferative effects were calculated as the relative ratio of the absorbance for treated cells (curcumin, piperine, cisplatin or gemcitabine) over the related absorbance for negative control (treated with PBS, treated with DMSO or cells in complete DMEM). Statistical analysis included calculations of mean, standard deviation and p-values using a two-tailed t-test.

## RESULTS

The *in vitro* anti-proliferative activity of curcumin and piperine compounds was assessed using a WST-1 assay, at different concentrations on H1299 and A549 cell lines (data not shown). From those results, concentrations having the highest anti-proliferative effect have been selected. An additional test has been performed using the previous selected concentrations (Figure 1) on both cell lines. The relative cell viability was calculated as the ratio of viable cells compared to untreated cells.

Cisplatin displays a greater effect than curcumin independently of the treatment concentration on both cell lines (Figure 1a). Furthermore, curcumin 20  $\mu$ M and curcumin 40  $\mu$ M display the similar antiproliferative effect than gemcitabine 20 nM (respectively  $P=0,12>0,05$  and  $P=0,35>0,05$ ) for the A549 cell line.

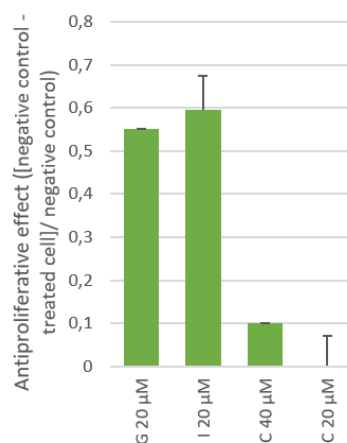
For the H1299 cell line, curcumin 20  $\mu$ M has a significantly lower antiproliferative effect than gemcitabine 20 nM ( $P=0,015<0,05$ ) and curcumin 40  $\mu$ M has a significantly greater effect ( $P=0,018<0,05$ ). Thus, a concentration of curcumin between 20 and 40  $\mu$ M could be used to exert a similar effect as a gemcitabine concentration of 20 nM. No significant difference has been shown between the effects of curcumin 20  $\mu$ M ( $P=0,21>0,05$ ) on A549 compared to H1299 and the same observation can be made for curcumin 40  $\mu$ M ( $P=0,40>0,05$ ) comparing between cell lines. Concerning piperine, it has a little effect on both cell lines when added alone in the media (data not shown).



**Figure 1.** Antiproliferative effect of several compounds and mixes after 30h treatment (G: Gemcitabine, I: Cisplatin, C: Curcumin, P: Piperine) on A549 (p53 active) and H1299 (p53 inactive) cell lines following 1h incubation with WST-1. a) Comparison of the effect of chemotherapeutic agents with curcumin. b) Effect of different combinations of curcumin and piperine. Statistical analysis at 5% of confidence using a two-tailed t-test.

The curcumin/piperine mixes show that curcumin 40  $\mu$ M + piperine 700 nM has a greater antiproliferative effect than curcumin 40  $\mu$ M on both A549 ( $P=0,049<0,05$ ) and H1299 ( $P=0,048<0,05$ ) cell lines. Furthermore, curcumin 40  $\mu$ M + piperine 350 nM has a greater effect than curcumin 40  $\mu$ M alone for A549 ( $P=0,031<0,05$ ). A concentration of at least 700 nM of piperine is required to enhance the curcumin effect on H1299 and at least a concentration of 300 nM for A549 (Figure 1b). It suggests that H1299 has a lower sensitivity to piperine than the A549 cell line. There is no significant differences between curcumin 20  $\mu$ M and curcumin 20  $\mu$ M + piperine 700 nM or curcumin 20  $\mu$ M + piperine 350 nM. Consequently, a minimum concentration of 40  $\mu$ M of curcumin is required for piperine to be a good enhancer.

The effect of curcumin on cell proliferation compared to chemotherapies was also assessed on BJ fibroblasts. These non-cancerous cells were considered as model for treatment side effects (Figure 2). As expected, gemcitabine and cisplatin show high anti-proliferative effect on BJ cells, respectively 55% and 59%. Curcumin has a relatively low antiproliferative effect on fibroblast with 10 % at 40  $\mu$ M and no effect at a concentration of 20  $\mu$ M (Figure 2).

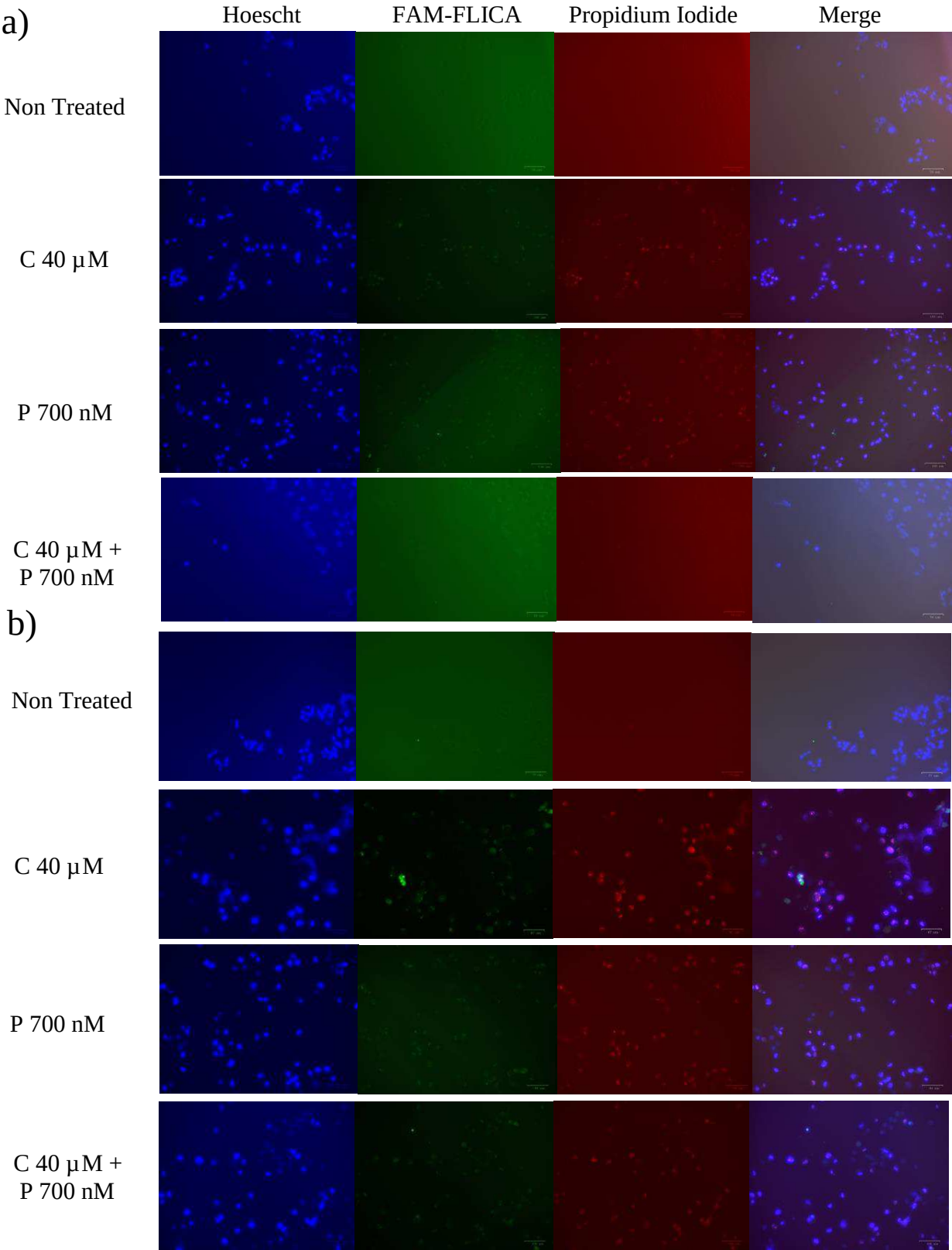


**Figure 2.** Antiproliferative effect of several compounds after 48h treatment (G: Gemcitabine, I: Cisplatin, C: Curcumin, P: Piperine) on BJ cell line (fibroblasts).

Based on the Figure 1. results, curcumin could have an equivalent effect at a concentration equal or higher than the concentration used for gemcitabine treatment. Thus, curcumin use will confer a safer environment for healthy surrounding cells. Further experiments of curcumin/piperine mixes are required on BJ cells due to solubilization issues encountered in the present experiment.

To better understand the mechanisms involved in the antiproliferative effect of curcumin and piperine, caspase 3 and caspase 7 activities were assessed by a FAM-FLICA assay (Figure 3). For the A549 line, cells treated with curcumin are positive for propidium iodide (PI), a marker of nuclear envelope degradation and cell death. Moreover, some of these cells display fluorescence for FAM-FLICA marker suggesting that the caspase 3/7 apoptosis pathway is involved. However, cells treated with the curcumin/piperine mix do not show any fluorescence for FAM-FLICA or propidium iodide, even though the WST-1 assay has shown a high antiproliferative effect for this treatment.

For H1299 cells, the curcumin, piperine as well as the mix treatment show a positive fluorescence for caspase 3/7 and for propidium iodide staining especially for curcumin at 40  $\mu$ M. These results suggest that curcumin and piperine also play a role in the caspase 3/7 apoptosis pathway for H1299 cell line.



**Figure 3.** Activation of caspase 3/7 in a) A549 cells b) H1299 cells after 24h treatment with curcumin, piperine and a combination of both. Cells were stained with FAM-FLICA, PI and Hoescht staining.

## DISCUSSION

Throughout this study, the antiproliferative and apoptotic effects of curcumin and piperine were tested on three cell lines: A549 and H1299 for non-small lung cancer cells, and BJ fibroblasts as a healthy cell model.

Our results indicate that curcumin at concentrations of 20  $\mu$ M and 40  $\mu$ M on A549 cell line have similar anti-proliferative effect than gemcitabine at 20  $\mu$ M. These results showed that curcumin could have an impact on cell proliferation in a dose independent manner, which was previously reported [16]. On the other hand, curcumin at a concentration of 20  $\mu$ M had a lower anti-proliferative effect than gemcitabine or curcumin at 40  $\mu$ M on H1299 cells, which is not in accordance with the literature. Indeed, it was reported that the effect of curcumin on H1299 cell line is both time and concentration dependent [16]. However, curcumin and piperine have not showed a prominent activity over cisplatin for both cell lines. This could be linked to the ability of cisplatin to bind DNA which facilitates its interference with repair mechanisms, and therefore the induction of cell death in a more efficient way [17].

According to the obtained results, the mix corresponding to curcumin at 40  $\mu$ M and piperine at 700 nM has a significantly greater effect than curcumin at 40  $\mu$ M alone, in both cell lines. This supports the idea that a mix could be more efficient than the compound alone. Furthermore, literature has reported that the administration of curcumin with piperine increases the curcumin bioavailability *in vivo* [18].

A fundamental aspect of this study was to compare the cytotoxicity of curcumin (with and without piperine), cisplatin, and gemcitabine on BJ fibroblasts given that current chemotherapies exert toxic side effects on healthy cells [18]. Gemcitabine and cisplatin induce a high anti-proliferative effect on BJ cells, respectively 55% and 59% (Figure 2). On the contrary, curcumin shows a low anti-proliferative effect on fibroblasts with a maximum of 10% for a concentration of 40  $\mu$ M. Studies have described the selectivity of curcumin towards cancer cells rather than normal cells [19].

Evidences highlight the curcumin induced cell death mechanism of action, which is mediated by both the activation of cell death pathways and by the inhibition of growth/proliferation pathways. Literature has reported that curcumin induces the initiation of both p53 active and p53 inactive G2/M phase cell cycle arrest, thereby restricting cell proliferation and tumor progression [18].

According to Lin et al. (2014), piperine also exerts a cytotoxic effect against A549 cells and exhibit the ability to cause cell cycle arrest in G2/M phase and to activate caspase-3 and -9 cascades in this cell line [20]. To confirm

apoptosis induction by curcumin and piperine, caspase 3 and caspase 7 activities were assessed using a FAM-FLICA assay. Both curcumin and piperine separately seem to activate caspase 3,7 pathway (Figure 3). According to Park et al. (2013), curcumin activates the caspase 3 apoptotic pathway in A549 and H1299 cells [21]. However, in contrast to this research, a second study indicate that curcumin influences the caspase 9 apoptotic signaling pathway [22]. It seems that more than one pathway can be triggered by curcumin. Our results indicate that piperine activates caspase 3 as well as caspase 7. Literature has previously reported that piperine has the ability to cause cell cycle arrest in G2/M phase and to activate caspase-3 and caspase-9 cascades in A549 cells [20]. An explanation could be that due to piperine's cell cycle arrest ability, caspase 3 is activated, yet no information on caspase 7 have been reported. In our results, A549 cells treated with the mixes did not show any fluorescence for caspase 3,7 and thus, caspase 7 is not the signaling pathway for cellular death.

On the other hand, propidium iodide staining is not observed for any mix, whereas WST-1 assay showed high cytotoxicity for these treatments. However, we have to consider that the amount of cells observed in the FLICA assay was very low and cells could have been detached during washing steps, which could explain these results. Indeed, FAM-FLICA staining seems not to be the most optimal test for 96-well plate testing. Moreover, microscopic pictures are not representative of the global behavior and the quantification of the fluorescence has not been performed considering these results. Results obtained in the wells were highly heterogeneous and that could be linked to a strong cell loss. Taking into account the numerous washing steps required, apoptotic cells were perhaps too weak to remain adherent and they got lost during the experiment.

For H1299 cells, curcumin, piperine and the mixes seems to show an interaction with caspase 3,7 pathway. Previously, it has been reported that there is an involvement of caspases in the curcumin-induced death of H1299 cells, with time-dependent increase of caspase 3, 6 and 7 activity observed after 16-24h incubation [22]. Thus, the results of caspase experiments suggest that curcumin by itself do have an effect on caspase 3,7 pathways. According to Jafri et al., (2019), piperine has the ability to activate caspase-3 in human cervical adenocarcinoma [23]. Piperine stimulates an excessive intracellular ROS production in HeLa cells after 12 h treatment period, which can suggest that piperine could have a similar effect on H1299 and thereby inducing apoptosis. However, no study has described the effects of piperine on H1299 cells.

Based on the obtained results, p53 seems not to be involved in the antiproliferation pathway. Indeed, no differences have been observed between the H1299 (p53 inactive) and A549 (p53 active) results. Curcumin and piperine have the same antiproliferative effects on both cell lines.

## CONCLUSION & PERSPECTIVES

Lung cancer is known as the most widely spread cancer in the world. Treatments benefit/harm ratio needs to be improved by reducing the cytotoxic side effects. In the present study, we demonstrated that curcumin and piperine combination could be a natural alternative to chemotherapeutic agents as it exerts a cytotoxic effect on cancer cells with less side effects on surrounding healthy cells. In the future, *in vitro* tests of curcumin and piperine mixes on spheroids may be interesting to perform to study their effect on a 3D-structure. Moreover, investigations on the impact of the mix on metastatic properties of the cancer through scratching assay could be conducted, along with experiments to better understand the cancer cell death pathway. Further investigations should be performed on the p53 implication on antiproliferation pathway as the present study shows that p53 is not involved.

## ACKNOWLEDGMENT

This work was supported by Sup'Biotech engineering school and especially Mrs. Perdiz, our supervisor and also researcher at CellTechs, a partner laboratory between Sup'Biotech and CEA (Commissariat à l'Energie Atomique et aux Energies Alternatives) which provided us the different cell lines used in the present study. We gratefully thank Mrs. Perdiz for her time and her expertise. We also thank Mrs. Jarray for initiating the project.

## REFERENCES

- [1] WHO. (n.d.) Latest global cancer data: Cancer burden rises to 18.1 million new cases and 9.6 million cancer deaths in 2018. [online] Retrieved November 15, 2019 from <https://www.who.int/cancer/PRGlobocanFinal.pdf>
- [2] Gately.K. "Lung Cancer, a comprehensive overview", *Nova Science Publisher*. New York, 2013.
- [3] Mayo Clinic. (n.d.) Lung cancer [online]. Retrieved November 15, 2019, from <https://www.mayoclinic.org/diseases-conditions/lung-cancer/diagnosis-treatment/drc-203746>
- [4] Liu, J. C., & Ridge, J. A. "What is Cancer?" *Abernathy's Surgical Secret*, N.C, 2018, 307–310.
- [5] Merimeskt, O. et al., "Introduction of apoptosis in non-small lung carcinoma cell line (H1299) by combination of anti-asthma drugs with gemcitabine and cisplatin", *International Journal of Oncology*, 26(2), 2005, pp.475–482.
- [6] Lichota, A., & G wozdzinski, K. "Anticancer Activity of Natural Compounds from Plant and Marine Environment.", *International journal of molecular sciences*, 19(11), 2018, pp.3533.
- [7] Yao, Q. et al, "Curcumin induces the apoptosis of A549 cells via oxidative stress and MAPK signaling pathways.", *International Journal of Molecular Medicine*, 36(4), 2015, pp.1118–1126
- [8] Lin, Y., Xu, J., Liao, H., Li, L., & Pan, L., "Piperine induces apoptosis of lung cancer A549 cells via p53-dependent mitochondrial signaling pathway." *Tumor Biology*, 35(4), 2014, pp.3305–3310.
- [9] Salem, M., Rohani, S., & Gillies, E. R., "Curcumin, a promising anti-cancer therapeutic: a review of its chemical properties, bioactivity and approaches to cancer cell delivery." *RSC Advances*, 4(21), 2014, 10815.
- [10] Prasad, S., Tyagi, A. K., & Aggarwal, B. B. "Recent Developments in Delivery, Bioavailability, Absorption and Metabolism of Curcumin: the Golden Pigment from Golden Spice." *Cancer Research and Treatment*, 46(1), 2014, pp.2–18
- [11] Liu F, Gao S, Yang Y, et al., "Curcumin induced autophagy anticancer effects on human lung adenocarcinoma cell line A549." *Oncol Lett.*, 14(3), 2017, pp.2775–2782.
- [12] Gorgani, L., Mohammadi, M., Najafpour, G. D., & Nikzad, M. "Piperine-The Bioactive Compound of Black Pepper: From Isolation to Medicinal Formulations.Comprehensive", *Reviews in Food Science and Food Safety*, 16(1), 2016, pp.124–140.
- [13] Rather, R. A., & Bhagat, M, "Cancer Chemoprevention and Piperine: Molecular Mechanisms and Therapeutic Opportunities", *Frontiers in Cell and Developmental Biology*, 6, 2019
- [14] Kakarala M, Brenner DE, Korkaya H, et al., "Targeting breast stem cells with the cancer preventive compounds curcumin and piperine". *Breast Cancer Res Treat.*;122(3), 2010, pp.777–785.
- [15] Hay, E., Lucariello, A., Contieri, M., Esposito, T., De Luca, A., Guerra, G., & Perna, A. « Therapeutic effects of turmeric in several diseases: An overview" *Chemico-Biological Interactions*, 2019, 108729.
- [16] Li, Y., Zhang, S., Geng, J.X., Hu, X.Y. "Curcumin inhibits human non-small cell lung cancer A549 cell proliferation through regulation of bcl2/bax and cytochrome c" *Asian Pacific Journal of Cancer Prevention*, 14(1), 2013, pp.4599–4602.
- [17] Dasari, S., Tchounwou, P.B. "Cisplatin in cancer therapy: molecular mechanisms of action" *European Journal of Pharmacology*, 740(1), 2014, pp.364–378.
- [18] Panda, A.K., Chakraborty, D., Sarkar, I., Khan, T., Sa, G. "New insights into therapeutic activity and anticancer properties of curcumin" *Journal of Pharmacology and Experimental Therapeutics*, 9(1), 2017, pp.31–45.
- [19] Ravindran, J., Prasad, S., Aggarwal, B.B. "Curcumin and cancer cells: How many ways can curry kill tumor cells selectively ?" *American Association of Pharmaceutical Scientists*, 11(3), 2009, pp.495–510.
- [20] Lin, Y., Xu, J., Liao, H., Li, L., Pan, L. "Piperine induces apoptosis of lung cancer A549 cells via p53-dependent mitochondrial signaling pathway" *Tumor Biology*, 35(4), 2014, pp.3305–3310.
- [21] Park, W., Amin, A.R.M.R., Chen, Z.G., Shin, D.M. "New perspectives of curcumin in cancer prevention" *Cancer Prevention Research*, 6(5), 2013, pp.387–400.
- [22] Amin, A.R.M.R., Haque, A., Rahman, M.A., Chen, Z.G., Khuri, F.R., Shin, D.M. " Curcumin induces apoptosis of upper aerodigestive tract cancer cells by targeting multiple pathways" *Plos One*, 10(4), 2015, pp.1–11.
- [23] Jafri, A., Siddiqui, S., Rais, J., Ahmad, M.D., Kumar, S., Jafar, T., Afzal, M., Arshad, M. "Induction of apoptosis by piperine in human cervical adenocarcinoma via ROS mediated mitochondrial pathway and caspase-3 activation" *Experimental and Clinical Sciences*, 18(1), 2019, pp.154–164.



# Curcumin and crocin as natural substitutes for current chemotherapeutic agents

Antoine RETHO, Léa LISIECKI, Sofia HERVAIS, Minerva ESPARZA, Tais ROMERO Sup'Biotech, Villejuif, France

Corresponding author: antoine.retho@supbiotech.fr

**Abstract** – Among the various kinds of cancer, lung cancer is the first one in terms of mortality. Currently, the main treatment line is chemotherapy, which involves drugs with severe toxic side effects, such as cisplatin. Its mode of action has been linked to its ability to crosslink with the purine bases on the DNA, interfering with DNA repair mechanisms, causing DNA damages and inducing apoptosis in cancer cells. However, it can produce drug resistance and several undesirable side effects (relapse, severe kidney problems, allergic reactions, decreased immunity, hemorrhage). Given this, there is a need for the development of natural non-toxic alternatives that could replace ongoing treatments. Curcumin, a natural compound derived from turmeric and crocin, found in saffron, show a high potential to fight cancer.

The purpose of this study is to decrease the dose (concentration) of cisplatin and to compensate it with the add of a mix of natural compounds (curcumin and/or crocin) in order to maintain or even increase the toxicity of cisplatin on cancer cells. Through this work and on a 2D model, higher cytotoxic effect was obtained when cisplatin was used with curcumin and crocin than when cisplatin was used alone at the same concentration. Furthermore, on a 3D model, curcumin and cisplatin decreased the size and induced fragmentation of the spheroids.

**Index Terms** – Cancer treatment, Cisplatin, Crocin, Curcumin, Lung Cancer

## INTRODUCTION

According to the World Health Organization, cancer is the second leading death cause worldwide

among both men and women and represents a major health issue, with 9,6 million death worldwide in 2018<sup>1</sup>. Cancer is defined by the National Cancer Institute as a disease caused by the transformation of cells that become abnormal and proliferate excessively. These deregulated cells form a mass called a malignant tumor. Cancer cells tend to invade neighboring tissues and detach themselves from the tumor. Then, they can migrate through blood vessels and lymphatic vessels to form another tumor, called metastasis<sup>2</sup>. Cancer can lead to numerous symptoms such as pain, coughing up blood, loss of weight, breath shortness among others<sup>3</sup>. Among all types of cancer, lung cancer is the 4th most common cancer, but the first one in terms of mortality, with 30 000 death per year<sup>4</sup>.

This cancer is divided in two subgroups<sup>5</sup>:

- Non-Small Cell Lung Cancer (NSLC): most common type of lung cancer with 85% of lung cancers. It includes different cell lines as H838, H2126, and also H1299 and A549, the 2 cell lines used for this experiment.
- Small Cell Lung Cancer (SLC): also called oat cell cancer which represent 10 to 15% of lung cancers.

Currently, some cancer therapies exist such as surgery in early stages, radiotherapy or chemotherapy<sup>6</sup>. Despite great advances, the incidence and mortality rate remain high. At present, the most common line for most cancers (including lung cancer) is the chemotherapy, with the use of different molecules such as cisplatin or gemcitabine. Cisplatin is a metallic coordination

compound with a square planar geometry, and it has been clinically proven to combat different types of cancers, including sarcomas, muscles, bones and lung cancer<sup>7</sup>. Despite its chemotherapeutic effect due to its capacity to interfere with DNA repair mechanisms and causing apoptosis in cancer cells, this type of molecule can lead to several side effects (relapse, severe kidney problems, allergic reactions, decreased immunity, hemorrhage)<sup>7,8</sup>.

To overcome those side effects, it is crucial to find alternatives. Thus, we aim to establish a novel therapeutic strategy for lung cancer using natural substitutes without stopping the use of cisplatin. For this experiment, we chose metabolites produced by plants: curcumin, principal curcuminoid of turmeric (*Curcuma longa*) and member of the ginger family; and crocin, a carotenoid chemical compound found in crocus and gardenia and responsible for the color of saffron.

These two components present different chemotherapeutic effects:

- Curcumin has therapeutic and anti-cancer properties, by acting on biological pathways involved in cell growth and cell survival through various pathways and has shown anti-proliferative and pro-apoptotic properties in many cancers such as colon, kidney or lung (H1299 and A549 cells) cancer<sup>9</sup>.
- Crocin inhibits cell proliferation and nucleic acid synthesis, induces apoptosis and enhances anti-oxidative system. This molecule can be used as an anticancer agent according to actual reports, either by itself or as a companion for cisplatin treatments<sup>10,11</sup>.

The purpose of this study is to decrease the dose (concentration) of cisplatin and to compensate it with the add of a mix of natural compounds (curcumin and/or crocin) in order to maintain or even increase the toxicity of cisplatin on cancer cells. The objective would be in the long term, to reduce the undesirable effect of anti-cancer treatments. First, we will determine the concentration giving 50% of cytotoxicity (IC50) of curcumin, crocin and cisplatin independently on two lung cancer cell lines, H1299 and A549. Based on the published IC50, different combinations will be established and use to test on the cell lines. As our experiments do not allow in first intention the distinction between a cytotoxic or an anti-proliferative (COAP) effect, results will be expressed in COAP effect percentage. Lastly, the most suitable concentrations combination will be tested on 3D models (spheroids) to see whether growth is altered or not.

Furthermore, we will provide a comprehensive review of cisplatin effect alone and with different combinations (cisplatin with either curcumin or crocin) for the treatment of Non-Small Cell Lung Cancer, H1299 and A459 and discuss its potential side effects.

## MATERIALS AND METHODS

**Cell lines and reagent.** Non-Small cell lung cancer H1299 (ATCC®CRL-5803™) and A549 (ATCC® CCL-185™) were obtained from CellTechs, a partnership laboratory of Sup'Biotech/CEA. Cells were cultured in high-DMEM (Sigma, D6546) containing 4500 mg/L of glucose, sodium pyruvate, and sodium bicarbonate, supplemented with 1% of L-Glutamine (ThermoFisher, 25030081) and enriched with 1% of PenStrep (Sigma, P4333) and 10% FBS (Sigma, F9665) in an incubator with 5% CO<sub>2</sub> at 37°C.

Curcumin (C1386-5G), Crocin (17304-5g), Cisplatin (P4394-25mg), Gemcitabine (G6423),

Dimethyl Sulfoxide (DMSO) (D4540-100ML) and Phosphate Buffered Saline (PBS) (P5493-1L) were purchased from Sigma-Aldrich and stored in a cupboard.

Cell proliferation reagent WST-1 (5015944001) was also provided from Sigma-Aldrich and store at 4°C.

For the experiments, curcumin and gemcitabine were dissolved in DMSO and then in complete DMEM. Cisplatin and crocin were dissolved in PBS1X.

Agarose solution was prepared by mixing 2g of ultrapure agarose with 0,9g of NaCl into 100ml of ultrapure water and then autoclave.

A MicroTissues® 3D Petri Dish® micro-mold (Z764019) purchased from Sigma-Aldrich was used in order to form 81 large-spheroids.

**WST-1 assay.** A549 and H1299 cells were seeded in a 96-well plate at a density of  $6 \times 10^3$  cells/well in a final volume of 200µl medium. After 24h in an incubator (37°C and 5% CO<sub>2</sub>), the culture medium was changed to fresh medium with different concentration of treatment. Each concentration of treatment was performed in triplicate. After 48h of treatment, culture medium with the different concentration of treatment was removed and 100µL of WST-1 was added in each well. 96-well plates were then placed in an incubator (5% CO<sub>2</sub> at 37°C) for 30 min/1h/1h30 and 2h. Absorbance (OD) at 450 and 600 nm were measure using a spectrophotometer. Corrected absorbance was then calculated with the formula: OD 450 – OD 600. The percentage of cell proliferation was calculated according to the manufacturer using the following formula:

$$\text{Cell proliferation} = 100 \cdot \frac{c.n - c.e}{c.n} (1)$$

*c.n.* corrected absorbance negative control

*c.e.* corrected absorbance experimental

This formula allows to normalize the absorbance value of the different products to the absorbance of their corresponding negative control.

Then, the percentage of COAP (Cytotoxicity or antiproliferative effect) was calculated with the formula:

$$\% \text{ COAP} = 100 - \text{Cell proliferation} (2)$$

Between-groups, differences were analyzed using Student's t-test.

**Mold formation.** Following the manufacturer instruction, a 2% agarose solution (containing 0,9% of NaCl) was melted in a microwave. Then, 600µL of this solution was poured into a micro- mold, taking care to not create air bubbles. The agarose was then quickly gelified by placing the micro-mold on ice. This allowed to form a 3D Petri Dish® which was then placed in a well of a 12-well plate. This operation was then repeated to fill each well with a 3D Petri Dish®. 2.5ml of complete DMEM was then added in each well to equilibrate the 3D Petri Dish®.

**Spheroids formation.** Medium was removed from the 3D Petri Dish® and H1299 cells were seeded at a density of  $81 \times 10^3$  cells/3D Petri Dish® in a final volume of 190 µL. After waiting 30min to let the cells settle into the features of the 3D Petri Dish®, an additional 2,5ml of medium was delicately added to the wells. After 48h in an incubator (5% CO<sub>2</sub> at 37°C), the culture medium surrounding outside the 3D Petri Dish® was changed to fresh medium with different concentration of treatment.

Spheroids were then observed thanks to the ZOE Fluorescent Cell Imager purchased from BIO- RAD and analyzed thanks to the ImageJ software.

## RESULTS

The anti-cancerous properties of the therapeutic agent cisplatin and natural products, curcumin and

crocin, were tested on both NSLC cell lines H1299 and A549.

In order to do so, the WST-1 assay was performed. This assay gives information on the cytotoxic effects of our compound. However, throughout the experiments, the term "COAP effect" is used because it is a broader term indicating that the products may have a cytotoxic or anti-proliferative effect on both cell lines.

For all the WST-1 assay, only results obtained after 1h30 WST-1 incubation, as they were showing the most relevant data for further analysis (data not shown).

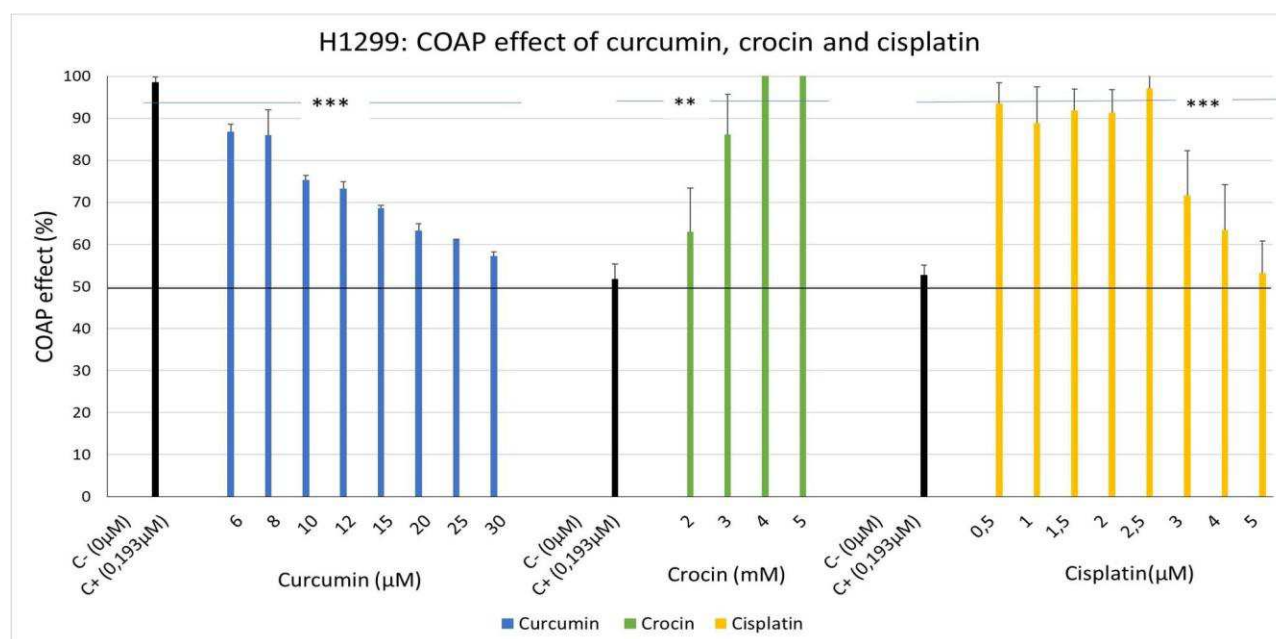
In addition, for all the WST-1 assay, gemcitabine was chosen as a positive control of COAP effect. Indeed, gemcitabine is a chemotherapeutic molecules better known as an antimetabolite specific for S phase of the cell cycle<sup>12</sup>.

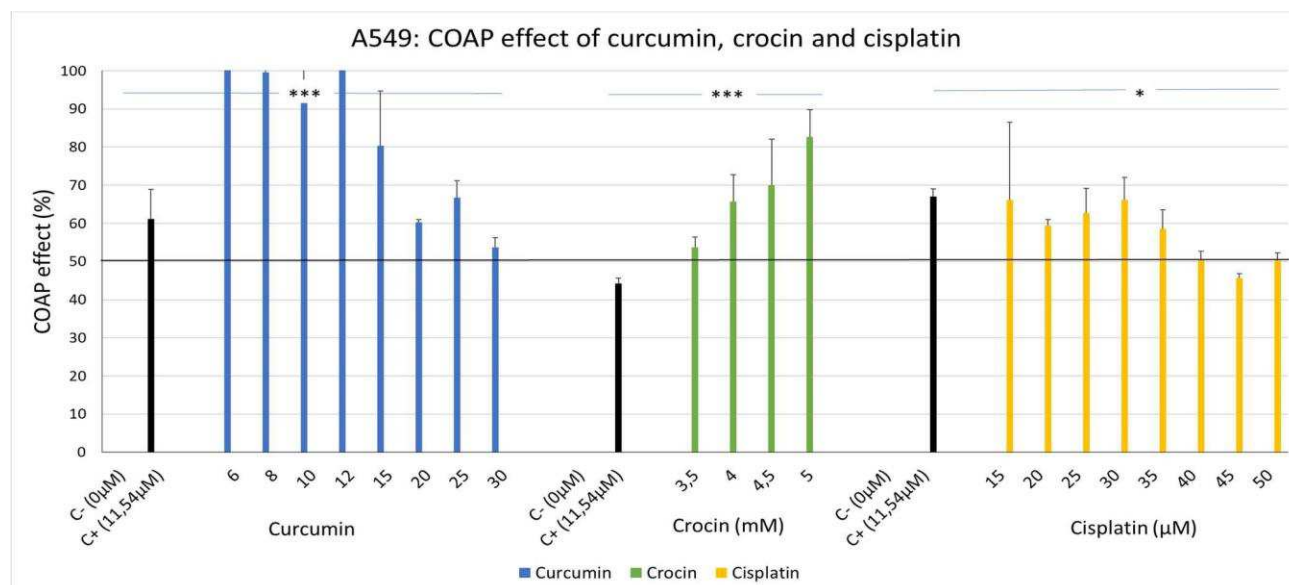
Thanks to this test, the IC50 (concentration giving 50% of cytotoxicity) of therapeutic agent cisplatin

and natural products, curcumin and crocin, were approximately estimated in both NSLC cell lines (black line in Figure 1)

For the H1299 cell line, a surprising decreasing dependent dose-effect was observed for curcumin (from 6 $\mu$ M to 30 $\mu$ M) and cisplatin (from 2,5 $\mu$ M to 5 $\mu$ M) compare to crocin which showed an increasing dependent dose-effect (from 2mM to 4mM).

Same observations could be done on the A549 cell line for curcumin and crocin with respectively a decreasing (from 12 $\mu$ M to 30 $\mu$ M) and increasing (from 3,5mM to 5mM) dependent dose-effect.





**Figure 1:** Determination of COAP (cytotoxic or anti-proliferative effect) and the approximative IC<sub>50</sub> (grey box) of curcumin, crocin and cisplatin on H1299 and A549 cells after 48h treatment incubation at 1h30 WST-1 incubation.

C- correspond to the negative control without any treatment.

C+ correspond to the positive control with a treatment of gemcitabine at a concentration found in the literature equal to the IC<sub>50</sub> for the corresponding cell line.

Results are obtained from one of our experiments.

Standard-deviations correspond to triplicates intra-manipulation.

\* $p < 0.05$ ; \*\* $p < 0.01$ ; \*\*\* $p < 0.001$

Estimated experimental IC<sub>50</sub> were then compared with theoretical IC<sub>50</sub> found in the literature<sup>11,13–16</sup> to verify the consistency of obtained results (Table 1). Firstly, the experimental IC<sub>50</sub> of curcumin for both cell lines were very different from those found in the literature (30 μM vs 6,03 μM and 30 μM vs 11,2 μM for H1299 and A549 respectively)<sup>13</sup>. However, experimental and theoretical IC<sub>50</sub> of crocin were similar on A549 cells (3,5 mM vs 4,2

mM respectively)<sup>11</sup>. Finally, same results were obtained for cisplatin on H1299 and A549 (5 μM vs 2,88 μM and 45 μM vs 35 μM for H1299 and A549 respectively)<sup>14,15</sup>. For all experiments, gemcitabine was used as a positive control (C+) at the concentration where the IC<sub>50</sub> was obtained on both cell lines (0,193 μM for H1299 and 11,54 μM for A549)<sup>14,16</sup>.

| Cell lines        | H1299                      |             |                             | A549                       |             |                             |
|-------------------|----------------------------|-------------|-----------------------------|----------------------------|-------------|-----------------------------|
| Compound          | Curcumin ( $\mu\text{M}$ ) | Crocin (mM) | Cisplatin ( $\mu\text{M}$ ) | Curcumin ( $\mu\text{M}$ ) | Crocin (mM) | Cisplatin ( $\mu\text{M}$ ) |
| Experimental IC50 | 30                         | 2           | 5                           | 30                         | 3,5         | 45                          |
| Theoretical IC50  | 6,03                       | ND          | 2,88                        | 11,2                       | 4,2         | 30 $\pm$ 5                  |

**Table1:** Experimental and theoretical IC50 of curcumin, crocin and cisplatin on H1299 and A549 cell lines

However, due to problems with crocin and cisplatin dissolution, these results must be taken carefully because of a lack of reproducibility (data not shown).

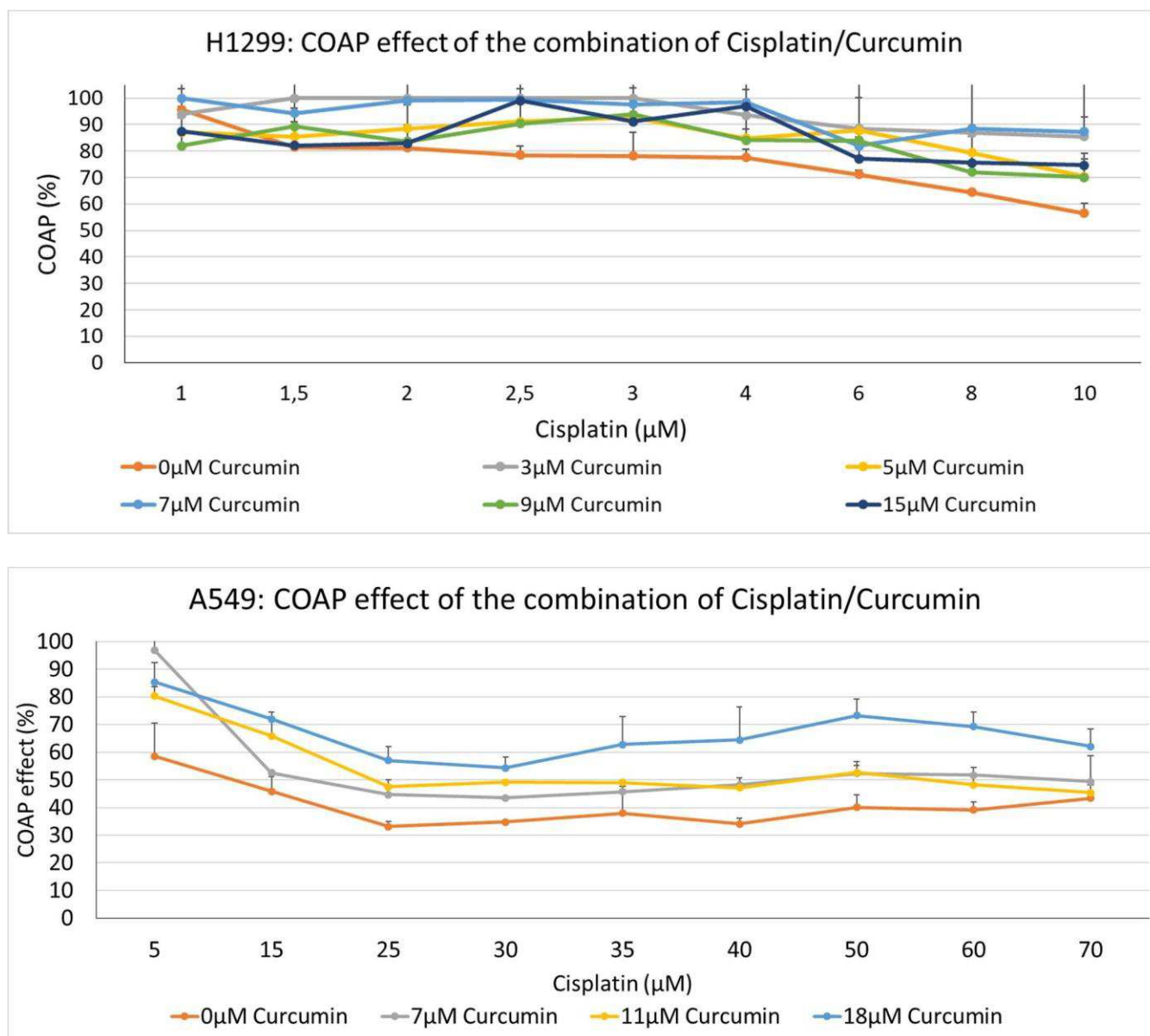
Thereafter, different combinations of compounds (cisplatin/curcumin (Figure 2) and cisplatin/crocin (data not shown)) were performed on both cell lines and the COAP effects were estimated thanks to the WST-1 assay. The aim of making combination was to determine whether it was possible to decrease the chemotherapeutic agent concentration by combining it with one of the natural compounds while maintaining the same COAP effect as if cisplatin was used alone at the same concentration (Figure 2).

On both cell lines, combinations of cisplatin with curcumin (3,5,7,9 and 15 $\mu\text{M}$  on H1299 and 7,11 and 18 $\mu\text{M}$  on A549) seemed to show a higher COAP than those obtained for cisplatin alone

(Figure 2). Moreover, for the same concentration of cisplatin, higher COAP were obtained when cisplatin was used in combination with curcumin than when cisplatin was used alone. Indeed, on H1299, for a concentration of cisplatin equal to 3 $\mu\text{M}$ , the COAP effect of cisplatin alone was 78%, compare to 90-92% when cisplatin was used with 5,9 and 15 $\mu\text{M}$  of curcumin and close to 100% when used with 3 and 7 $\mu\text{M}$  of curcumin.

Similarly, on the A549 cell line, for a concentration of cisplatin equal to 40 $\mu\text{M}$ , the COAP effect of cisplatin alone was 35%, compare to 48% when cisplatin was used with 7 and 11 $\mu\text{M}$  of curcumin and close to 65% when used with 18 $\mu\text{M}$  of curcumin.

As a conclusion, the use of curcumin to compensate the decrease of cisplatin seemed to induce higher COAP effect on both cell lines than when cisplatin is used alone.



**Figure 2:** COAP Effect of the combination of Cisplatin/Curcumin on H1299 and A549 cells after 48 hours treatment incubation at 1h30 WST-1 incubation.

Orange curve correspond to cisplatin only (without any curcumin concentration).

Results are obtained from one of our experiments.

Standard-deviations correspond to triplicates intra-manipulation.

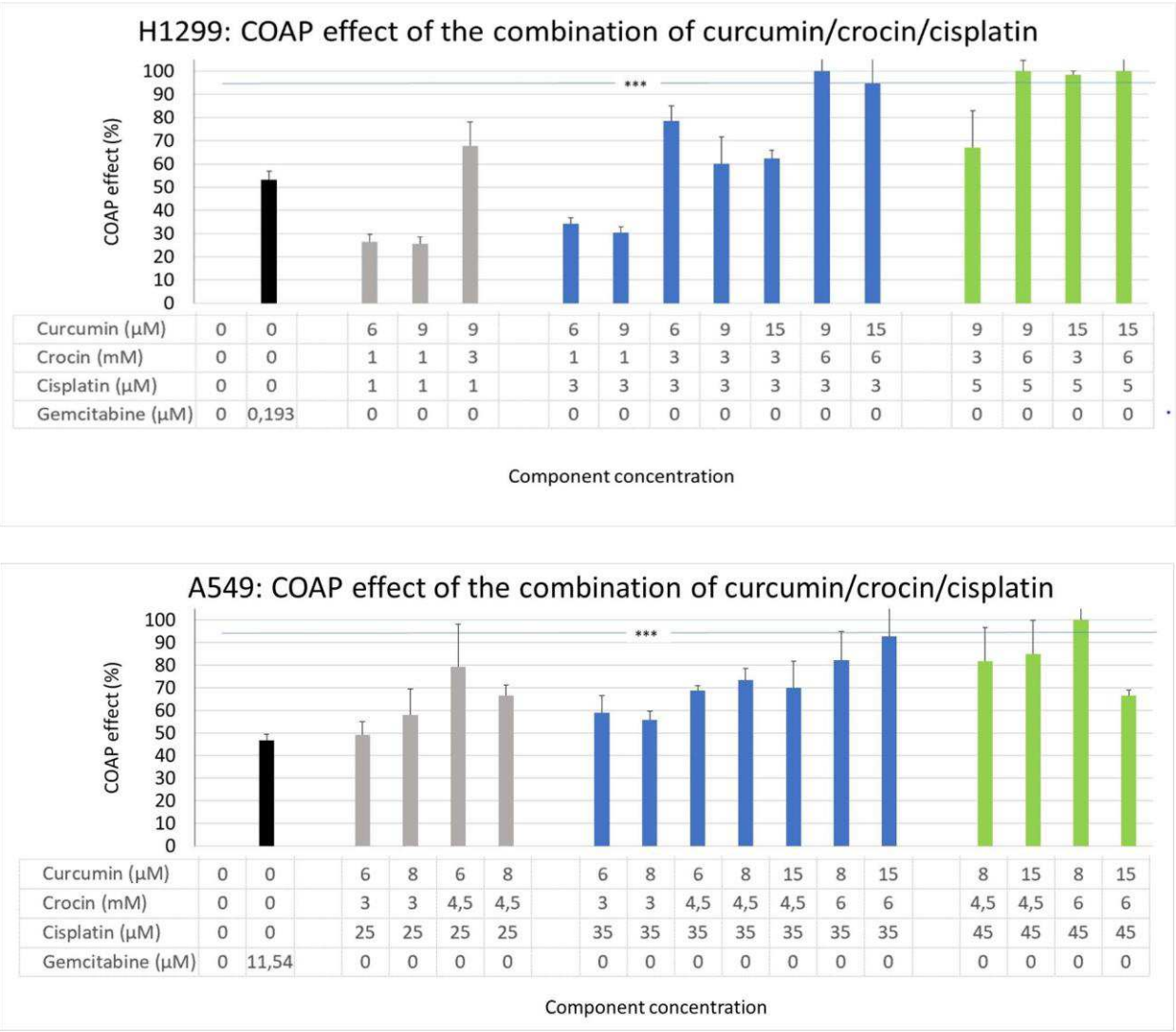
\* $p < 0.05$ ; \*\* $p < 0.01$ ; \*\*\* $p < 0.001$

Finally, a mix of curcumin/crocin with cisplatin was assessed on both cell lines and the COAP effect were determined (Figure 3) thanks to the WST-1 assay.

The aim was to determine whether it was possible to decrease the chemotherapeutic agent concentration by combining it with both natural compounds, while maintaining the same COAP

effect as if cisplatin was used alone at the same concentration.

On the H1299 cell line, the less efficient COAP effects were obtained with low concentration of Cisplatin (1 and 2,5µM) and Crocin (1mM) with a COAP max effect at 30%.



**Figure 3:** Determination of the COAP max effect of Curcumin/Crocin/Cisplatin on H1299 and A549 after 48 hours treatment incubation at 1h30 WST-1 incubation.  
C- correspond to the negative control without any treatment.  
C+ correspond to the positive control with a treatment of gemcitabine at a concentration found in the literature equal to the IC50 for the corresponding cell line.  
Results are obtained from one of our experiments.  
Standard-deviations correspond to triplicates intra-manipulation.  
\* $p < 0.05$ ; \*\* $p < 0.01$ ; \*\*\* $p < 0.001$



When comparing the COAP effect obtained with the mix curcumin/crocin/cisplatin (Figure 3) to cisplatin alone (Figure 1) on both cell lines, some relevant data could be outlined.

On the H1299 cell line, when cells were treated with cisplatin alone at a concentration equal to  $3\mu\text{M}$ , a COAP effect of 72% was obtained (Figure 1). In comparison, when cells were treated with a mix of curcumin/crocin/cisplatin at the same concentration of cisplatin ( $3\mu\text{M}$ ), higher or lower COAP effect were obtained. Indeed, when the concentration of curcumin/crocin into the mix of curcumin/crocin/cisplatin were respectively at  $6\mu\text{M}/3\text{mM}$ ,  $9\mu\text{M}/6\text{mM}$  and  $15\mu\text{M}/6\text{mM}$ , higher COAP effects were obtained (78%, 98%, 92%). Similar observations could be made regarding cisplatin concentration at  $5\mu\text{M}$ . The COAP effect was at 50% when cisplatin alone was used compare to 68%, 98%, 100% and 98% when cisplatin was used at  $5\mu\text{M}$  with the respective  $9\mu\text{M}/3\text{mM}$ ,  $9\mu\text{M}/6\text{mM}$ ,  $15\mu\text{M}/3\text{mM}$ ,  $15\mu\text{M}/6\text{mM}$  concentrations of curcumin/crocin. However, when cisplatin was used at  $1\mu\text{M}$ , COAP effect of cisplatin alone was higher than when cisplatin was used in combination with curcumin/crocin.

On the A549 cell lines, similar results were obtained. Indeed, COAP effects obtained into the mix of 3 products with a concentration of cisplatin equal to  $35\mu\text{M}$  were in majority higher (curcumin/crocin concentrations at  $6\mu\text{M}/3\text{mM}$  and  $8\mu\text{M}/3\text{mM}$  excluded) than when cisplatin was used alone. Moreover, all the COAP effect obtained with the mix at a concentration of cisplatin equal to

$45\mu\text{M}$  were higher than cisplatin alone was used. As a conclusion, the use of curcumin/crocin to compensate the decrease of cisplatin seemed for some concentration to induce higher COAP effect on both cell lines than when cisplatin is used alone.

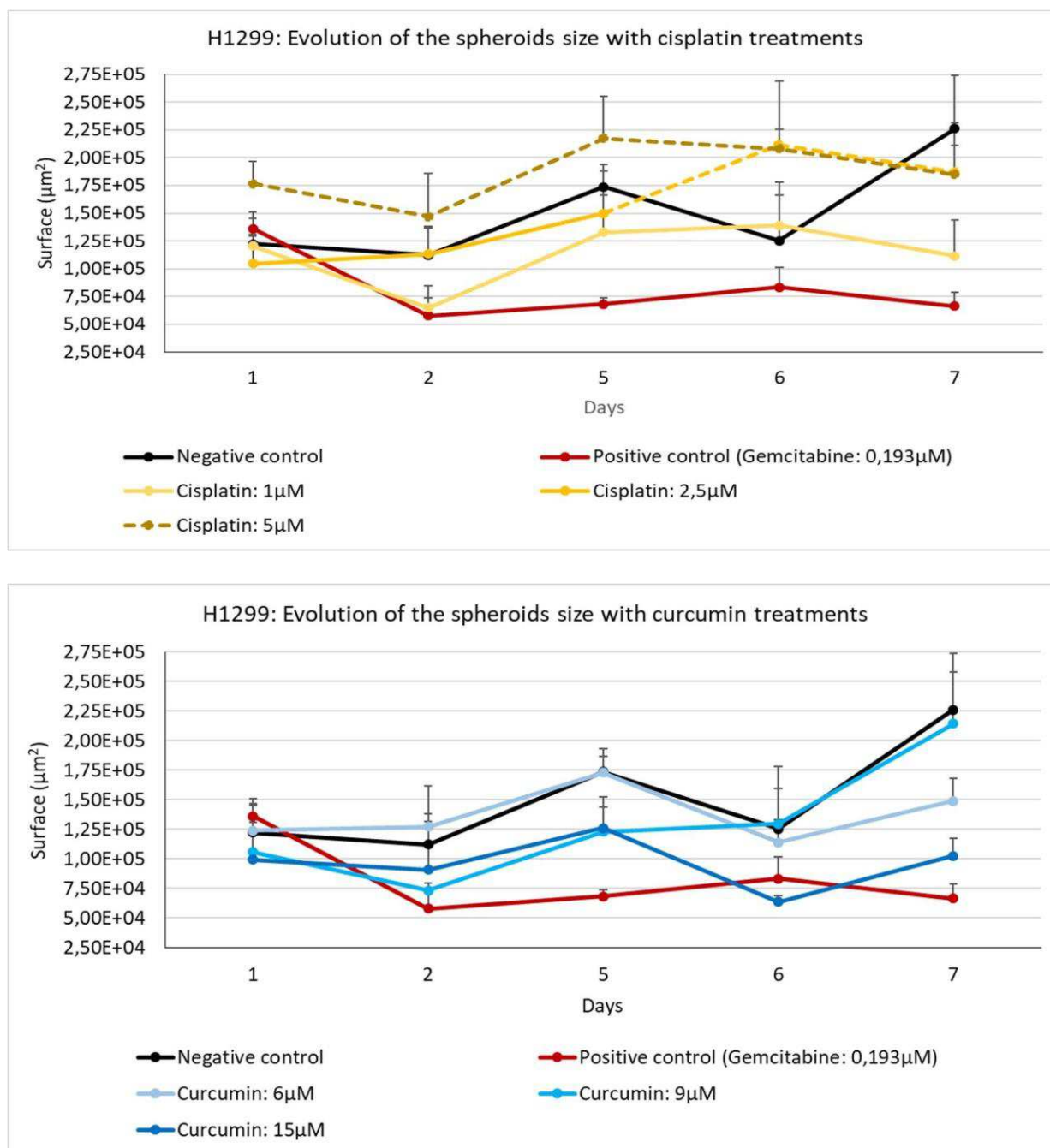
However, due to the short time period to make the experiment and problem of cisplatin and crocin dissolutions, all the results on the combinations (curcumin/cisplatin or curcumin/crocin/cisplatin) must be taken carefully because of a lack of reproducibility (data not shown).

After the assessment of the COAP effect of the different products on H1299 and A549 on 2D model, the COAP effects of the most suitable combination concentrations were assessed on a 3D cell model using spheroids formed from the same cell lines.

For the spheroids, only results obtained from the H1299 cell lines will be showed. No spheroids were obtained with the A549 cell lines.

First of all, spheroids are formed and grown for 48 hours before adding treatment. Once the treatment is added and in order to analyze their impact, the evolution of the size of the spheroids was followed-up during 7 days by taking cross-sectional plane images thanks to the ZOE microscope. Areas of these cross-sectional plane images are then evaluated using the ImageJ software.

Since spheroids are relatively spherical, it is expected that their size will be directly correlated to the area of a slice plane.



**Figure 4:** Evolution of the spheroid size made with H1299 cells through 7 days under several range of concentration of Curcumin and Cisplatin. Results are obtained from one of our experiments. Standard-deviations correspond to triplicates intra-manipulation. Negative controls correspond to non-treated spheroids. Positive control corresponds to spheroids treated with gemcitabine at a concentration found in the literature equal to the IC<sub>50</sub> for the H1299 cell line.

Spheroids formed from H1299 cells are treated with gemcitabine as positive control, and with different concentrations of curcumin and cisplatin (Figure 4).

For both series, spheroids treated with gemcitabine had the lowest tendency of surface compare to spheroids treated with cisplatin or curcumin or non-treated (day 2:  $5,7 \times 10^4 \mu\text{m}^2$ , day 5:  $6,8 \times 10^4 \mu\text{m}^2$ ).

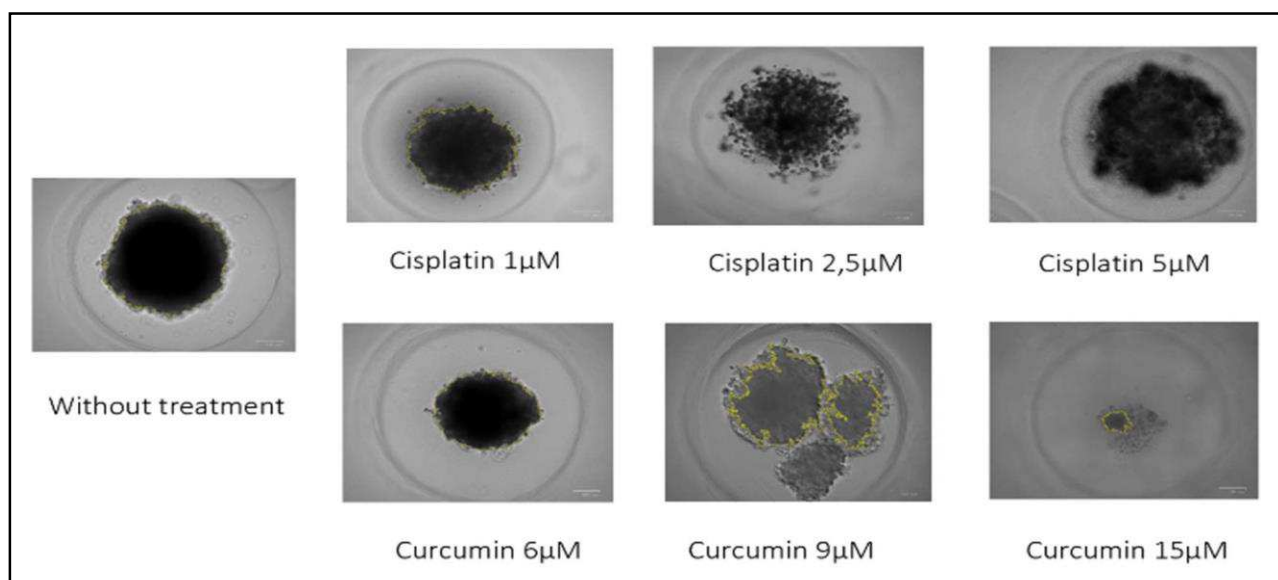
For the spheroids treated with different concentration of cisplatin (1 and  $2,5 \mu\text{M}$ ), their surfaces were smaller than those of the non-treated spheroids from day 1 to 5. Indeed, at day 5, the spheroids treated with 1 and  $2,5 \mu\text{M}$  of cisplatin had a smaller surface than those non-treated ( $1,3 \times 10^5$  vs  $1,5 \times 10^5$  vs  $1,6 \times 10^5 \mu\text{m}^2$  respectively). Moreover, for the spheroids treated with  $1 \mu\text{M}$  of cisplatin, their surface remained lower than those of the non-treated spheroids during everyday of follow-up. Nevertheless, the surface of the spheroids treated with  $5 \mu\text{M}$  of cisplatin was higher than the surface of the spheroids non-treated during the majority of the follow-up (at day 2:  $1,4 \times 10^5$  vs

$1,1 \times 10^5 \mu\text{m}^2$ , at day 5:  $2,1 \times 10^5$  vs  $1,7 \times 10^5 \mu\text{m}^2$ ). Similar observation could be done for spheroids treated with  $2,5 \mu\text{M}$  of cisplatin for day 6.

Similarly, spheroids treated with different concentrations of curcumin (6, 9 and  $15 \mu\text{M}$ ) showed a lower surface than those obtained for the non-treated spheroids. Indeed, at day 5, the spheroids treated with 6, 9 and  $15 \mu\text{M}$  had a smaller surface than those non-treated ( $1,6 \times 10^5$  vs  $1,2 \times 10^5$  vs  $1,2 \times 10^5$  vs  $1,7 \times 10^5 \mu\text{m}^2$  respectively). In addition, all the spheroids treated with curcumin showed a surface lower than those of the spheroids non-treated for the 7 days of follow-up.

As a conclusion, it can be concluded that for most of the treatment with cisplatin (cisplatin  $5 \mu\text{M}$  excluded) and for all the treatment with curcumin, a reduction size of the spheroids was obtained.

Thereafter, imaging of the spheroids was performed thanks to the ZOE at different day in order to analyze the structure of the spheroids with the added of treatments.



**Figure 5:** Pictures from the fluorescent cell imager ZOE after analysis with the ImageJ software at day 5 for different concentration of treatment on H1299

First of all, with the addition of treatments, an evolution of the roundness of the spheroids could be observed (Figure 5). Indeed, by increasing the concentration of the compound, the roundness of the spheroids decreases.

For the cisplatin, the size remained constant through the increasing concentrations but a spheroid fragmentation into cells is obtained and is more important for high dose of cisplatin.

On the other hand, for the curcumin, a diminution of the size of the spheroids was obtained through increasing concentration. Moreover, a fragmentation in larger part was obtained for spheroids with 9  $\mu$ M of curcumin.

As a conclusion, the addition of treatments to the spheroids induced a diminution of the size of the spheroids and tended to induce a fragmentation of the spheroids either in cells for cisplatin or in smaller spheroids for curcumin.

## DISCUSSION AND CONCLUSION

The objective of this study is to demonstrate that a decrease in the dose (concentration) of cisplatin compensated by its association with natural compounds (curcumin and/or crocin) would maintain or even increase the toxicity of cisplatin on cancer cells. The long-term goal would be to reduce the adverse effects of cancer treatments.

To assess this hypothesis, the cell proliferation WST-1 assay was used to analyze the cytotoxic or anti-proliferative (COAP) effect of cisplatin, curcumin and crocin, either independently or either in combination (cisplatin/curcumin, cisplatin/crocin and curcumin/crocin/cisplatin). Experimental IC<sub>50</sub> values for cisplatin and crocin were close to those found in the literature for both H1299 and A549 cell lines. However, in our hands,

those results weren't reproducible probably because of problem of dissolution of the products (cisplatin and curcumin), and have to be performed again.

Thereafter, the COAP effect of cisplatin in combination with either curcumin or a mix of curcumin/crocin were assessed.

All tested combinations gave positive dose-dependent effects with increase effects upon high concentration on both cancer cell lines.

Thanks to the WST-1 assay that was used on both cell lines with cisplatin and curcumin, higher COAP effect were obtained when the chemotherapeutic agent cisplatin was combining with curcumin than when cisplatin was used alone at the same concentration.

These results suggested that on both cell lines, combinations of cisplatin with curcumin could give better anti-cancerous effects than when cisplatin is used alone at the same concentration of cisplatin.

Then, a WST-1 assay on both cell lines with cisplatin and both natural compounds was made. Similarly, to the previous results obtained with the mix curcumin/cisplatin, a combination of cisplatin with both natural compounds showed higher COAP effect than when cisplatin was used alone at the same concentration (for cisplatin concentration equal to 3 and 5  $\mu$ M on H1299 and 35 and 45  $\mu$ M on A549). These results also suggested that on both cell lines, combinations of cisplatin with curcumin and crocin could give better anti-cancerous effects than when cisplatin is used alone at the same concentration of cisplatin.

However, because of the short time period accorded to make the manipulation, we were not able to reproduce our results and test their reproducibility. So, these results need to be taken carefully.

Subsequently, based on the COAP effect determined with the WST-1 assay on the H1299, the most suitable concentrations of curcumin and cisplatin were assessed on the 3D model (spheroids). No products were tested on the A549 cell line because we were not able to make spheroids with them.

After several days of treatment on the spheroids, a decreasing tendency of the spheroids size was observed, meaning that products combinations (curcumin, crocin and cisplatin) had a positive impact on spheroids size.

ImageJ analysis of the images obtained with the ZOE shows that the decrease in the size of the spheroids comes from the periphery of the spheroids. Cisplatin tends to induce cell fragmentation compared to curcumin, which tends to convert spheroids into smaller ones.

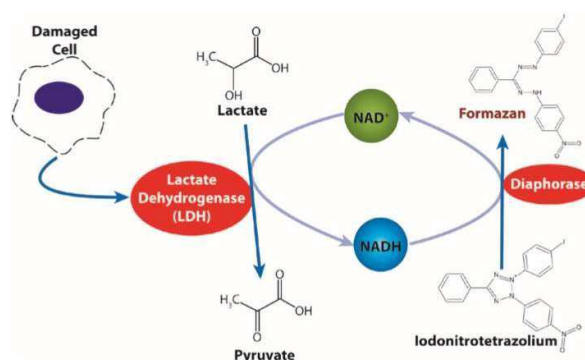
Similarly, because of the short time period accorded to make the spheroids manipulation, we were not able to test the reproducibility of these results. So, these results need to be taken carefully. Moreover, none combinations of compounds were assessed on the spheroids and need to be performed in order to see if the higher COAP obtained with the 2D model would be also obtained on the 3D model.

In order to completely respond to our hypothesis, other experiments need to be performed. Indeed, none of the tests realized in the present study gave information about a potential reduction of the negative effect of the cisplatin when used with one or 2 natural compounds.

In order to evaluate the negative effect of cisplatin alone compare to the combination of cisplatin with natural compounds, a WST-1 assay could be performed on non-cancerous cells (fibroblast).

With this test, if a lower COAP is obtained when cisplatin and the natural compound are combined rather than when the cisplatin is used alone, the adverse toxic effects of the cisplatin could, then, be assessed.

Moreover, further assays could be planned to determine which mechanism is involved in the COAP: cytotoxicity or anti-proliferation. Indeed, to know if our products only induce cytotoxic effect on cells, a lactate dehydrogenase (LDH) assay could be performed. LDH is a stable cytoplasmic enzyme present in all cells and rapidly released after damage to the plasma membrane. Then, LDH activity can be determined with a coupled enzyme reaction (figure 6). The increase in the amount of formazan produced in the culture supernatant is directly related to the increase in the number of lysed cells. The formazan dye is water soluble and can be detected spectrophotometrically at 500 nm<sup>17</sup>.



**Figure 6:** Principle of the LDH cytotoxic assay<sup>17</sup>

On the other hand, a flow cytometry analysis could give information on the anti-proliferative properties of compounds on cells. Indeed, the use of fluorochromes (propidium iodide staining<sup>18</sup>) and flow cytometry would allow to determine in which phases of the cell cycle the cells are after treatment with the different compounds and therefore deduce a possible alteration of the cell cycle

Finally, performing another cell viability assay with the MTT test could improve the reproducibility of the results obtained with the WST-1 assay.

## REFERENCES

1. Cancer. <https://www.who.int/fr/news-room/fact-sheets/detail/cancer>.
2. Définition cancer. <https://www.e-cancer.fr/Dictionnaire/C/cancer>.
3. Signes précurseurs de cancer - Cancer - Manuels MSD pour le grand public. <https://www.msmanuals.com/fr/accueil/cancer/présentation-des-cancers/signes-précurseurs-de-cancer>.
4. Les chiffres clés du cancer en France | Ligue contre le Cancer. [https://www.ligue-cancer.net/article/26089\\_les-chiffres-cles-des-cancers](https://www.ligue-cancer.net/article/26089_les-chiffres-cles-des-cancers).
5. Tumeurs cancéreuses du poumon - Société canadienne du cancer. <https://www.cancer.ca/fr-ca/cancer-information/cancer-type/lung/lung-cancer/cancerous-tumours/?region=on>.
6. Les traitements et soins de support | Fondation ARC pour la recherche sur le cancer. <https://www.fondation-arc.org/traitements-soins-cancer>.
7. Dasari, S. & Bernard Tchounwou, P. Cisplatin in cancer therapy: Molecular mechanisms of action. *European Journal of Pharmacology* vol. 740 364–378 (2014).
8. Résumé des caractéristiques du produit - CISPLATINE TEVA 1 mg/1 mL, solution pour perfusion - Base de données publique des médicaments. <http://base-donnees-publique.medicaments.gouv.fr/affichageDoc.php?specid=68377552&typedoc=R>.
9. Pillai, G. R., Srivastava, A. S., Hassanein, T. I., Chauhan, D. P. & Carrier, E. Induction of apoptosis in human lung cancer cells by curcumin. *Cancer Lett.* **208**, 163–170 (2004).
10. Curcuma | Fondation contre le Cancer. <https://www.cancer.be/complementsalimentaires/c>urcu ma.
11. Chen, S. *et al.* Crocin inhibits cell proliferation and enhances cisplatin and pemetrexed chemosensitivity in lung cancer cells. *Transl. Lung Cancer Res.* **4**, 775–783 (2015).
12. Pili, B. *et al.* Interaction of a new anticancer prodrug, gemcitabine-squalene, with a model membrane. Coupled DSC and XRD study. *Biochim. Biophys. Acta - Biomembr.* **1798**, 1522–1532 (2010).
13. Amin, A. R. M. R. *et al.* Curcumin Induces Apoptosis of Upper Aerodigestive Tract Cancer Cells by Targeting Multiple Pathways. *PLOS ONE* **10**, e0124218 (2015).
14. Tsai, C. M. *et al.* Cytotoxic effects of gemcitabine-containing regimens against human non-small cell lung cancer cell lines which express different levels of p185neu. *Cancer Res.* **56**, 794–801 (1996).
15. Baharuddin, P. *et al.* Curcumin improves the efficacy of cisplatin by targeting cancer stem-like cells through p21 and cyclin D1-mediated tumour cell inhibition in non-small cell lung cancer cell lines. *Oncol. Rep.* **35**, 13–25 (2016).
16. Li, J., Wang, S., Su, Z. F. & Yuan, Y. Synergistic effects of sorafenib in combination with gemcitabine or pemetrexed in lung cancer cell lines with K-ras mutations. *Wspolczesna Onkol.* **20**, 33–38 (2016).
17. Cytoscan™-LDH Cytotoxicity Assay. A Colorimetric Assay for Cellular Cytotoxicity (Cat. # 786-210, 786-324). 12.
18. Riccardi, C. & Nicoletti, I. Analysis of apoptosis by propidium iodide staining and flow cytometry. *Nat. Protoc.* **1**, 1458–1461 (2006).

# Adapting a microfluidic device for applications in 3D cell culture

Pierre-Emmanuel CORBET, Mary-Amélie MASSON\*, Pauline MEGALLI, Thibaud METAIS, Julie SIX;

Project supervised by Dr Frank YATES, Dr Vanessa PERDIZ and Dr Ferid NASSOR

Sup'Biotech, Villejuif, France

\*Corresponding author: Mary-Amélie MASSON, [mary-amelie.masson@supbiotech.fr](mailto:mary-amelie.masson@supbiotech.fr)

**Abstract -** Microfluidic devices are increasingly used to help modelize organs or diseases. However, in the case of spheroids or organoids, the supply of oxygen and nutrients is not efficient enough in the center, and cells have a tendency to undergo necrosis. The creation of a vascularization into a spheroid remains one of the main challenges in the field. Several methods are developed for vascularization. One of them consists in using endothelial cells on a microchip to favorize a vascularization towards the inside of the spheroid. For that, the question asked is: how can a microchip support favorize three-dimensional cell culturing? We based our work on the previous publications of Nashimoto and his colleagues.

A method to cultivate spheroids, and more generally cells on a microchip, managing several parameters such as the design of the chip, is under development.

*Index terms* - Microfluidics, Spheroid, Vascularization, Necrosis, 3D Cell Culture, Engineering Problems

## INTRODUCTION

In order to study diseases, *in vitro* three-dimensional (3D) tissues, such as cerebral [1], [2], hepatic [3] or intestinal [4] organoids, are developed to mimic the function of organs in a physiological or physiopathological context. These 3D structures can be obtained from induced pluripotent stem (iPS) cells which possess the properties to be self-renewable and pluripotent (capable of differentiating into the three germ lineages: mesoderm, ectoderm, endoderm).

The advantages of using organoids compared to animal models, two-dimensional (2D) cell culture or perfusable organs [5], are that they are able to better recapitulate some aspects of the human pathologies. Also, the use of *in vitro* assays contributes to the respect of the 3Rs regulation (Replacement, Reduction, Refinement) [6] as they can provide a complement to animal experiments.

Briefly, the first step to obtain organoids from iPS cells consists in generating embryoid bodies (EBs). EBs are 3D aggregates of cells into compact and spherical structures. The microenvironment surrounding the embryoid body and internal cellular mechanisms such as the maintenance of E-cadherin junctions [7] allow these aggregates to achieve a three-dimensional structure. These EBs can then be differentiated into specific tissues by using specific growth factors.

Several techniques exist to form 3D EBs [8] or spheroids (similar to organoids but composed of tumor or

transformed cells). The “hanging drop” method consists in self-aggregation of cells into small droplets. The “well formation” method consists in using non-adhesive micro-well molds, with diameters comprised between 300 and 800  $\mu\text{m}$ . Suppliers such as SIGMA-ALDRICH (Microtissue 3D Petri Dish 12-81 or 12-256) or STEMCELL TECHNOLOGIES (AggreWell 400 or 800) have developed an array of products to achieve this. Moreover, a group of colleagues at Sup'Biotech (Mauviel *et al.*, Sup'Biotech Projects, 2020) developed a reusable system made of PolyDiMethylSiloxane (PDMS) with pyramid-shaped wells of 800  $\mu\text{m}^2$ , and a mesh allowing medium change without compromising the spheroids in formation.

However, one of the main issues met during organoid culture is that oxygen and nutrients cannot reach the inside of organoids to feed the inner cells [8], and waste products cannot leave the organoids. This lack of oxygen at the center of organoids, which has been mathematically modeled [9], causes necrosis inside the structure.

Several tools exist [10] to increase the nutrients and oxygen inputs in the center of organoids. These solutions include regular shaking using bioreactors [11], 2D compartments compressing the organoids [12], *in vivo* engraftment [13], bioprinting artificial vasculature [14] as well as the use of microfluidic chips [15]. The main advantages of microfluidic devices are the versatility of the technology, the reproducibility and the possibility of automatization. Microchips are generally made of PDMS. This elastomer is the most used liquid precursor for nanofabrication. It is inexpensive, durable, unreactive toward other materials, chemically resistant to solvent, transparent above a wavelength of 280 nm and biocompatible [16]. Nevertheless, PDMS is a hydrophobic material, which is not suitable for adhesive cell culture. To overcome this, a plasma oven can be used to introduce polar functional groups [17], converting PDMS into a hydrophilic polymer, but the hydrophilicity is lost when the chip dries. Another obstacle usually encountered with this type of device is the nanoscale of the channels. Indeed, it creates resistance forces to the medium inserted inside the microchip, and therefore, air bubbles emerge. Consequently, microfluidic chips are, most of the time, used in a dynamic setting.

Vascularization of the organoid could provide a solution to the necrosis problem inside 3D organoids and spheroids. Endothelial cells are the cells implicated in the vascularization of organs and tissues [18]. Thus Human Umbilical Vein Endothelial Cells (HUVECs) [10], [19], could contribute to the survival and oxygenation of the tissue. We decided to focus on the use of a PDMS



microfluidic device to facilitate the co-culture of 3D structures with HUVEC cells. By culturing HUVECs with a spheroid or an organoid on the same surface, we expect to form functional angiogenic sprouts from the initial location of HUVECs to the spheroid / organoid. This strategy was already validated by a Japanese team in 2017 and 2018 [19], [20] where they used HUVECs to make angiogenic sprouts to a spheroid made of a co-culture of human lung fibroblasts (hLFs) with HUVECs. A continuous lumen was obtained from the merging of angiogenic sprouts with the vessel-like structures already obtained inside the spheroid.

The work presented in the following pages tried to take advantage of these prior researches to solve the necrosis problem by creating a vascularization.

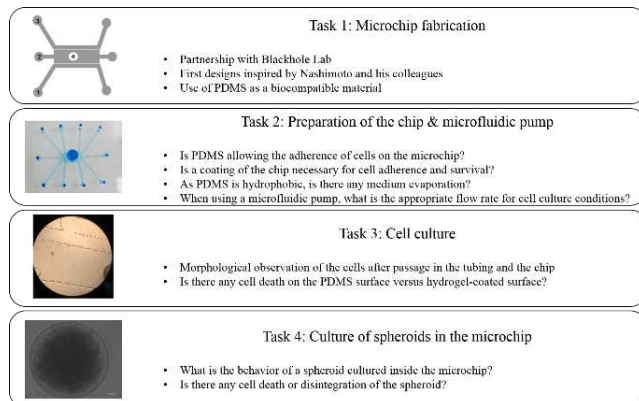


FIGURE 1  
SUMMARY OF THE EXPERIMENTS PERFORMED

However, as previously stated, difficulties appeared when working at the nanoscale and especially when PDMS is used. Our objective was thus to deal with engineering problems such as the coating of the chip, determining medium flow and cell culture conditions inside the microchip (FIGURE 1).

In this work, the results were obtained with tumoral cells, in order to obtain an easier and faster modelization than by using iPS cells or primary endothelial cell culture. An interaction between tumoral cells and a spheroid will be a proof of concept that in this type of microfluidic device, it is possible to have peripheral cells which will invade a spheroid or an organoid in order to create a vascularization.

## MATERIAL AND METHODS

### I. Microchip Fabrication

Microchips in PDMS were produced and provided by BlackHole Lab<sup>1</sup>. The chip model was adapted from the article of Nashimoto and his coworkers [20]. Microchips were filled with PBS, sealed with glass, and then they were put in a bath of PBS for transport. This allows the microchips to remain hydrophilic.

The central well (n°4 in FIGURE 2) is designed to host the spheroid. The top wells (n°1, 2 and 3 in FIGURE 2) are the entrance of cells and medium. The non-numbered bottom wells (FIGURE 2) are the exit for medium. At the

beginning of the experiment, the central well (n°4 in FIGURE 2) should only contain the spheroid. The side channels are for the cell culture (endothelial or other type).

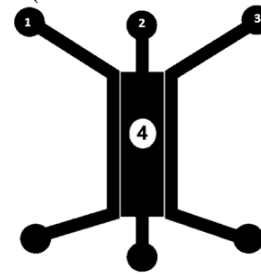


FIGURE 2  
PLAN OF THE MICROCHIP USED

At the end of the experiment, there should be growth of the cells from the side channels towards the spheroid. Indeed, the microposts that are located at the interface between the central compartment and the channel will favorize the growth of the cells from their initial location to the spheroid. In the case of endothelial cells, they will form angiogenic sprouts to the inside of the spheroid (FIGURE 3) [19].

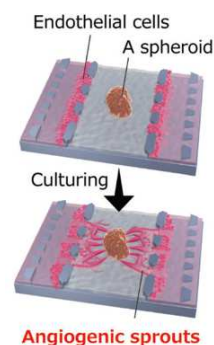


FIGURE 3  
SCHEMATIC DIAGRAM OF AN OVERVIEW OF THE CULTURE EXPERIMENT ON THE MICROFLUIDIC CHIP PRESENTED BY NASHIMOTO AND HIS COWORKERS [19]

The dimensions of the microfluidic chip used for the experiments are presented in FIGURE 4. These dimensions were found to be optimal after working on both models proposed by Nashimoto and colleagues [19], [20].

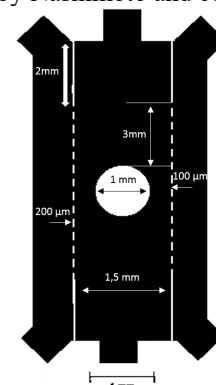


FIGURE 4  
ZOOM OF THE CENTRAL PART OF THE FIGURE 2 AND DETAILED PLAN WITH THE DIMENSIONS OF THE MICROCHIP USED

<sup>1</sup> <https://www.blackholelab-soft-lithography.com>



## II. Microchip Sterilization

An assay based on DH5 $\alpha$  bacteria culture was devised in order to verify the efficiency of the sterilization techniques used in this study.

### 70% ethanol sterilization

DH5 $\alpha$  bacteria were added into the channels of the microchip to induce contamination and then the chip was put in a 70 % ethanol bath for 15 minutes. After an overnight drying step, Dulbecco's Phosphate Buffered Saline (PBS) (SIGMA, D8537) was passed through the chip and PBS was dropped off on an LB agar Petri dish specific for bacterial culture. After an overnight incubation at 37°C, Petri dish was observed.

### UV sterilization

DH5 $\alpha$  bacteria were added into the channels of the microchip to induce contamination, then the chip was incubated under the UV lamp of a basic bacteriological hood. After 1h of incubation, the same protocol than for ethanol sterilization was followed.

## III. Flow Set-up in the Microchip

Also, a membrane pump system from Fluigent (FIGURE 5) was used in order to have an autonomous system and a constant renewal of medium. The set-up of the pump was adapted from the installation protocol of the manufacturer (Fluigent). In this system, it is possible to work with pressures up to 2 bars. This membrane pump can be used for several types of experiments such as organ-on-a-chip culture or droplets and particles manipulation. The most suitable pressure for cell culture will be determined by calculating the time for the consumption of 0.5 mL of liquid for different pressures.

1/32" (0,8 mm for outer diameter) tubing was used to put culture medium into the microchip.

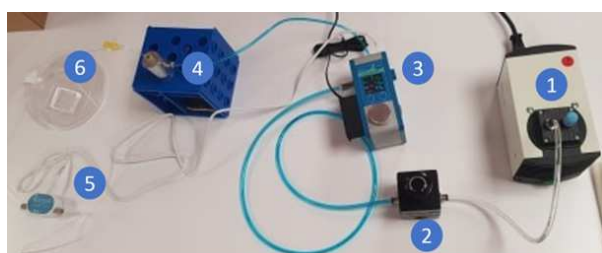


FIGURE 5  
MEMBRANE PUMP FOR A CONTINUOUS FLOW SYSTEM  
(FLUIGENT)

(1) Membrane pump N 86 KN18 (Laboport, P065034); (2) Precision regulator R27-200 (IMI Norgren, R27-200-RNG); (3) Flow EZ (LU-FEZ-1000); (4) Medium reservoir (15 mL Falcon tube); (5) Flow unit (size S) (CTQ\_Kit\_LQ); (6) Microchip

## IV. Cell Culture

### Culture Medium Preparation

The culture medium for tumoral cells is prepared with Dulbecco's Modified Eagle's Medium (DMEM) – High Glucose (SIGMA, D6546) supplemented with 10 % of Fetal

Bovine Serum (FBS) (GIBCO, 10500-054), 1 % HEPES buffer 1M (SIGMA, 83264), 1 % of Non-Essential Amino Acids 100X (GIBCO, 11140-035), 1 % of L-Glutamine solution 200 mM (SIGMA, 59202C) and 1 % of Penicillin Streptomycin 100X (GIBCO, 15140-122).

### Passaging Tumoral Cells

HeLa cell line [21] was used. For the cells to be ready for the experiments, it is necessary to passage them when they are at 70 – 80 % confluence. Cell medium is removed, and cells are rinsed with 4 mL of PBS. 1.5 mL of Trypsin EDTA 0,05%, phenol red (GIBCO, 25 300-054) are added to the cells which are incubated for 5 to 10 minutes at 37°C. Then, the flask is flushed to recover all the cells and they are put into a 15 mL Falcon tube containing at least one volume of warm complete DMEM. 20  $\mu$ L of the cellular mix are taken in order to count the number of cells using a Malassez cell counting slide. Finally, the desired number of cells is placed in a cell culture dish previously filled with 10 mL of complete DMEM. Cells are incubated at 37°C, 5% of CO<sub>2</sub>.

### Cell Culture in the Microfluidic Device

The microfluidic device was coated with Matrigel (Corning, 354263, batch n° 3330625 with an initial concentration of 18,06 mg/mL) [22], using the membrane pump. 1h of incubation at room temperature is necessary to let the Matrigel jellyify and cover correctly the surface.

In order to facilitate the observation of the cells in the microchip, green genetically modified HeLa cells were used. These cells stably express the green fluorescent protein (GFP) and were provided by CellTechs laboratory (Sup'Biotech/CEA DRF IBFJ SEPIA).

HeLa cells previously grown were harvested and seeded in the microfluidic device thanks to a flow provided by a membrane pump. A pressure of 25 mBar was applied, allowing a provision of medium from a 15 mL Falcon tube for more than 21h without refilling.

## V. Spheroid Formation

HeLa spheroids were made using 3D Petri Dish Microtissues kit (Microtissues 3D, SIGMA, 12-81), according to the manufacturer's instructions. Briefly, a 2X agarose (SIGMA, A9539) 0.9% NaCl (SIGMA, S8776) mix was prepared and autoclaved, and then 600  $\mu$ L of this preparation were used to create agarose micro-molds using the 3D Petri Dish.

To generate spheroids of 500  $\mu$ m or 600  $\mu$ m diameter, respectively 1,200,000 cells/190  $\mu$ L or 2,200,000 cells/190  $\mu$ L were seeded on the 3D Petri Dish. This corresponds to respectively 14,815 and 27,160 cells per microwell composing the mold. Two days of incubation at 37°C, 5% of CO<sub>2</sub> were necessary to obtain spheroids with an optimal size.

## VI. Spheroid Culture in a Microfluidic Device

A spheroid was harvested with a micropipette and placed in the central well of the microchip. A constant flow was then applied to continuously provide the spheroid with fresh medium. This flow was produced with a membrane pump.

Pressure applied is the same than in cell culture with the microfluidic device (25 mBar).

## RESULTS

### I. Microchip Sterilization

Even if the microchip was produced under sterile conditions and firstly sterilized by BlackHole Lab thanks to a plasma oven, we preferred sterilizing it again to avoid contamination in our microfluidic chip before culturing cells in it. Several methods are described in the literature to sterilize PDMS devices, and here two different techniques were compared and evaluated.

After the 70 % ethanol sterilization, the Petri dish was observed. Some colonies of bacteria were still present (FIGURE 6). However, the Petri dish treated with UV sterilization did not contain any bacterial colony (data not shown). Even though the testing conditions were harsh (inserting bacteria directly in the channels of the microchip, which is highly unlikely in reality), both sterilization methods showed promising results. However, ethanol sterilization appeared to be less convenient because it is difficult to totally wash it: even after letting the microchip dry overnight, there were still residues of ethanol inside the microchip, which could interfere with cell culture.



FIGURE 6  
DH5α BACTERIAL CULTURE AFTER 70% ETHANOL  
STERILIZATION

### II. Flow Set-up in the Microchip

Since the chip can contain a maximum volume of 30  $\mu$ L, we wanted to find a flow rate that would allow medium renewal constantly and for a relatively long period of time. As explained previously, the flow rate was calculated compared to the pressure. The pressure of 25 mBar was selected. According to our calculations, the corresponding flow is around 12  $\mu$ L per minute. With a reservoir of medium of 15 mL, the autonomy is at least of 21h, which is acceptable for our purpose. Our goal here was to see whether a pressure and the use of a tube to seed cells would damage and kill the cells. Cells were put in a medium reservoir and a pressure of 50 mBar was applied for the cells to pass through the tubing. They were then seeded in a Petri dish and observed after 24h and 36h. No significant change was noticed, and cells were still viable (data not shown). This pressure of 50 mBar did not seem to damage the cells.

### III. Cell Culture in a Microfluidic Device

HeLa cells were injected in the microchip through the tubing, under a 50-mBar pressure. Right after the seeding step, HeLa cells were observed under the ZOE Fluorescent Cell Imager (Bio Rad, 1450031). After the cell seeding step, the flow had to be stopped for at least 6 hours in order to let the cells adhere to the matrix.

24h after seeding, HeLa cells were observed under the ZOE microscope. Most of them did not adhere to the microchip, even after coating (shown with arrowheads on FIGURE 7). However, some of them adhered which corresponds on the figure to cells with a deformation (shown with arrows on FIGURE 7).

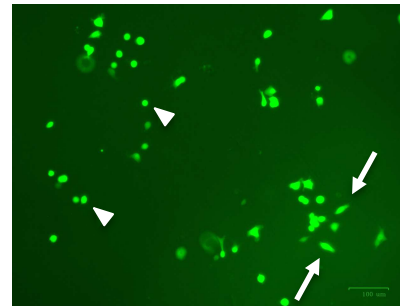


FIGURE 7  
HELA CELLS SEEDDED ON THE MICROFLUIDIC CHIP (T = 24H),  
SCALE = 100  $\mu$ m  
Arrows show adherent HeLa cells on the coated surface of the PDMS microfluidic chip. Arrowheads show HeLa cells that did not adhere to the coating, but which stuck to it.

### IV. Spheroid Culture in a Microfluidic Device

Two-day-old spheroids were harvested from the agarose Microtissues wells and an optical microscope was used to collect one single big and non-dissociated spheroid. This spheroid was then inserted into the central well (n°4 on FIGURE 2) of the microchip.

The spheroid put in culture into the chip (FIGURE 8) kept its round shape and just a few cells from the edge dissociated from the core of the spheroid.

Nevertheless, when the culture medium flow was applied with the membrane pump, the spheroid did not remain in the central well; it moved inside the chip, and was washed away into the different channels.

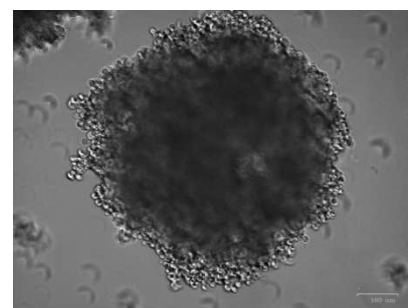


FIGURE 8  
SPHEROID OF HELA CELLS SEEDDED ON THE MICROFLUIDIC  
CHIP, SCALE = 100  $\mu$ m

## CONCLUSION

Several problems were identified when using a PDMS microfluidic chip for cell culture.

Firstly, we determined that the UV sterilization is easier and more convenient than the ethanol sterilization. Secondly, we found the best cell culture medium flow set-up according to the size and the design of our chip, using the membrane pump manufactured by Fluigent. This flow set-up was finally also used for the coating of the chip and the seeding of the cells in order to minimize the formation of bubbles into the channels. At the end, we also put a spheroid into the chip, but we did not have enough time to observe any interactions between cells and the spheroid as we expected (FIGURE 3). As a reminder, an interaction between cells and our spheroid will prove that in this type of microfluidic device, it is possible to have peripheral cells which will invade (in the case of tumoral cells) or vascularize (in the case of endothelial cells) the spheroid or the organoid placed in the central well. The vascularization by endothelial cells of an organoid could be a solution to the necrosis problem.

## DISCUSSION

Through a collaboration with the company BlackHole Lab we obtained two versions of microfluidic chips based on the work published by Nashimoto *et al.* [19], [20] (first version not shown; second version shown in FIGURE 2 with dimensions in FIGURE 4). We mainly worked with the second version as it was the most convenient for us.

In the first version of the microfluidic chip [19], there were two more channels and they were also larger. Throughout our experiments with the first version of microfluidic chip (100  $\mu$ L), air bubble formation constituted a major problem when working in a static setup (i.e. without a microfluidic pump). Air bubbles were probably due to the fact that PDMS is a hydrophobic material. Moreover, the fact that this material is hydrophobic prevents the adhesion of cells and therefore their potential survival inside the device.

The second version of the microfluidic chip (presented in FIGURE 2 with its detailed dimensions in FIGURE 4) is similar to the chip presented by Nashimoto *et al.*, [20] but according to our previous observations with the first design that we used, we decided to design a microchip with much smaller dimensions; it can contain up to 30  $\mu$ L of medium. A microfluidic pump was used in order to avoid the formation of air bubbles by applying a constant stream of medium. This method was used to perform both coating and cell seeding. In order to reduce medium evaporation and to maintain the hydrophilicity of the material, microfluidic chips were immersed into cell culture medium.

We still encountered minor bubble problems with this second version of our microchip. Therefore, we have devised a new design avoiding sharp angles, and minimizing the number of channels (FIGURE 9). Indeed, these two parameters are thought to promote the formation of air bubble in microfluidic devices. In this new design, only one channel subsists, with one flow inlet and one flow outlet at the opposite side. Cells will be added with the

medium flow and should surround the central well where the spheroid or the organoid will be inserted. The advantage of using a microfluidic chip composed of just one channel is that the flow will now be mainly laminar as the number of bifurcations is decreased. With this type of flow inside a chip, we hope for a decrease in the number of bubbles that appear.

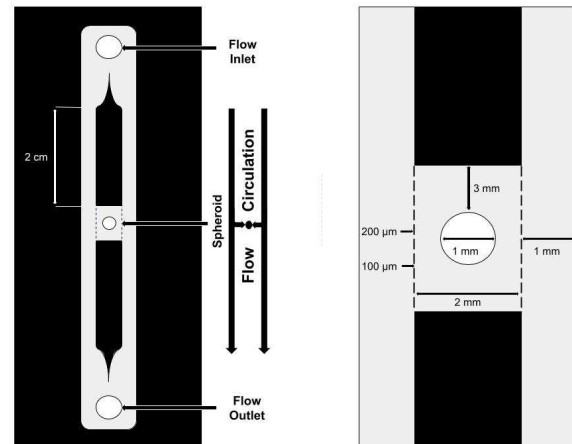


FIGURE 9  
NEW DESIGN OF THE MICROCHIP THAT WE WOULD LIKE TO  
USE FOR FUTURE EXPERIMENTS

As a perspective, it would be possible to test other coating agents, like Bovine Serum Albumin (BSA) or polylysine. Polylysine is already widely used in the literature and is a positively charged homopolymer. This would facilitate the adhesion of cells inside the microchip, as the cell membranes are negatively charged.

Here, our final goal was to co-culture endothelial cells with an organoid in order to vascularize it and reduce the number of cystic structures in the center of the 3D structure. The results obtained with tumoral cells like HeLa lineage allow us to solve some of the engineering problems met towards this goal.

However, other applications are possible with our microchip. For example, as cerebral organoids do not possess any microglial cells, we could imagine adding this cell type into the chip to insert them in a mini brain.

Neuronal inflammation or glioblastoma could also be modeled and studied. In this last example, Ogawa and colleagues [23] developed a human cerebral organoid to model this cerebral cancer thanks to the CRISPR-Cas9 system. After a few weeks, all the healthy tissue of the organoid was replaced by tumoral cells. The use of a microfluidic device would allow to vascularize the organoid and therefore, to obtain a bigger mini brain. This could allow to conduct longer studies on this disease by maintaining some healthy brain organoid cells for a longer period.

## ACKNOWLEDGMENT

We would like to thank our tutors, especially Dr Frank YATES and Dr Vanessa PERDIZ, for their numerous advice as well as the other members of the staff of Sup'Biotech: Dr Ferid NASSOR, Dr Elise DELAGE. We

also would like to thank Dr Estelle MOGENSEN and the three lab technicians for their help.

We also would like to thank BlackHole Lab and more precisely Sébastien CARGOU and Jules HERAUD, who provided us several microchips with the design we wanted.

We would like to thank the INOContest staff and AVIESAN team (ITMO cell biology, development and evolution) for allowing us to present our project the meeting “From 3D culture to organoids” and for rewarding our project with the silver medal.

Finally, we would like to thank Dr Josquin COURTE for his advice concerning 3D culture and microfluidics.

## REFERENCES

- [1] M. A. Lancaster *et al.*, “Cerebral organoids model human brain development and microcephaly,” *Nature*, vol. 501, pp. 373–379, 2013.
- [2] M. A. Lancaster and J. A. Knoblich, “Generation of cerebral organoids from human pluripotent stem cells,” *Nat. Protoc.*, vol. 9, no. 10, pp. 2329–2340, 2014.
- [3] L. Broutier *et al.*, “Culture and establishment of self-renewing human and mouse adult liver and pancreas 3D organoids and their genetic manipulation,” *Nat. Protoc.*, vol. 11, no. 9, pp. 1724–1743, 2016.
- [4] C. Flatres, É. Loffet, M. Neunlist, and M. M. Mahé, “Façonner l’intestin à partir des cellules souches pluripotentes humaines,” *médecine/sciences*, vol. 35, pp. 549–555, 2019.
- [5] N. Noor, A. Shapira, R. Edri, I. Gal, L. Wertheim, and T. Dvir, “3D Printing of Personalized Thick and Perfusible Cardiac Patches and Hearts,” *Adv. Sci.*, vol. 6, no. 11, 2019.
- [6] W. M. S. Russell and R. L. Burch, “The Principles of Humane Experimental Technique,” *Methuen Co., Ltd.*, 1959.
- [7] A. M. Bratt-Leal, R. L. Carpenedo, and T. C. McDevitt, “Engineering the embryoid body microenvironment to direct embryonic stem cell differentiation,” *Biotechnol. Prog.*, vol. 1, pp. 43–51, 2009.
- [8] R. Z. Lin and H. Y. Chang, “Recent advances in three-dimensional multicellular spheroid culture for biomedical research,” *Biotechnology Journal*, 2008.
- [9] C. Magliaro, A. Rinaldo, and A. Ahluwalia, “Allometric Scaling of physiologically-relevant organoids,” *Sci. Rep.*, vol. 9, 2019.
- [10] T. Osaki, V. Sivathanu, and R. D. Kamm, “Vascularized microfluidic organ-chips for drug screening, disease models and tissue engineering,” *Curr. Opin. Biotechnol.*, vol. 52, pp. 116–123, 2018.
- [11] X. Qian *et al.*, “Brain Region-specific Organoids using Mini-bioreactors for Modeling ZIKV Exposure,” *Cell*, vol. 165, no. 5, pp. 1238–1254, 2017.
- [12] E. Karzbrun, A. Kshirsagar, S. R. Cohen, J. H. Hanna, and O. Reiner, “Human brain organoids on a chip reveal the physics of folding,” *Nat. Phys.*, vol. 14, no. 5, pp. 515–522, 2018.
- [13] A. A. Mansour *et al.*, “An in vivo model of functional and vascularized human brain organoids,” *Nat. Biotechnol.*, vol. 36, no. 5, pp. 432–441, 2018.
- [14] J. S. Miller *et al.*, “Rapid casting of patterned vascular networks for perfusable engineered three-dimensional tissues,” *Nat. Mater.*, vol. 11, no. 9, pp. 768–774, 2012.
- [15] S. R. Adam Kratz, G. Höll, P. Schuller, P. Ertl, and M. Rothbauer, “Latest trends in biosensing for microphysiological organs-on-a-chip and body-on-a-chip systems,” *Biosensors*, vol. 9, no. 110, 2019.
- [16] B. D. Gates, Q. Xu, M. Stewart, D. Ryan, C. G. Willson, and G. M. Whitesides, “New approaches to nanofabrication: Molding, printing, and other techniques,” *Chem. Rev.*, vol. 105, no. 4, pp. 1171–1196, 2005.
- [17] S. H. Tan, N. T. Nguyen, Y. C. Chua, and T. G. Kang, “Oxygen plasma treatment for reducing hydrophobicity of a sealed polydimethylsiloxane microchannel,” *Biomicrofluidics*, vol. 4, no. 30, 2010.
- [18] M. Potente, H. Gerhardt, and P. Carmeliet, “Basic and

therapeutic aspects of angiogenesis,” *Cell*, vol. 146, pp. 876–887, 2011.

- [19] Y. Nashimoto *et al.*, “Integrating perfusable vascular networks with a three-dimensional tissue in a microfluidic device,” *Integr. Biol. (United Kingdom)*, vol. 9, no. 6, pp. 506–518, 2017.
- [20] Y. Nashimoto *et al.*, “Perfusible vascular network with a tissue model in a microfluidic device,” *J. Vis. Exp.*, vol. 9, no. 6, pp. 506–518, 2018.
- [21] W. F. Scherer, J. T. Syverton, and G. O. Gey, “Studies on the propagation in vitro of poliomyelitis viruses. IV. Viral multiplication in a stable strain of human malignant epithelial cells (strain HeLa) derived from an epidermoid carcinoma of the cervix,” *J. Struct. Biol.*, vol. 97, no. 5, pp. 695–710, 1953.
- [22] K. C. Chaw, M. Manimaran, F. E. H. Tay, and S. Swaminathan, “Matrigel coated polydimethylsiloxane based microfluidic devices for studying metastatic and non-metastatic cancer cell invasion and migration,” *Biomed. Microdevices*, vol. 9, no. 4, pp. 597–602, 2007.
- [23] J. Ogawa, G. M. Pao, M. N. Shokhirev, and I. M. Verma, “Glioblastoma Model Using Human Cerebral Organoids,” *Cell Rep.*, vol. 23, no. 4, pp. 1220–1229, 2018.

## AUTHOR INFORMATION

**Pierre-Emmanuel CORBET**, 5<sup>th</sup> year student, R&D, Health and Pharmaceuticals, Sup’Biotech, Villejuif, France  
**Mary-Amélie MASSON**, 5<sup>th</sup> year student, R&D, Health and Pharmaceuticals, Sup’Biotech, Villejuif, France  
**Pauline MEGALLI**, 5<sup>th</sup> year student, R&D, Health and Pharmaceuticals, Sup’Biotech, Villejuif, France  
**Thibaud METAIS**, 5<sup>th</sup> year student, R&D, Health and Pharmaceuticals, Sup’Biotech, Villejuif, France  
**Julie SIX**, 5<sup>th</sup> year student, R&D, Health and Pharmaceuticals, Sup’Biotech, Villejuif, France

**Dr Ferid NASSOR**, Sup’Biotech/CEA DRF IBJF SEPIA, Villejuif/Fontenay-aux-Roses, France

**Dr Vanessa PERDIZ**, Sup’Biotech/CEA DRF IBJF SEPIA, Villejuif/Fontenay-aux-Roses, France

**Dr Frank YATES**, head of the Research department at Sup’Biotech and head of CellTechs laboratory, Sup’Biotech/CEA DRF IBJF SEPIA, Villejuif/Fontenay-aux-Roses, France

# Microwell strategy for high throughput uniform microtissue generation

## *Developing a versatile prototyping approach for low-cost novel labware*

Maxime Mauviel\*, Elodie Rosa\*, Younes Bakhalek, Ulysse Godet  
Sup'Biotech, Villejuif, France

\*Corresponding authors: maxime.mauviel@supbiotech.fr, elodie.rosa@supbiotech.fr

**Abstract** - 3D cell cultures replicate *in vivo* conditions more accurately and provide physiologically more relevant data than 2D culture systems. Among such cultures, spheroid, “embryoid body” (EB) or microtissue are 3D aggregates of cells that exhibit structural and functional features of biological materials, such as tumours. Tumour spheroids have become an important tool for studying basic tumour biology and cancer therapy as well as serving as a first step in many pluripotent stem cell (PSC) differentiation protocols. Unfortunately, conventional formation methods result in EBs which are heterogeneous in size and shape, leading to inefficient and uncontrolled differentiation or characterization. Microwell platforms are frequently described for the efficient and uniform manufacture of such 3D microtissues. Each well contains many microwells of defined size, making it easy to produce large numbers of highly uniform spheroids and ensures reproducibility of differentiation experiments. Such equipment remains highly unaffordable for student’s laboratory daily use and commercial solutions don’t allow any functionalization of these platforms. We introduce here a low-cost novel labware for high throughput culturing cells in 3D and allowing easy media exchanges.

**Index Terms** – Microwells, PDMS, SLA 3D printing, tissue engineering, rapid prototyping, micro-mesh, molding.

### INTRODUCTION

2D cell culture is widespread and provides a good control over the cellular environment and an easy observation and manipulation. 2D culture thus makes it possible to control the quantity of nutrients and growth factors delivered to the cells and especially ensures that those quantities are similar for each cell. This allows homogeneous growth and proliferation of cells while being simple and effective. However, 2D culture systems’ disadvantage is related to their incapacity to reflect *in vivo* situation, where cells grow in a complex 3D micro-environment<sup>1</sup>.

To tackle this difficulty, 3D cell culture offers a higher degree of structural complexity and homeostasis, which is closer to tissues and organs. Researchers are recognizing the need to transition to a more physiologically relevant 3D screening systems that better replicate the cell-cell interactions and *in vivo* microenvironment<sup>2,3</sup>. However, improving physiological relevance lead to an increase of heterogeneity within cells that can trouble cell cultures and confound standard assay readouts.

3D culture systems are increasingly being employed to better replicate the tumour microenvironment for cancer research, drug development, and toxicity screening. These types of culture can also be used as effective simulators of tumour characteristics such as latency, hypoxia, anti-apoptotic behavior and receive stimuli of the local environment<sup>4</sup>. Many recent publications highlight the use of

3D culture methods as being more predictive than traditional 2D culture methods in determining drug efficacy *in vivo*<sup>3,5,6</sup>.

Spheroids are 3D structures formed by tumour cells under certain *in vitro* conditions. This structure has the advantage to avoid complexity, heterogeneity and low transparency of whole tissue while representing cell communication, cell proliferation and the action of drug on *in vivo* tumour<sup>4,7</sup>. However, 3D culture and spheroid generation techniques can be laborious, time consuming and low throughput.

An increasingly favored 3D culture approach in cancer cell biology, organoid culture and Pluripotent Stem Cell (PSC) differentiation is the formation of multicellular aggregates, termed microtissues. Microtissues are less representative of actual organ structure but serve as powerful platforms for studying human tissues and disease. This work focuses on tackling the 3D culture issues namely reproducibility, ease of use, costs, scaling of microtissues production and media exchanges<sup>6</sup>.

### CONTEXT AND STRATEGY

Microtissues have been manufactured for decades by aggregating hundreds-to-thousands of cells in either a single droplet (hanging drop) or in non-adherent tissue culture vessels<sup>1</sup>. Microtissue could be defined by a 3D arrangement of cells introducing a higher degree of complexity and biological relevance. While often favored over conventional 2D cultures, these relatively labor-



intensive methodologies have delayed the widespread adoption of 3D microtissue culture. There is a real need for spheroid high throughput generation at low cost (financial- and labor-wise).

Recent technological advances and the commercial availability of products that enable high throughput microtissue manufacture (microfluidic-based hanging drop, micro-patterned surfaces or microwell platforms) are increasingly bringing 3D microtissue culture into mainstream practice<sup>8</sup>. The culturing method in soft agar solution was the first developed 3D cultures carried out by Hamburg and Salmon in 1970. Today distributors like Sigma Aldrich offer solutions like the 3D Petri Dish<sup>1</sup> mold for agarose substrate enabling the production of up to 256 microtissues in microwells<sup>9</sup>.

Microwell platforms such as AggreWell<sup>2</sup> or Nunclon Sphera<sup>3</sup> are devices enabling 3D culture and high throughput cell aggregation, as well as their easy manipulation. The fundamental motivation for using a microwell platform to manufacture microtissues is its capacity to reproducibly generate thousands of uniform microtissues. This is achieved either through forced aggregation (centrifugation) or gravity settling of a cell suspension into an array of microwells placed at the bottom of a larger tissue culture well.

AggreWell plates<sup>10,11</sup>, are microwell-based 3D cell culture devices that produce many size-controlled spheroids per well from a single pipetting step. Within each well, every spheroid can be individually identified, indexed, and analyzed to provide multiple repeated measures per treatment condition to yield more representative and predictive data. This product is furnished as a sterile cell culture plate, treated for ultra-low cell adhesion, single use and thus expensive for daily lab experiment (50 to 60 euros for one plate).

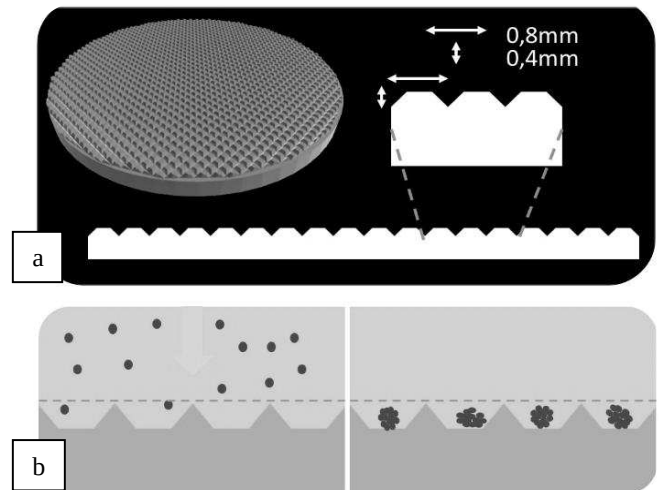
The work consisted in developing a new microwell platform that mimics this technology using only widespread and low-cost materials. The approach relies on 3D printed resin molds with a patterned surface on which are casted cell culture compatible substrates such as PDMS<sup>12</sup> or agarose<sup>13</sup>. Once casted these substrates solidify or polymerise and the microwell patterns are fixed. This technique could be assimilated to soft lithography<sup>14</sup> (technique for fabricating or replicating structures using elastomeric stamps, molds, and photomasks). Cells can then be grown in the short or long term on these substrates microwells as spheroids, Embryoids (EBs) or microtissues.

This device should allow a high yield of uniformly sized spheroids, at low production cost-per-spheroid, ease of use, reproducible and compatible with a variety of cell types and 3D culture application and easy media exchange for long termed culture.

## MATERIAL AND METHODS

### Computer-aided 3D pattern design (CAD)

The first step was to design a 3D pattern mimicking the AggreWell technology with the help of a Computer Aided Design (CAD) software. CAD is the use of computers (or workstations) to aid in the creation, modification, analysis, or optimization of a design. CAD software are used to increase the productivity of the designer, improve the quality of design, improve communications through documentation, and to create a database for manufacturing. The design chosen is an array of truncated pyramidal structures in which cells should aggregate after sedimentation by gravity. This pyramidal array is presented in Figure 1 below.



**Fig. 1: 3D design of the mold pattern by CAD (a).** The surface once patterned (PDMS or agarose replica of the mold) (b) is seeded with cells that will sediment and aggregate to form uniform 3D structures. A micro sized mesh is playing the role of a selective membrane allowing cell to sediment through and spheroids to be retained once formed for easy media exchange.

### High resolution 3D printing

3D printing is a relatively new technology in the world of architecture, design and manufacturing. Also known as rapid prototyping<sup>15</sup>, 3D printing is a type of additive manufacturing technology where a 3D object is created by laying down subsequent layers of material at the mm, or even  $\mu\text{m}$  scale. 3D printers print objects by reading a CAD design file, the model will then be ‘sliced’ into layers and converted into a printer-readable file. The first commercial 3D printer was offered for sale in January 2009<sup>16</sup>. It was at the time highly costly.

The most used technology is called ‘Fused Deposition Modeling’. The FDM technology uses thermoplastic resins as raw materials. It is the simplest 3D additive manufacturing technology, in which the thermo-plastic resin softens when heat is applied. The first developed prototype (no data shown due to a printing failure) was to be printed by this technique with a 0.1mm lateral resolution. Indeed, while inexpensive this technique is limited in axial resolution (100  $\mu\text{m}$ , up to 50

<sup>1</sup> Sigma Aldrich, Z764086, 3D Petri Dish® - MicroTissues

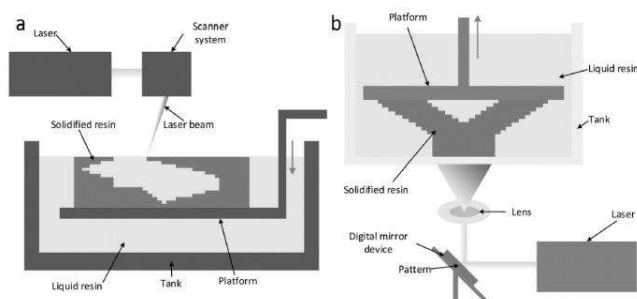
<sup>2</sup> StemCell Technologies, # 34421, AggreWell™400/800

<sup>3</sup> ThermoFisher, 174925, Nunclon™ Sphera™ Microplates, 96U-Well Plate

and 25 nm for industrial printers) and therefore is not usable for high precision micrometer sized designs.

One of the most commonly used high resolution 3D printing technology is the stereolithography (SLA) technology<sup>17</sup>. This technology utilizes photocuring resins as raw material polymerized in a layer by layer fashion using photochemical processes by which light (often in the UV spectra) causes chemical monomers to cross-link together to form polymers. The resolution of the printing is therefore determined by the laser wavelength and fluence and reaches 25  $\mu\text{m}$ . New resins that are being developed will combine transparency, heat resistance and toughness.

3D printing has revolutionized our society from providing medical advances; to scalable production of everything from product parts to buildings. The flexibility and precision of 3D printing as a manufacturing technique naturally lends itself to tissue engineering. These techniques use high-precision desktop stereolithography or digital light processing (SLA or DLP) 3D printers to enable the production of a variety of geometries (Fig. 2). Commercially available printers range from the thousand to a few thousand euros making them affordable.



**Fig. 2: High resolution 3D printing using stereolithography (SLA) (a) or digital light processing (DLP) (b).**<sup>18</sup>

#### Rapid and versatile prototyping (RP)

Rapid prototyping systems claim fast processing at low operating costs with stringent tolerance. Biologists often use soft lithography as it has become widely used for micropatterning and microfluidics for prototyping. It is a family of techniques for fabricating or replicating structures using elastomeric stamps, molds, and photomasks<sup>14</sup>. It is called "soft" because it uses elastomeric materials, most notably Polydimethylsiloxane (PDMS) which is used to generate a negative imprint of the patterned surface. PDMS is widely used because it has the following advantages; a high transparency, good mechanical properties (MPa range), easily sterilisable and functionalizable (coating with antifouling materials such as pluronic acids, PEG, Sigmacote, coating with ECM like materials Matrigel, Geltrex....) and is biocompatible<sup>19</sup>.

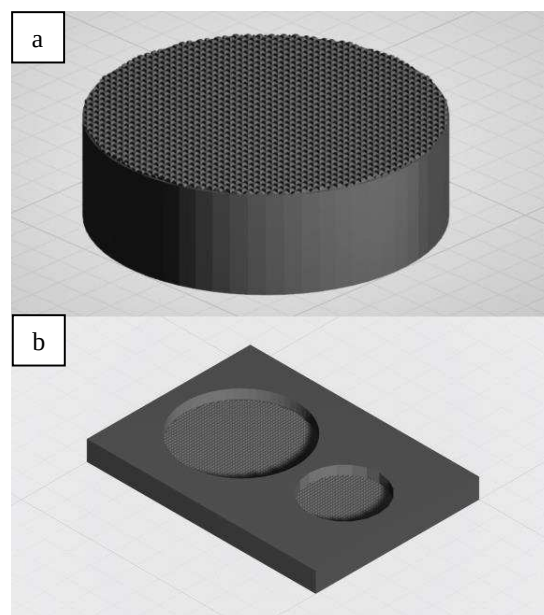
Soft lithography is generally used to construct features measured on the micrometer to nanometer scale. Development of soft lithography expanded rapidly from 1995

to 2005<sup>20</sup>. Soft lithography tools are commercially available making it affordable for rapid prototyping but still requires skills and expertise.

Computer-aided design and 3D printing at high resolution can help to bypass the costly design of photomasks traditionally used for lithography and therefore makes it accessible for daily prototyping.

#### Microwell stamp and molds manufacture

AggreWell plates are available in 400 $\mu\text{m}^2$  and 800 $\mu\text{m}^2$  microwell sizes to accommodate spheroids of up to 3,000 or 20,000 cells, respectively, meaning approximately 28,000 spheroids per plate, thus much beyond the scale possible with other methods including 96- or 384-well V-bottom or hanging drop methods. Hence, we designed two different arrays of patterns based on these parameters (Fig 3).



**Fig. 3: Designs strategies for micropatterning of the wells.** The design provides an easy method to produce large numbers of 3D spheroids. Each well contains a standardized array of 1800 microwells (0.8mm<sup>2</sup> and 0.4mm depth or 0.4mm<sup>2</sup> and 0.2mm depth), allowing the production of large numbers of spheroids in a single well. The size of spheroids generated is highly uniform and can be easily modified by adjusting the cell seeding concentration. (a) **Stamp design** for patterning under pressure, (b) **mold design** for substrate casting.

Designs were then sent to 3DHubs<sup>4</sup>, an online 3D printing platform, to be printed using printing SLA or DLP technologies (SLA printed by Formlabs<sup>5</sup> Form 2 reaching 25 $\mu\text{m}$  resolution x,y and 50  $\mu\text{m}$  z resolution and DLP by Zortrax Inkspire<sup>6</sup> reaching 50  $\mu\text{m}$  resolution x,y and 50  $\mu\text{m}$  z resolution). The delivery time was around 1 week, costs of printing are detailed in the Table 2 (Appendix).

<sup>4</sup> 3D Hubs | On-demand Manufacturing: <https://www.3dhubs.com>

<sup>5</sup> Formlabs Form 2™: <https://formlabs.com/fr/>

<sup>6</sup> Zortrax Inkspire: <https://zortrax.com/3d-printers/inkspire/>

### PDMS and Agarose microwell discs fabrication

Two strategies were developed in order to prepare micro-patterned substrate discs. The first one lays on a 3D printed stamp in resin with a pyramidal array at its surface allowing the formation of microwells. PDMS and agarose are the raw materials permitting the preparation of the discs. Agarose 2%, NaCl 0.9% was prepared and PDMS (ratio 7:1, silicone elastomer base Sylgard 184: curing agent) was degassed 5 min using a vacuum pump. Both were poured on a 6-well plate and cured at 90°C for 1h30 (PDMS) or gelled after 30 min (agarose 2%) at room temperature with the stamp pattern directly in contact with both components. Finally, the stamp is removed gently, and the discs obtained remain inside the well.

The second strategy uses a mold in the form of a resin plate incorporating wells with the same pyramidal array as the stamp from the first strategy at the bottom of each well. Substrate are poured in the wells and discs were punched from the plate after curing or gelation (under aseptic condition for Agarose). PDMS discs are then autoclaved on a short cycle before use.

PDMS (Dow, Sylgard 184 10:1 or 7:1) or 2% ultrapure agarose, 0.9% NaCl were used to create microwells patterns. Once the PDMS or agarose has polymerized, microwells were removed from the dishes and placed in a 6 or 12-wells plate.

### Microwell-mesh for easy media exchange

The use of a mesh (porous nylon layer) that allows single cells to pass through the micro-pores and remain trapped into the microwells because of the size they reach when they grow as organoids has also been investigated. It allows an easier changing of the medium and harvesting of the microtissues<sup>21</sup>.

Discs fitting 6 or 12 well plate have been inserted in each well. Polyamide Nylon 6,6 (two monomers each containing 6 carbon atoms) mesh was selected as it is a non-toxic biocompatible material commonly used in clinical applications (F035N-08C, Itblox). Mesh was glued using a small amount of PDMS applied at its periphery, on the open face of the microwell. After curing time, this system allowed the nylon mesh to stick (Fig. 6). The sterilization of the microwell-mesh is performed by autoclaving (121°C for 1 hour).

### Cell culture

HeLa cells lines, A549, adenocarcinomic human alveolar basal epithelial cells, and H1299, human non-small cell lung carcinoma cell line, (good model because those lines are used daily for many experiments and are provided by the SEPIA laboratory, CEA, Fontenay-aux-Roses) were used to generate spheroids as a proof of concept for this microwell platform. Cells were cultured in Dulbecco's modified Eagle's medium (DMEM) high glucose 4500mg/L (Sigma, D6546) supplemented with 10% Fetal bovine serum (FBS) (Dutscher, P 30-8500), 1% L-Glutamine (Sigma, G7513), 10 mM HEPES (Sigma, H0887), 1% MEM-non

essential amino acids (Sigma, M7145), 1% penicillin/streptomycin (Sigma, 11074440001).

Microwells were equilibrated twice with 2.5 mL complete DMEM for 15 minutes to provide nutrients to the cells when seeded.

At 60-70% confluence, cell cultures were trypsinized and counted with Malassez Chamber to know the concentration of cells. According to the AggreWell 400 and 800 protocols, size of spheroids depends on the number of cells seeded in micropattern (Table 1). Then, cells in agarose or PDMS microwells were incubated for 24h at 37°C and 5% of CO<sub>2</sub>.

**Table 1: Required number of cells per well for AggreWell 800 plates<sup>10</sup>.**

| 6-WELL PLATES                          |                                   |
|----------------------------------------|-----------------------------------|
| Desired number of cells per microwell* | Required number of cells per well |
| 3000                                   | $5.4 \times 10^6$                 |
| 4000                                   | $7.2 \times 10^6$                 |
| 5000                                   | $9.0 \times 10^6$                 |
| 10,000                                 | $1.8 \times 10^7$                 |

### Characterization technique

Light optical microscope was used to identify eventual damages or impurities on the material used for the fabrication of the microwell-mesh. It was also helpful to analyse cell aggregation after seeding and the pattern on PDMS or agarose discs.

### Adhesion test of the cells

Cells usually tend to stick to untreated PDMS in culture besides of its hydrophobicity. PDMS is often treated within a plasma chamber to increase its hydrophilicity. Unfortunately, such equipment was unavailable in the laboratory. Poloxamer 188 Poly-ethylene glycol (PEG block-copolymer)<sup>22</sup>, poly-hydroxyethyl-methacrylate (poly-HEMA)<sup>23,24</sup> and saturation with bovine serum albumin (BSA) are reagents often used in literature for the coating of PDMS to decrease cell adhesion. PDMS surface were coated with these reagents and incubated at 37°C for 1h30. Glass coverslip and cell culture plates were used as a control to test the efficiency of these cell adhesion lowering reagents.

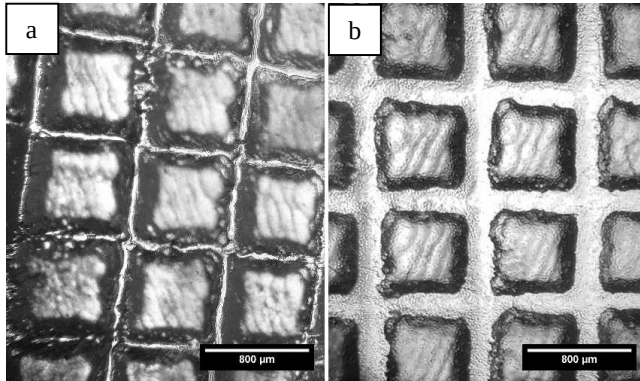
PolyHEMA substrate can modulate cell expression of E-cadherin when grown as spheroids. Such spheroids are usually more resistant to drug treatments (6 to 10-fold). However, no cytotoxicity of PolyHEMA was found in literature (silicone based surface treatment showed a significant toxicity on cancerous cell lines)<sup>25</sup>. Poloxamers interact with multidrug-resistant tumours resulting in drastic sensitization of these tumours when used as treatment but PEG treated surfaces have a relatively low cytotoxicity for most experiments<sup>26</sup>.

## RESULTS

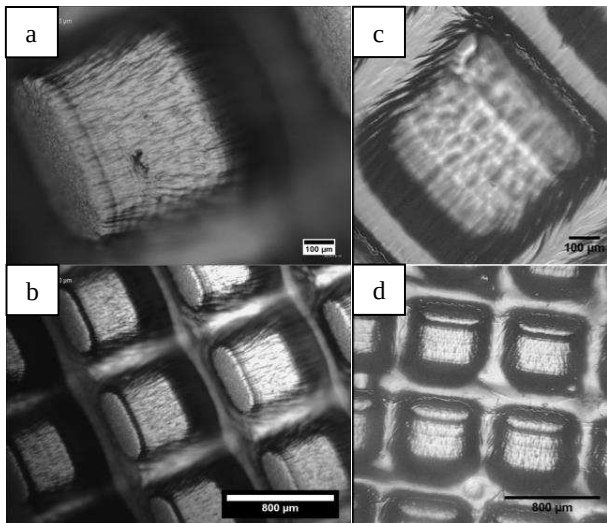
### Mold patterns, PDMS and agarose microwells resolution



The first issue in this project was the choice of printing technology for the prototyping of the patterned surfaces as well as the easiest approach to cast it on the substrates. Figure 4 shows the results obtained by DLP printing and stamping approach while Figure 5 shows the mold pattern obtained by SLA printing. Both techniques have approximately the same costs (Table 2, Appendix), only DLP allows a faster printing (1h30 to 2h).



**Fig. 4: a) Mold pattern obtained from DLP 3D printing (40x).** The mold was designed in CAD then printed with a 50 µm resolution on a Zortrax Inkspire. **b) PDMS Microwells (40x)** obtained after casting and curing at 80°C for 1h and autoclaving.



**Fig. 5: a) Mold obtained from SLA 3D printing (100x).** The mold was designed in CAD then printed with a 25 µm resolution on a Formlabs Form 2. **b) Pattern at 40X magnification.** (c-d) Agarose 2% NaCl 0.9% patterned surface obtained with SLA printed molds.

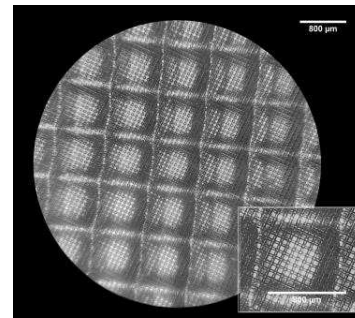
These results show that DLP 3D printing at 50 µm resolution is enough to obtain a patterned surface with some important shape variation. The main concern was regarding the surface aspect of the microwells that should exhibit minimal deformation in order to lower cell interaction and to prevent sticking at the surface. SLA clearly offers a more

suitable resolution (25µm) for these patterns even though the surface aspects are still a bit brushy.

Figure 5 shows that the microwell pattern can also be cast with a good resolution on a softer material such as agarose 2% that exhibit no adherence to cells.

#### Nylon mesh

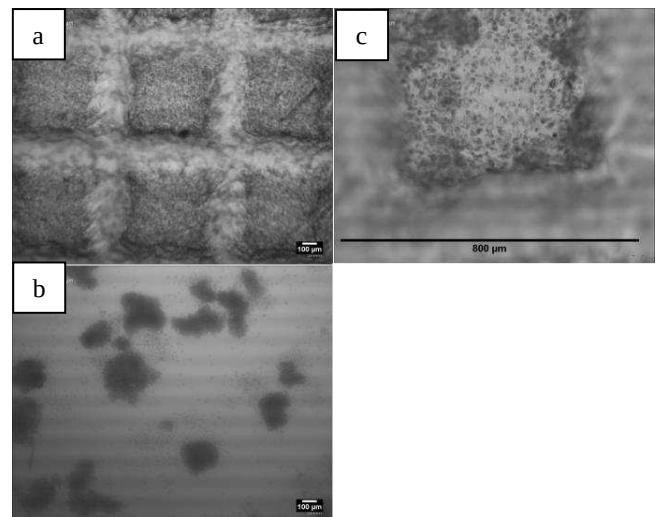
Light optical microscopy was used to characterize the shape and dimension of the cavities in the nylon mesh in order to be sure that there were no impurities or damages before we use it for cell culture (Fig. 6). Then, our system has been studied with the same technique. The results showed that the mesh was perfectly glued to the disc and that its open area was enough to welcome the cells in the microwells.



**Fig. 6: Micro-mesh pasted on the PDMS microwell pattern.** No imperfection was observed and the prototype is "ready to use". **Nylon mesh** with an appropriate pore size over the microwell openings allows single cells to pass through the mesh into the microwells during the seeding process. Nylon 6/6 mesh, ½ inches, 36 microns, 28% open area.

#### Comparison of culture protocols before and after treatment of anti-adherence solution on the mold

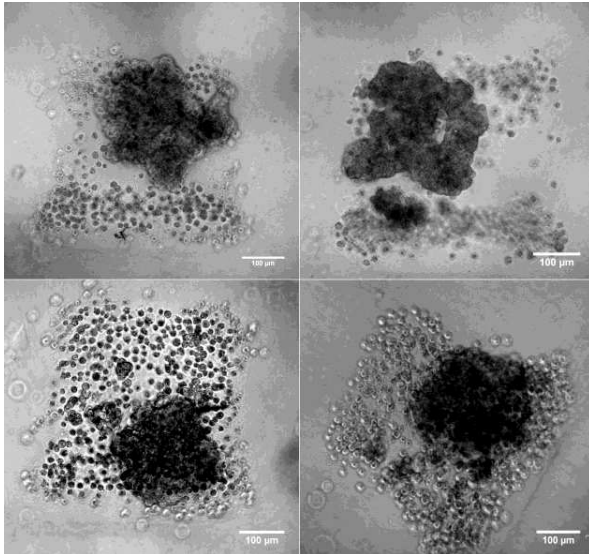
First results suggested that cells tend to stick to untreated PDMS patterned surface (as shown in Fig. 7).



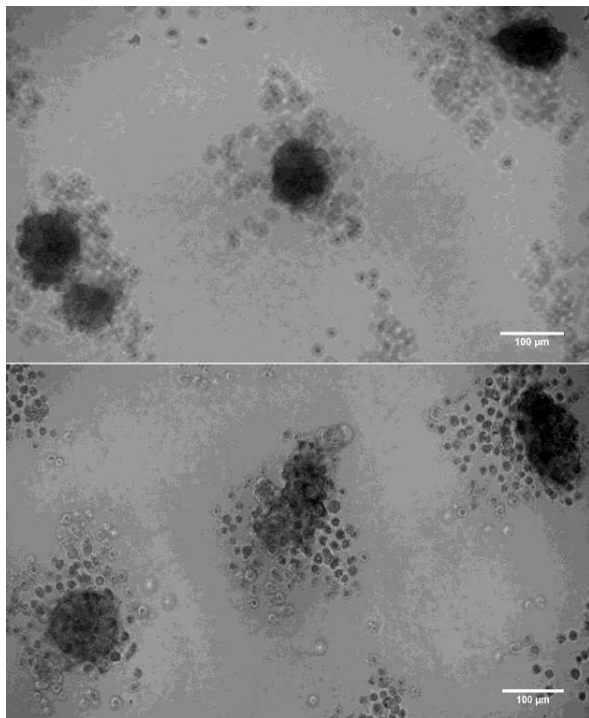
**Fig. 7: a) H1299 cells stuck on the patterned surface (40x).** Cells stuck to PDMS surface with little aggregation. **b) H1299 cells aggregates seeded on agarose surface (40x).** Stamped patterned surface totally faded on agarose 2% surface and cells didn't attached

and formed distinct aggregates. **(c) Z project of a single microwell stack (800µm 100x).** Cell density is higher in the corners of the microwells.

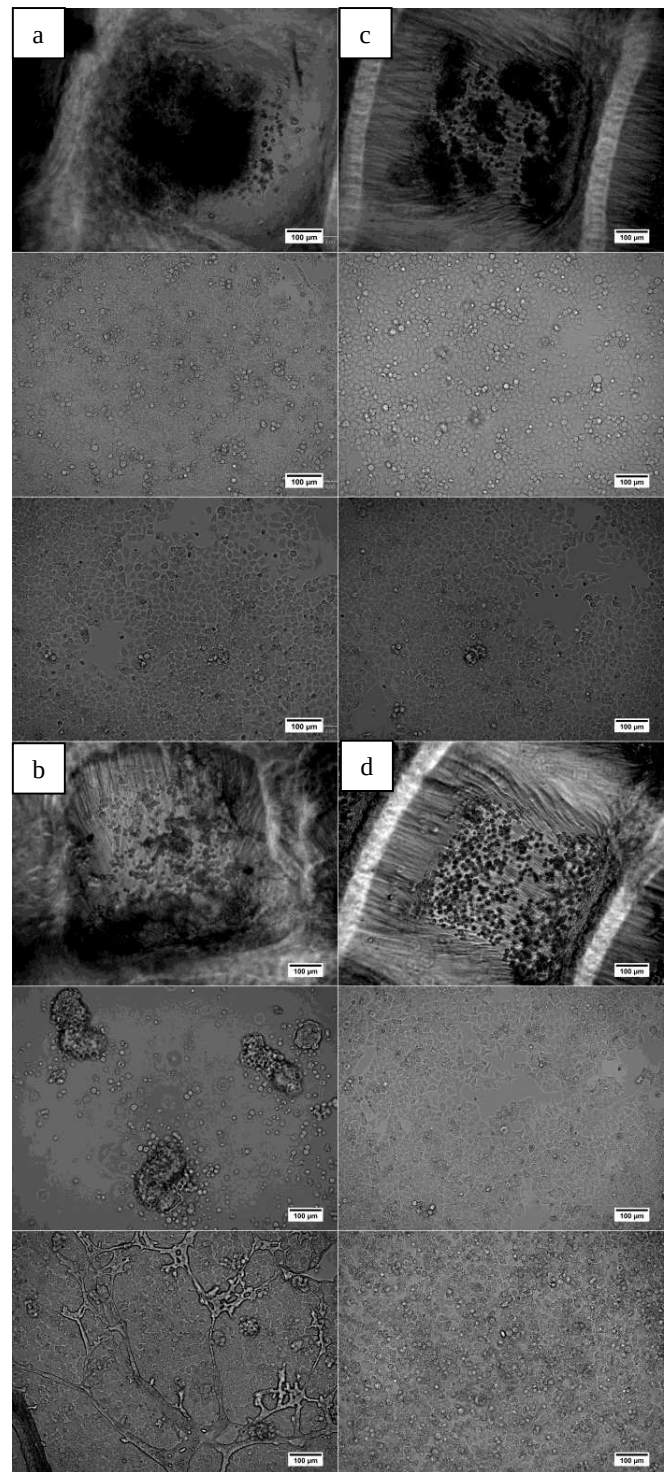
However, Agarose 2%, NaCl 0.9% pattern obtained by molding seemed to conserve its structure, even though the pattern slowly faded, cell aggregates formed (Fig. 8 - 9).



**Fig. 8:** Agarose 2%, NaCl 0.9%, patterned on 800 µm<sup>2</sup> by molding on SLA printed mold (HeLa cell line).



**Fig. 9:** Agarose 2%, NaCl 0.9%, patterned on 400 µm<sup>2</sup> by molding on SLA printed mold (HeLa cell line).



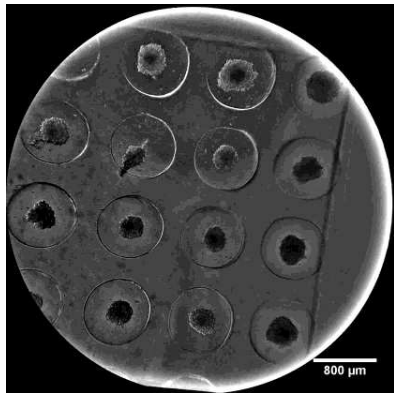
**Fig. 10:** (a) Poloxamer 188 and Poly-HEMA (50/50) coating of PDMS SLA molded pattern, Glass coverslip, culture well plate seeded with HeLa cells. (b) Poly-HEMA coating of PDMS SLA molded pattern, Glass coverslip, culture well plate seeded with HeLa cells. (c) Poloxamer 188 coating of PDMS SLA molded pattern, Glass coverslip, culture well plate seeded with HeLa cells. (d) BSA saturation of PDMS SLA molded pattern, Glass coverslip, culture well plate seeded with HeLa cells.

Figure 10 recapitulates the results obtained by the coating tests to lower cell adhesion to the PDMS patterned surface obtained by molding. It seems that the best coating would be a mixture of poloxamer 188 and Poly-HEMA (1:1 ratio) as the cells tended to aggregate and didn't stuck to the walls of the wells neither its corners. A striking result is that controls all exhibit cell adhesion on coated surfaces either glass or cell treated culture plastic. Microwells in agarose allows the cells not to attach to the matrix and thus form a sphere during growth.

## DISCUSSION AND CONCLUSION

### *Comparison to the tissue mimic technique*

Spheroid generation was used with 3D Petri dishes by the Sup'Biotech group (Retho et al., published in the same issue) using the same cell lines as a control. The technique of microtissue is fast and easy to implement for the generation of a high number of spheroids per well in a petri dish using agarose gel, however the number of produced spheroids is very low compared to this approach and the collection of spheroids can be a limiting step.



**Fig 11: H1299 Spheroids obtained with 3D Petri Dish (800μm microwell) (Retho et al., published in the same issue).**

### *Critics regarding the design*

The design exhibits several issues that would need improvement. While working at seeding concentration recommended by Stem Cell Technologies for the 6-well plates ( $5.4 \cdot 10^6$ ) formed spheroids (around 3000 cells) seemed small compared to the microwell size (also depending on the cell line used). Moreover, one critical issue was the fact that some cells went between the substrate discs and the cell culture plate. To solve this issue discs were glued to the plate, but no perfect results were obtained.

The truncated pyramidal shape pattern is also not optimized for spherical microtissues as some cells tend to stick to the corners of this pattern.

The huge number of cells required per well is also not consistent with cost effective daily lab routine. A smaller pattern on 12-well plates was thus designed with a minimal seeding requirement of  $1.10^6$  cells.

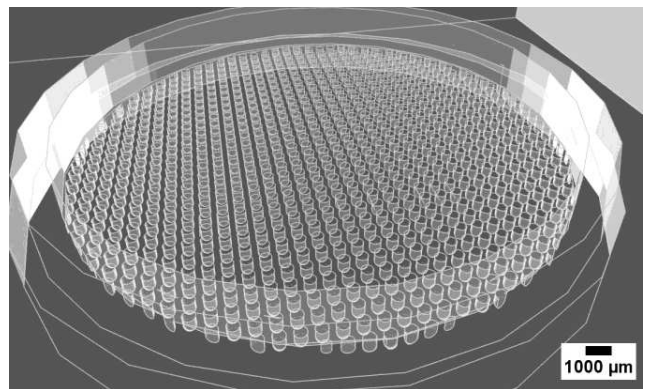
Surface treatments of the PDMS wells did not show great efficacy for lowering cell adherence, pluronic acid F-

127 or pure PEG coating should be investigated before further development of this technique.

In the future, rapid prototyping of micro-patterned surfaces can be realised through structured illumination to pattern photosensitive resins (SU8 photoresist) to create reusable molds on which you can polymerize material in the desired shape. This technique is available by companies as Alveole, with the PRIMO technology, which have kindly invited us to visit their laboratory.

### *Reverse engineering for ideal design*

Learning from these initial design issues, the following final design (Fig. 12) was realised to offer new perspectives. The microwell array is now composed of  $0.2 \mu\text{m}$  radius spherical shapes at the end of a cylindrical tube allowing better sedimentation of the cells and improved patterning for spheroid generation. A 2mm high rim is also introduced around the microwells in order to contain a defined number of cells without losing them outside the pattern design from a single pipetting step. Finally, this design can be sized for different microwell sizes ranging from 0.4mm to 0.8mm.



**Fig. 12: Ideal design for high throughput microtissue generation.**

We validated during these experiments that using PDMS or Agarose as a material to pattern specific models is cost-effective and simple. We demonstrated that by printing a single mold, the cost-effectiveness is reached as fast as the production of 2 6-well plates. These results suggest the capability of versatile rapid prototyping for biologists at low operating costs using 3D high resolution printing and substrate casting.

## ACKNOWLEDGMENT

The authors thank Dr. Frank Yates, Dr. Ferid Nassor and Dr. Vanessa Perdiz from the CellTechs (CEA SEPIA/Sup'biotech) laboratory for the support provided for this project.

The assistance of Dr. Josquin Courte for participating as our guest to the related symposium and his advices was also greatly appreciated.

The advices of Matthieu Optiz from the company *Alveole* for providing us a good insight of opportunities offered by micro-fabrication and micropatterning techniques.

The insights provided by Dr. Yannick Coffinier from IEMN were helpful regarding the choice of coatings for the microwell platform.

### CONFLICT OF INTERESTS

The authors declare no conflicts of interest. The products used as models in this work are commercialized by Stem Cell Technologies (AggreWell) technology and Sigma Aldrich (3D Petri Dish) and were used for research purpose only.

### REFERENCES

- Kuo, C. Te *et al.* Three-dimensional spheroid culture targeting versatile tissue bioassays using a PDMS-based hanging drop array. *Sci. Rep.* **7**, 1–10 (2017).
- Ho, B. X., Pek, N. M. Q. & Soh, B. S. Disease modeling using 3D organoids derived from human induced pluripotent stem cells. *Int. J. Mol. Sci.* **19**, (2018).
- Mroczek, M. *et al.* Abstract 5015: Drug screening and phenotypic analysis in a microwell-based 3D cell culture system. *Cancer Res.* **78**, 5015–5015 (2018).
- Laschke, M. W. & Menger, M. D. Life is 3D: Boosting Spheroid Function for Tissue Engineering. *Trends Biotechnol.* (2017). doi:10.1016/j.tibtech.2016.08.004
- Eglen, R. M. & Reisine, T. Human iPS Cell-Derived Patient Tissues and 3D Cell Culture Part 2: Spheroids, Organoids, and Disease Modeling. *SLAS Technol. Transl. Life Sci. Innov.* **24**, 18–27 (2019).
- Chambers, K. F., Mosaad, E. M. O., Russell, P. J., Clements, J. A. & Doran, M. R. 3D cultures of prostate cancer cells cultured in a novel high-throughput culture platform are more resistant to chemotherapeutics compared to cells cultured in monolayer. *PLoS One* **9**, (2014).
- Ozolat, I. T. & Hospodiuk, M. Current advances and future perspectives in extrusion-based bioprinting. *Biomaterials* **76**, 321–343 (2016).
- Saraswat, R., Ratnayake, I. & Wood, S. CellWell: A micropatterned biphasic nanocomposite platform for culturing chondrocytes. (2019). doi:10.1101/790030
- Sigma Aldrich. The 3D Petri Dish® A New Cutting-Edge Culture Format.
- Bodies, U. E. AggreWell™ Reproducible, Uniform Embryoid Bodies Reproducible Production of Uniform Embryoid Bodies Advantages of AggreWell™ AggreWell™. 1–2 (2016).
- Culture, T. C. AggreWell™ Easy, Reproducible Spheroid Production Production AggreWell™ Selected AggreWell™ Publications AggreWell™ has been used in a. 3–4 (2017).
- Xiao, Y. *et al.* Engineered Microvasculature in PDMS Networks Using Endothelial Cells Derived from Human Induced Pluripotent Stem Cells. *Cell Transplant.* **26**, 1365–1379 (2017).
- Ravi, M. & Paramesh, S. R. K. V. Culture phases, cytotoxicity and protein expressions of agarose hydrogel induced Sp2 / 0, A549, MCF-7 cell line 3D cultures. *Cytotechnology* **68**, 429–441 (2016).
- Rogers, J. A. & Nuzzo, R. G. Recent progress in soft lithography. *Mater. Today* **8**, 50–56 (2005).
- Buch-månson, N. *et al.* Rapid Prototyping of Polymeric Nanopillars by 3D Direct Laser Writing for Controlling Cell Behavior. 1–9 (2017). doi:10.1038/s41598-017-09208-y
- Fang, E. H. C. & Kumar, S. The Trends and Challenges of 3D Printing. *Encycl. Inf. Sci. Technol. Fourth Ed.* 4382–4389 (2017). doi:10.4018/978-1-5225-2255-3.ch380
- Fuchs, A. *et al.* Development of a stereolithographic approach to build soft tissues Principle of Digital Mask Stereolithography. (2019).
- Valino, A. D. *et al.* Advances in 3D Printing of Thermoplastic Polymer Composites and Nanocomposites. *Prog. Polym. Sci.* doi:10.1016/j.progpolymsci.2019.101162
- Mata, A. & Fleischman, A. J. Characterization of Polydimethylsiloxane (PDMS) Properties for Biomedical Micro / Nanosystems. **2**, 281–293 (2005).
- Chadly, D. M. *et al.* Full Factorial Microfluidic Designs and Devices for Parallelizing Human Pluripotent Stem Cell Differentiation. *SLAS Technol.* (2018). doi:10.1177/2472630318783497
- Futrega, K. *et al.* The microwell-mesh: A novel device and protocol for the high throughput manufacturing of cartilage microtissues. *Biomaterials* **62**, 1–12 (2015).
- Matsumoto, M., Sawamoto, M. & Terashima, T. Orthogonal Folding of Amphiphilic/Fluorous Random Block Copolymers for Double and Multicompartment Micelles in Water. *ACS Macro Lett.* (2019). doi:10.1021/acsmacrolett.9b00078
- Bhat, A., Smith, B., Dinu, C. & Guiseppe-elie, A. Molecular engineering of poly (HEMA-co-PEGMA) -based hydrogels: Role of X =. *Mater. Sci. Eng. C* **98**, 89–100 (2019).
- Steadman, K. *et al.* resistance of the intractable solid tumors ES SC RIB. **4101**, (2008).
- Lombello, C. & Malmonge, S. M. PolyHEMA and polyHEMA-poly (MMA-co-AA) as substrates for culturing Vero cells. (2000). doi:10.1023/A
- Liu, G. *et al.* RSC Advances Cytotoxicity study of polyethylene glycol derivatives. *RSC Adv.* **7**, 18252–18259 (2017).

### SUPPLEMENTARY INFORMATION

3D CAD models designed for this project are available at the following link for Sup'Biotech students use. Access granted upon request for academical research only. [https://supbiotechfr.sharepoint.com/:f/t/OrganodesGB/EizU hUlkBMFPjz3uTwvlsyoB40wIGn\\_NsnXRso38--bkQw?e=PEzizC](https://supbiotechfr.sharepoint.com/:f/t/OrganodesGB/EizU hUlkBMFPjz3uTwvlsyoB40wIGn_NsnXRso38--bkQw?e=PEzizC)

### APPENDIX

**Table 2: Cost comparison between Aggrewell, 3D Petri Dish and rapid prototyping (prices in euros).**

| Costs in Euros               | Commercial approaches |                | Rapid prototyping        |              |
|------------------------------|-----------------------|----------------|--------------------------|--------------|
|                              | Aggrewell             | 3D Petri Dish® | PDMS coating pluronic 5% | Agarose 2%   |
| Commercial <b>mold</b>       | -                     | 81,83          | -                        |              |
| 1 3D SLA printed <b>mold</b> | -                     | -              | 53,09                    |              |
| 6-well plate coated (5x)     | 271                   | 26,5           | 90,82                    | 26,5         |
| <b>Total costs</b>           | <b>271</b>            | <b>108,33</b>  | <b>143,91</b>            | <b>79,59</b> |



# Light Sheet Fluorescence Microscopy: a qualitative comparison of four different clearing methods in plants

Julien Ballbé Y Sabaté\*, Anne Fauvel\*, Agathe Claude, Sarah Pierre, Angéline Mole, Yannaëlle Rousseau\*  
Sup'Biotech, Villejuif, France

\*Sup'Biotech, co-first authors: julien.ballbe-y-sabate@supbiotech.fr, anne.fauvel@supbiotech.fr, last and corresponding authors: yannaëlle.rousseau@supbiotech.fr

**Abstract** - Light-Sheet Fluorescence Microscopy (LSFM) allows to reconstruct three dimensional isotropic structures from two dimensions images. This technology, compare to other microscopy technics, uses sheets of light to illuminate large surfaces but on a thin layer, making it particularly convenient for the study of thick samples. Plus, LSFM lowers phototoxicity and photobleaching, allowing thin and transparent living organisms imaging. However, plant tissue opacity, makes LSFM imaging challenging, and necessitates a previous clearing step. While mammal cleared samples are widely studied, fixed plants imaging applications with LSFM are truly underdeveloped. Also, natural autofluorescent of plants which allows a direct imaging of their anatomy by LSFM is an advantage to avoid the use of costly labelings. Taking these data into account, we investigated whether (i) clearing protocols developed for animal tissues are suitable for plant samples, (ii) clearing processes preserve the plant structure and integrity and (iii) LSFM is suitable for providing relevant information about plant complex structures, rich in auto-fluorescent molecules. We tested both aqueous (TDE and CUBIC) and organic solvent-based (ECi and u-DISCO) protocols on various plant samples: Romanesco cabbage for its highly complex fractal structure, Tulip and lily pistil and anthers for reproductive organs 3D visualization, Climbing Ivy for its rhizoid-secreted nanocomposite adhesive and *Arabidopsis Thaliana*, as it is the main model in plant biology. Our results showed that solvent-based techniques seem to clear and preserve efficiently the tissue structure. Indeed, 3D images accurately showed internal complex structures up to the cellular level. Surprisingly, aqueous protocols seem to have not properly clear, and even CUBIC protocol spoiled our Tulip pistil sample, while most published results used aqueous protocols (TDE and CUBIC) to clear plant tissues.

**Index Terms** – Light Sheet Fluorescence Microscopy, Plant anatomy, Tissue clearing.

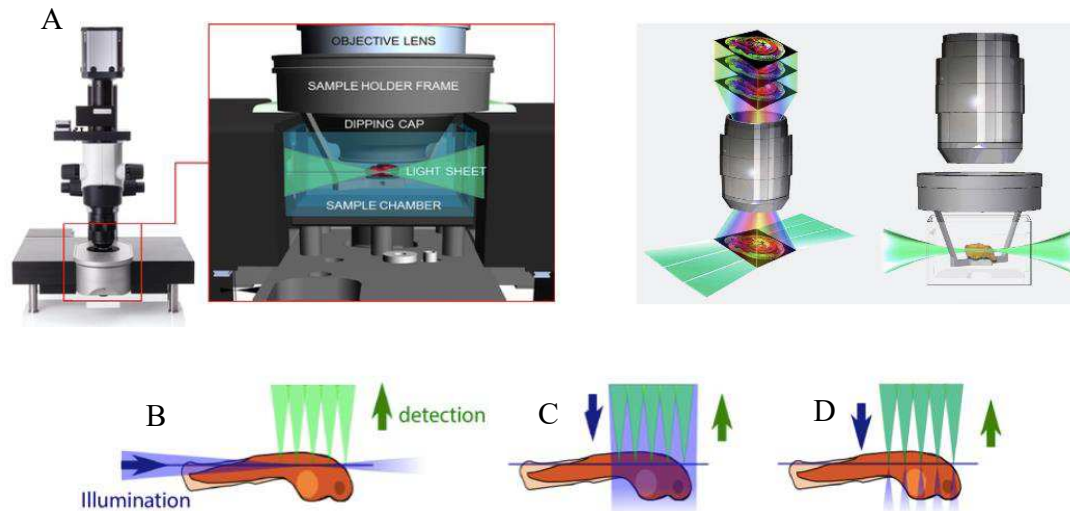
## ABBREVIATIONS LIST

**BABB** : Benzyl alcohol and benzyl benzoate  
**CUBIC** : Clear, Unobstructed Brain Imaging Cocktails  
**ECi** : Ethyl Cinnamate  
**GFP** : Green Fluorescent Protein  
**LSFM** : Light Sheet Fluorescence Microscopy  
**TDE** : 2,2' Thiodiethanol  
**U-DISCO** : Ultimate 3D Imaging of Solvent-Cleared Organs  
**YFP** : Yellow Fluorescent Protein

## INTRODUCTION

Techniques to observe animal or plant samples under Light Sheet Fluorescent microscope have been widely studied over the past ten years. Although, this concept was developed one hundred years ago, there was no real developed protocol to make it a useful tool for a daily use of imaging (GIRKIN et al., 2018). Indeed, there is a poor penetration of light in biological tissues due to light-scattering. Notably, in plants, the difference between refractive indices of the cell wall (1.42) and of the cytosol and vacuole (around 1.36) makes the samples opaque (Helmchen et al., 2005; Kurihara et al., 2013; Timmers, 2016) making them difficult to clear compared to animal samples. Thus, the necessity to make the samples transparent for light is challenging and during the last few years, optical clearing methods have been widely used on animal samples, notably with mice brains (Hama et al., 2015), but are still truly underdeveloped in the field of plants.

Fundamentally, LSFM observation is made possible by decoupling the illumination and detection optical pathways. A light sheet is formed by laser lights illuminating both lateral directions of the sample. Importantly, detection is performed along an orthogonal direction in order to maximize detection efficiency by minimizing fluorescence from out-of-focus



**Figure 1 | Principle of the light sheet fluorescent microscopy (A) and comparison with other fluorescent microscopy techniques (B, C, D)**

B- Light sheet fluorescent microscopy, C- Widefield microscopy, D- Confocal microscopy

Image A: adapted from LaVision Biotec website

Images B, C, D: adapted from "Comparison of different microscopy illumination" by J. Krieger

features (Fig.1A). Compare to other fluorescent microscopy techniques like Widefield or Confocal microscopy, the illumination of the focal plane is planar and comes from the sides (Fig.1B,C,D) which allows the reduction of photobleaching and phototoxicity, allowing for long-term imaging of the sample over many days, or even weeks (Ding et al., 2018).

Developed clearing procedures are classified into two categories. One is done by removing the cytoplasmic content making the sample opaque and the other by uniformizing refractive indices. Most techniques based on the second method are broadly applied for the study of plants (Herr, 1993; McBryde, 1936). Different treatments and their combinations were developed for plant imaging like sodium chloride, hydrogen peroxide, strong alkali or acids, phenol, lactic acid and chloral hydrate (Soukup et al. 2013). However, these techniques are not completely compatible with highly pigmented and highly optical dense tissues. Hence, concerning our study, we decided to perform four other clearing methods on plant samples, two aqueous (TDE & CUBIC) and two solvents (ECi & u-DISCO).

The TDE method is an aqueous passive immersion technique. The samples are immersed into TDE which matches the refraction indexes (Aoyagi et al., 2015). We chose to perform this clearing method because it is a simple technique, fast to realize and we do not need a lot of material. Plus, it is the only method among all the ones we selected which was published for plants and seemed to give good results (Musielak et al., 2016). The CUBIC method is our second technique based on aqueous medium. It removes the lipids with detergents (CUBIC-R1) and then urea and sucrose clear the samples (CUBIC-R2). It is a hyper-hydration technique (Tainaka et al., 2014). We wanted to perform this clearing method

because it is one of the most common clearing techniques we could perform in our laboratory.

The ECi is an organic solvent based method. It implies the samples dehydration in ethanol and then an immersion into Ethyl Cinnamate (ECi) for refractive index homogenization (Klingberg et al., 2006). We wanted to perform this clearing method because it is known to give a better transparency in animal samples compared to others technics. Moreover, it allows to clear without losing fluorescence. Therefore, the use of this protocol could allow the study of the localization of protein-fusion (GFP, YFP, etc.) within plant tissues. Moreover, compared to clearing methods using water-based agent, this method gives a better transparency and more resistant samples. The u-DISCO method is our second solvent-based technique. It consists in dehydration in Tert-butanol and then in BABB-D4 (Pan, 2016). Process of clearing and subsequent imaging is fast (hours to days depending on sample size) compared, for example, to CUBIC clearing (weeks). Above that, u-DISCO works on many types of tissues (lung, spleen, lymph nodes, mammary gland, tumors) and this technique also preserves the fluorescence of GFP.

In order to determine which protocols of clearing fit better with plants samples, we had to choose different types of relevant plants. The technology will allow us to observe morphological characteristics. We tried to find plants with interesting morphological characteristics, which would be observable with the LSM, and which could be of interest to the scientific community. These plants should be seen as models, which can be used to study other issues. Therefore, we choose to work on reproductive organs, pistil and anther of the tulip and lily flower. Indeed, this must be pertinent as it could be useful for certain studies about plant reproduction.

We also, worked on the climbing ivy because the clearing of this plant can be useful to study the morphology of its root hairs secreting a powerful glue made of polysaccharides and spherical nanoparticles. This glue is widely studied by the scientific community for industrial purposes (Burris et al., 2012). Moreover, we choose the Romanesco cabbage for its particular and complex structure made of fractals. It can be useful to see the impact of our clearing protocols on this particular type of structure. Finally, we choose to work on *Arabidopsis Thaliana* as it is the main model in the field of plant biology.

As a beginning, we chose to observe the plant samples without any staining to assess the efficacy of the clearing protocols. Thus, the observations made were based on the emission of auto fluorescent molecules naturally present in plant cells (Muller Sara M. et al., 2013).

In this study, we will perform a qualitative comparison between two aqueous-based and two solvent-based clearing techniques in order to provide a better protocol for the observation of plants with the LSFM.

## MATERIAL & METHODS

### I. Chemicals

The following chemicals were used for the following protocols : TDE solution (2,2'-Thiodiethanol) (Sigma, Ref: 166782), PBS powder (Sigma, Ref: P3813), Formol 4% (CEA), Direct Red 23 (Sigma, Ref: 212490), Ethanol (Sigma, Ref: 51976), ECi (Ethyl Cinnamate) (Sigma, Ref: W243000), Tert-Butanol (Sigma, Ref: 8222641000), BABB-D4 (mix of benzyl alcohol, benzyl benzoate, vitamin E) (CEA), NaCl (Sigma, Ref: S8776), Triton X-100 (Sigma, Ref: X100), Quadrol (N,N,N',N'-Tetrakis(2-Hydroxypropyl)ethylenediamine) (Sigma, Ref: 122262), Urea (Sigma, Ref: U4883), Sucrose (Sigma, Ref: S7903) and triethanolamine (2,2',2''-Nitrilotriethanol) (Sigma, Ref: 90279).

### II. Fixation of the samples

The samples were totally immersed in formol 4% for 2-4 hours. Then, the samples were washed 3 times with PBS.

### III. Staining procedure

Direct red 23, a staining agent of pectocellulosic wall was used to stained the plant samples.

For Aqueous protocols, 0.1% of direct red 23 was directly mix with the clearing solutions.

For Solvent based protocols, 0.1% of direct red 23 was mix with PBS 1X and samples were immersed during 4h before the clearing.

### IV. Clearing procedures

TDE protocol was prepared according to Musielak et al., 2016 and Aoyagi et al., 2015. A 50% TDE solution was prepared

with 50% of distilled water. The samples were incubated in the TDE solution and under agitation at room temperature.

CUBIC protocol was done according to Nehrhoff et al., 2016. PBS 1X was prepared by dissolution of a PBS tablet in 1L of distilled water (dH<sub>2</sub>O). Reagent CUBIC-R1 was prepared by mixing triton X-100 (15 wt% final concentration), quadrol (25 wt% final concentration) urea (25 wt% final concentration) and dH<sub>2</sub>O until dissolution. Then 1/200 volume 5M NaCl (final 25mM) was diluted. Finally, the reagent was degassed with centrifuge (4500g, 10 minutes) to remove bubbles. Reagent CUBIC-R2 was prepared by mixing urea (25 wt% final concentration), sucrose (50 wt% final concentration) and dH<sub>2</sub>O until dissolution. Then 10 wt% triethanolamine was added to the solution. Finally, the reagent was degassed with centrifuge (4500g, 10 minutes). ½ CUBIC-R1 was prepared by mixing 1:1 of CUBIC-1 and dH<sub>2</sub>O. The samples were immersed in ½ water-diluted CUBIC-R1 with rotation at 37°C overnight. Then, the samples were immersed in CUBIC-R1 for 8 hours and CUBIC-R1 was replaced by fresh CUBIC-R1 for 20 hours. This action had been repeated for 2 days. All bathes were done under gentle agitation at 37°C. ½ CUBIC-R2 was prepared by mixing 1:1 of CUBIC-R2 and dH<sub>2</sub>O. After 3 washes of 2 hours in PBS 1X, the samples were immersed in ½ CUBIC-R2 overnight with gentle shaking at room temperature. Then, ½ CUBIC-R2 was replaced by CUBIC-R2 for 8 hours and then replaced by fresh CUBIC-R2 for 3 days. All bathes were done under gentle agitation at room temperature. At the end, samples were immersed into 80% TDE.

ECi protocol is adapted from Klinberg et al., 2017. Fixed samples were dehydrated in 30% EtOH (vol/vol) overnight, 50% EtOH for 4h, 70% EtOH for 24h and 100% EtOH for 24h and overnight with agitation at room temperature. Then, dehydrated samples were transferred into ECi solution and incubated at room temperature until they become transparent.

u-DISCO protocol is adapted from Pan et al., 2016. Fixed samples were dehydrated in 30% (vol/vol) Tert-butanol overnight, 50% Tert-butanol for 4h, 70% Tert-butanol for 4h, 80% Tert-butanol for 20h, 90% Tert-butanol for 20h, 96% Tert-butanol for 8h and 100% Tert-butanol overnight with agitation at 37°C. Then, dehydrated samples were immersed in BABB-D4 during 8h and transferred in ECi solution until imaging.

### V. Imaging

The slightest difference in refractive index can be seen when imaging with the LSFM, therefore, it is better to prepare bigger amount of storage solution to be able to image it. Consequently, for aqueous-based protocols, the light sheet microscope tank was filled with the 50% TDE solution for sample cleared with TDE protocol and 80 % TDE solution for samples cleared with CUBIC protocol. For solvent-based protocols, the light sheet microscope tank was filled with the ECi solution.



The imaging parameters were almost the same for all the samples. Only the light sheet width was adapted for each sample. However, for the thickness, we choose 4 or 5  $\mu\text{m}$ , which allowed to obtain a good resolution. We usually used an excitation wavelength of 488nm and an emission filter of 525nm with a 50 nm bandwidth. When we had samples with dye to image, we added an emission filter at 620nm with a 60nm bandwidth. The step used between the take of each image of the Z-stack was 2 $\mu\text{m}$ .

VI. Imaging software

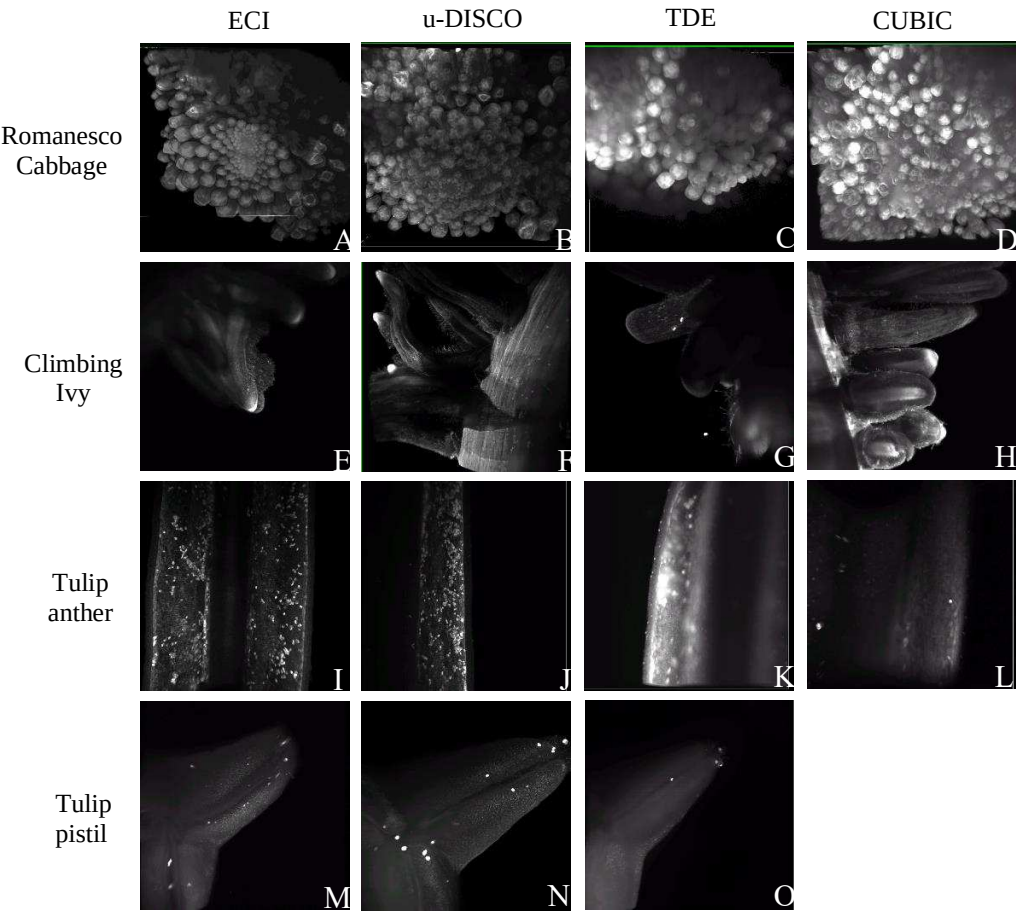
We used two software to process our images: ImageJ version 1.52r and VAA 3D v3.200. It can display multiple stacks in a single window, allowing Z-projection and, therefore, a 3D reconstruction.

Vaa3D is an image visualization and analysis system for bioimages. It is powerful for visualizing large- or massive-scale 3D image stacks, therefore very suitable in our case as the LSM images are very heavy.

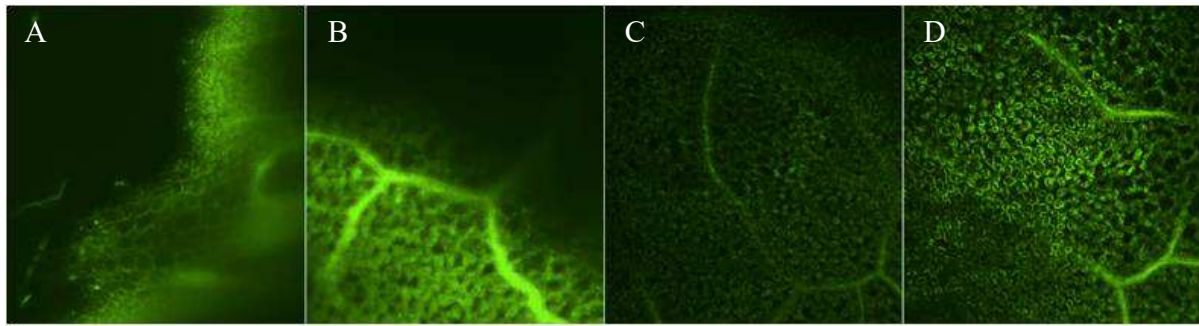
RESULTS

The different clearing protocols have been applied on each plant, to allow a comparison of their respective effectiveness. Firstly, we noticed that generally, all the protocols conserve the general macroscopic structure of the samples. Notably, we observed that the complex structure of the Romanesco Cabbage was well preserved, and the fractal organization of its flowers was not altered. This was notably the case for the two solvent-based protocols, and especially u-DISCO protocol (Fig.2). Same observation could be made on the rhizoid of the climbing ivy (Fig.2 ). However, the Tulip sample cleared with CUBIC was destroyed due to an error of manipulation.

The image quality seemed to vary between the protocols. Indeed, during the acquisition, we noticed that the clearness of the image was better in the solvent based-protocols (Fig.2) than in the aqueous-based ones (Fig.2 last two columns). However, none of the protocols we used was able to completely clarify the sample. Also, the imaging depth varied between the protocols and the samples. Notably, the Romanesco flowers were more easily observable and distinguishable in the ECI and u-DISCO protocols (Fig.2 A



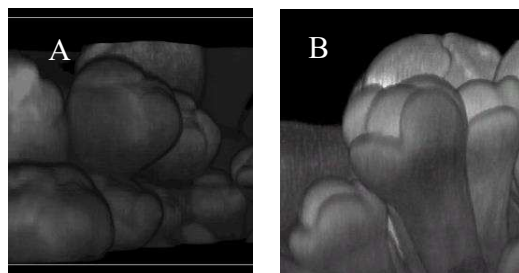
**Figure 2 | Plant samples imaging after various clearing protocols.**  
Light-sheet imaging of Romanesco Cabbage, Climbing Ivy, Tulip Anther and Tulip Pistil samples cleared with ECI, U-DISCO, TDE and CUBIC clearing protocols.  
Magnification 3.2



**Figure 3 | Light-sheet imaging of *Arabidopsis thaliana* leave**  
Comparison of 4 different clearing protocols A-TDE, B-CUBIC, C-ECi, D- u-DISCO.  
Magnification 12,6

and B) than in the TDE and CUBIC (Fig.2 C and D). Similarly, we obtained clearer images of the tulip anther in solvent-based (Fig.2 I and J) than in aqueous protocols (Fig.2 K and L). We repeated these experiments and obtained the same results (Sup Fig.7), confirming the hypothesis that ECi and u-DISCO protocols allow better observation of samples.

Based on these observations, we suggest that the solvent-based protocols (ECi and u-DISCO) are more suitable for plants clearing than aqueous-based protocols (TDE and CUBIC). We made further investigation to assess the effectiveness of the solvent-based protocols. Effectively, we tried to image the samples at maximum magnification (12,6x), to better assess the structural conservation, and the clearness of the image. We so observed that leave of *Arabidopsis thaliana* and Romanesco cabbage flowers were very well observable, without any noise or structural alteration (Fig.3 and 4). In the same way, the other samples were very well conserved and observable (data not shown).



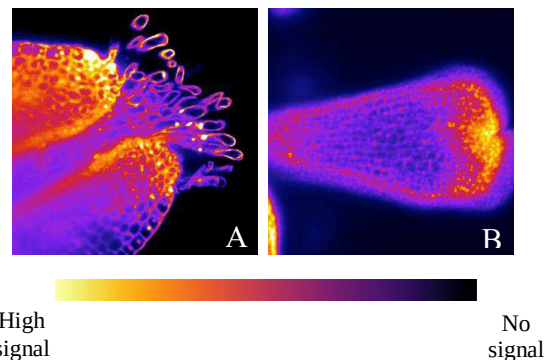
**Figure 4 | Romanesco Cabbage flowers at maximum magnification (12,6x)**  
A:ECi protocol; B: u-DISCO protocol

We then wondered if the clearing process allowed us to distinguish cellular structures, notably if the cell wall was

clearly distinguishable from cellular contents. To achieve it, we needed to have better contrasts on the images we obtained.

To do so, we generated heat maps of signal density on the most promising samples to have more contrast between the structures. Interestingly, we were able to clearly distinguish

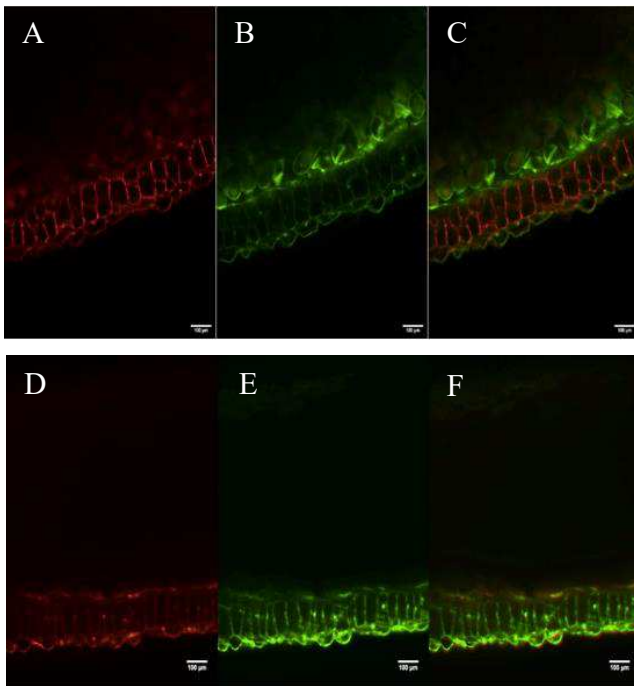
cellular structures in the tulip pistil and the flower of the Romanesco Cabbage (respectively Fig.5 A and Fig.5 B).



**Figure 5 | Heat map revealing cellular resolution in two samples**  
A: Tulip pistil cleared with u-DISCO protocol (12,6x)  
B: Romanesco Cabbage flower cleared with u-DISCO protocol (12,6x)

Actually, the clearest structures seem to be the ligno-cellulosic walls of the plant cells, whereas the dark area inside the cells seem to be the vacuole. To confirm this hypothesis, we chose to repeat these experiments, but adding a cell-wall staining with Direct Red 23 before the clearing. Interestingly, we observed a clear staining of the cell wall in Lily's anther cleared with u-DISCO protocol compared to non-stained sample (Fig.6). If the results were not as remarkable in the other samples, we noticed a higher signal during the image acquisition in the stained samples, suggesting that clearing protocols could be associated to staining methods.

In the same way, extra-cellular structures were observable in the tulip in the form of white structures in the ECi and u-DISCO treated samples that we supposed to be pollen grains. Also, we may be able to observe cell nucleus in the tulip anther. Nevertheless, these structures could barely be seen in the TDE treated sample and are not observable in the CUBIC one. However, further experiments (e.g. observations with specific stainings) should be made to confirm these structure's nature.



**Figure 6 | Light-sheet imaging of lily's anther previously cleared with u-DISCO protocol**  
A,B,C - Stained samples with Direct Red 23 and D, E, F- Samples without staining  
Bare scale: 100  $\mu$ m, Magnification 12.6

## DISCUSSION

The development of confocal microscopy opened the avenue to image large samples, or even entire organs, in 3D with high spatial resolution. However, imaging of tissues in 3D is still challenging. The main reason for that is the opaqueness of tissues that limits imaging in depth (Weiglin et al., 2016). Spalteholz introduced clearing of tissues already a century ago (Spalteholz, 1911), but only in the last decade clearing of tissues was the topic of intense research. The availability and continuous improvements of imaging techniques such as 1-photon and 2-photon microscopy and light-sheet microscopy were a strong stimulus of this research in clearing techniques.

The clearing methods that are available now are mainly used on animal samples. This study brought information on the protocols that are today used to clear animal samples, but that could be apply to plants. Based on this observation, we chose to observe plant structures with LSM. Our objective was, then, to find the most efficient clearing protocols among those that already exist, to obtain the best images of plant samples with the LSM, and to test the compatibility of the clearing with the staining methods.

Contrary to the literature, in which we found that TDE was the most widespread technic to clear tissues, our study showed that solvent based protocols were the most suitable for plants as it preserves the structure of the plants and we

could image deeper in the sample than with aqueous ones. By comparing out both solvent protocols, ECI and u-DISCO, we found that u-DISCO seems to be the best clearing protocol as it preserves better the structure of the plant sample than the ECI protocol. Also, cell wall had been successfully stained in u-DISCO cleared sample, suggesting that clearing is compatible with staining methods in plant tissues.

To pursue this work, experiment could be made to quantify the percentage of the sample that have been cleared, to be able to compare the efficacy numerically and not only visually. This study could be extended to other clarification technics that are currently used. If our work showed that staining was compatible with clearing, we could also wonder if endogenous marker such as GFP-linked protein would be conserved and show similar properties after clearing protocols.

To conclude, in this study, we demonstrated that clearing with solvent protocols seems to better clear plant samples than aqueous ones. ECI and U-DISCO immersed plant samples showed enhanced penetration depths as we observed it thanks to the imaging (images less blurry). We also noticed that the ECI protocols seem to further alter the structure of the samples.

## ACKNOWLEDGMENT

We thank Mrs Elise Delage for her support and her help in the development of the project. We also thanks Mrs Perdiz and Mrs Morgensen for their help in the laboratory. We are grateful to Maxime Mauviel for his help with the images processing.

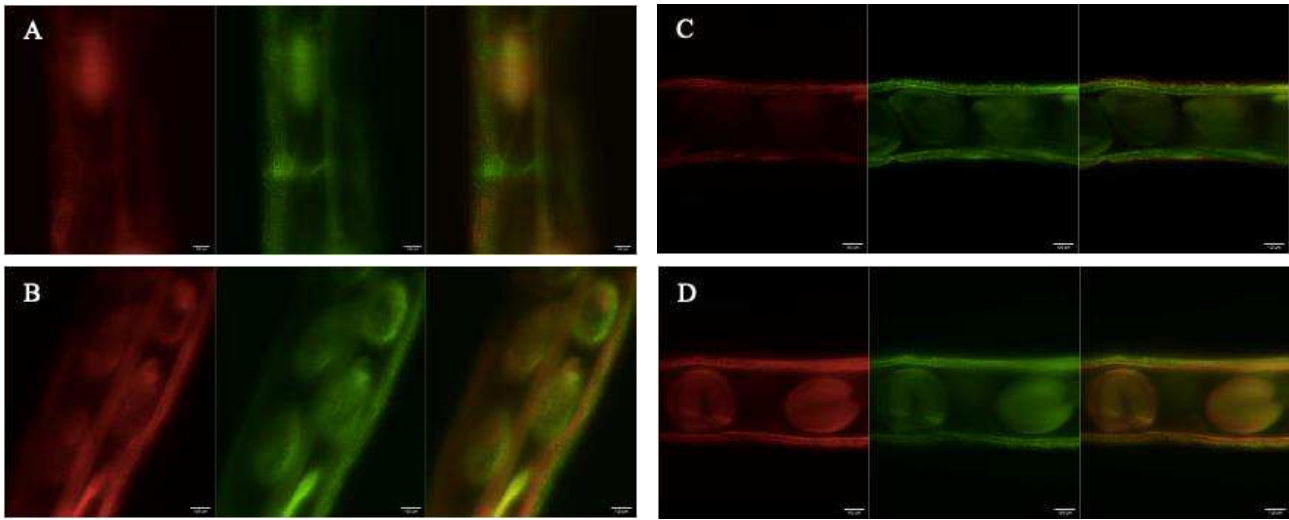
## REFERENCES

- [1] Aoyagi Yuka et al., "A Rapid Optical Clearing Protocol Using 2,2'-Thiodiethanol for Microscopic Observation of Fixed Mouse Brain", *PLOS One*, 2015
- [2] Burris J., N., et al., "Nanoparticle biofabrication using English ivy (*Hedera helix*).", *Journal of nanobiotechnology*, 2012
- [3] DING Yichen et al., « Multiscale light-sheet for rapid imaging of cardiopulmonary system », *JCI insight*, 2018
- [4] Eliceiri Kevin W. et al., "Biological Imaging Software Tools", *Nat Methods.*, 9(7), 697–710, 2013
- [5] Feijo JA, Moreno N., "Imaging plant cells by two-photon excitation", *Protoplasma*, 223(1), 2004, 1–32
- [6] Girkin, J. M., and Carvalho, M. T. "The light-sheet microscopy revolution." *Journal of Optics*, 20(5), 2018
- [7] Hama H, Kurokawa H, Kawano H, Ando R, Shimogori T, et al., "Scale: a chemical approach for fluorescence imaging and reconstruction of transparent mouse brain", *Nat Neurosci*, 14, 2011, 1481–1488
- [8] Helmchen F, Denk W., "Deep tissue two-photon microscopy", *Nat Methods*, 2(12), 2005, 932–40

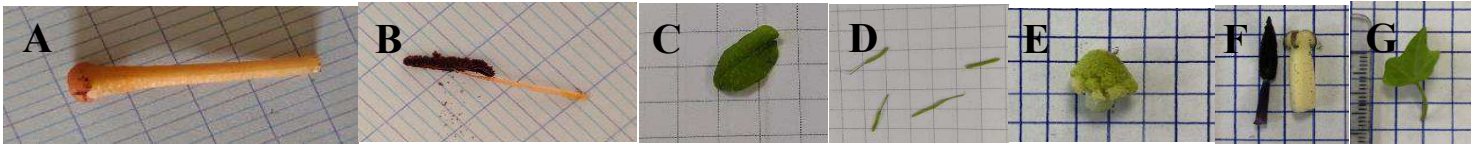
- [9] Herr J. M. Jr., "Clearing Techniques for the Study of Vascular Plant Tissues in Whole Structures and Thick Sections", *Tested studies for laboratory teaching*, vol.5, 1993, 63-84
- [10] Klinberg et al., "Fully Automated Evaluation of Total Glomerular Number and Capillary Tuft Size in Nephritic Kidneys Using Lightsheet Microscopy.", *Journal of the American Society of Nephrology*, 28, 2006, 452-459.
- [11] Kurihara D, Hamamura Y, Higashiyama T., "Live-cell analysis of plant reproduction: live-cell imaging, optical manipulation, and advanced microscopy technologies.", *Dev Growth Differ*, 55(4), 2013, 462-73
- [12] McBryde MC., "A method of demonstrating rust hyphae and haustoria in unsectioned leaf tissue.", *American Journal of Botany*, 23(10), 1936, 686-9
- [13] Müller Sara M., Galliardt Helena, Schneider Jessica, Barisas B. George and Seidel Thorsten, "Quantification of Förster resonance energy transfer by monitoring sensitized emission in living plant cells", *Frontiers in plant science*, 2013
- [14] Musielak T., J., et al, "A Versatile Optical Clearing Protocol for Deep Tissue Imaging of Fluorescent Proteins in Arabidopsis thaliana", *PLoS ONE* 11(8): e0161107, 2016
- [15] Nehrhoff Imke, Bocancea Diana, Vaquero Javier, Vaquero Juan José, Ripoll Jorge, Desco Manuel, and Gómez-gavira Maria Victoria, "3D imaging in CUBIC-cleared mouse heart tissue: going deeper", *Biomedical optics express*, vol. 7 (9), 2016
- [16] Pan C, Cai R, Quacquarelli FP, Ghasemigharagoz A, Loubopoulos A, Matryba P, Plesnila N, Dichgans M, Hellal F, Ertürk A, "Shrinkage-mediated imaging of entire organs and organisms using uDISCO", *Nat Methods*, vol.13(10), 2016, 859-67
- [17] Soukup A., and Tylová E., "Essential Methods of Plant Sample Preparation for Light Microscopy", *Plant Cell Morphogenesis*, 1-26, 2013
- [18] Spalteholz, W., "Über das Durchsichtigmachen von Menschlichen und Tierischen Präparaten und Seine Theoretischen Bedingungen", *S. Hirzel, Leipzig*, 1911
- [19] Tainaka K, Kubota SI, Suyama TQ, Susaki EA, Perrin D, Ukai-Tadenuma M, et al., "Whole-body imaging with single-cell resolution by tissue decolorization.", *Cell*, 159(4), 2014, 911-24
- [20] Weigel, B., Bakker, G.-J., Friedl, P., "Third harmonic generation microscopy of cells and tissue organization", *J. Cell Sci*, 129 (2), 2016, 245-255.



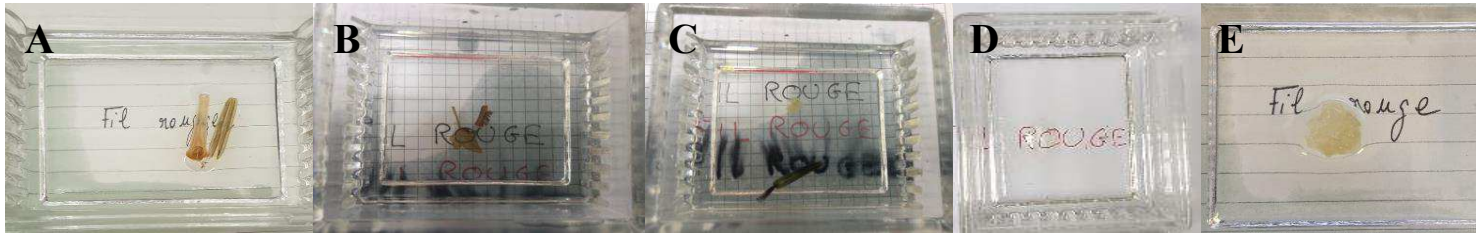
SUPPLEMENTAL DATA



**Figure 7 | Light-sheet imaging of *Arabidopsis thaliana* silique previously cleared with TDE (A-B) or u-DISCO (C-D) protocols**  
A, C – Samples without staining and B, D- Stained samples with Direct Red 23  
Bare scale: 100 µm, Magnification 12.6



**Figure 8 | Samples before clearing**  
A - Lily pistil, B – Lily anther, C – *Arabidopsis thaliana* leaf, D – *Arabidopsis thaliana* silique, E – Romanesco cabbage, F – Tulip anther and pistil, G – Climbing ivy leaf



**Figure 9 | Samples after clearing**  
A - Lily pistil and anther after U-DISCO clearing, B – Climbing ivy leaf after ECI clearing, C – Tulip anther and pistil after ECI clearing, D – *Arabidopsis thaliana* leaf and silique after U-DISCO clearing, E – Romanesco cabbage after U-DISCO clearing

# Cinergy: Degradation of Cigarette Butts to Produce Electricity Thanks to a Microbial Fuel Cell

Ambre Leleu\*, Lucas Bourmancé, Mélissa Chagnaud, Caroline Suant, Chloé Doizelet, Eva Zanditenas, Julie Breiner, Marina Firpion, Mathilde Barbier, Sarah Duquesne  
Sup'Biotech, Villejuif, France

\*Corresponding author: ambre.leleu@supbiotech.fr

**Abstract** - Cigarette butts filters are composed of Cellulose Acetate (CA), a plastic material. Once in the oceans, this material degrades into plastic nanoparticles, threatening both humankind and biodiversity. The aim of Cinergy is to add value to CA, by producing electrical energy using a Microbial Fuel Cell (MFC). In our system, cigarette butts are shredded to separate the CA fibers. They will then be reduced into glucose and acetate by two key enzymes: endo-1,4- $\beta$ -glucanase I (EG1) and cellulose acetate esterase (CAE), secreted by modified bacteria. Glucose and acetate are then metabolized into lactate, which will be secreted, and used by modified *Shewanella oneidensis* to produce electrons. They are transported to the anode and the anode-cathode system to produce an electrical current. Three main steps had to be demonstrated in the Cinergy process: the degradation of CA fibers, the optimized production of lactate, and the generation of electricity by *S. oneidensis*. Our experiments demonstrated the successful construction and integration of recombinant plasmids, also known as BioBricks, expressing the following enzymes: ACS, LDHA, LLDP, CAE, EG1, Tev protease, and DLD-II. We demonstrated that each step of the process was functional in a separate way. More research and replicates are required to strengthen the efficiency and stability of our device.

**Index Terms** – Cigarette butts, synthetic biology, cellulose acetate, electricity

## Abbreviation

ACS: Acetyl-CoA Synthetase  
CA: Cellulose Acetate  
CAE: Cellulose Acetate Esterase  
CET: CAE, EG1, Tev  
DLD-II: Operon  
*E. coli*: *Escherichia coli*  
EG1: Endo-1,4- $\beta$ -Glucanase I  
GGA: Golden Gate Assembly  
GOI: Gene Of Interest  
LDHA: Lactate Dehydrogenase A  
LLDP: L-Lactate Permease  
MFC: Microbial Fuel Cell  
OpACS: Operon ACetyl Synthase  
*S. oneidensis*: *Shewanella oneidensis*

Tev: Tev protease

## Introduction

Each year, around 44 billion cigarettes are sold in France, which makes it one of the world's largest tobacco consumers[1].

In addition to the harmful effects on smokers' health, cigarettes are also highly harmful to the environment. Indeed, more than 30 billion cigarettes are thrown each year on the ground in France and take 10 to 15 years to fully degrade, producing a very polluting waste: the cigarette butt. This represents a major environmental issue as one cigarette butt is composed of more than 4,000 toxic substances that are discharged in the soil, groundwater tables, and oceans and one cigarette butt can pollute up to 500L of water[1,2].

Moreover, one cigarette butt and more precisely its filter is composed of a plastic material: Cellulose Acetate (CA)[3,4]. Once in the oceans, this material is released into the water as plastic nanoparticles that enter the food chain and become very harmful to the marine biodiversity as well as to environment and mankind. This triple threat is an important issue that still rises too little concern[5].

In this context, the aim of our project Cinergy is to prevent the dispersion of the cigarette butts. To encourage people to recycle them in a controlled manner the system will produce electricity that could be used by everyone. This electrical energy production is done through the use of a Microbial Fuel Cell (MFC). This process will be done in a circular economy strategy and for the preservation of the environment and the well-being of future generations.

The power generation system is based on the functioning of a MFC[2]. Indeed, a MFC or microbial battery is a method that allows to generate an electrical current from microorganisms (e.g. bacteria). By degrading nutrients during their growth, bacteria produce electrons that are then transported to an anode to generate an electrical current, which can then either be used or recovered.

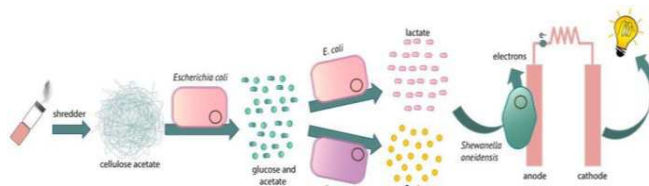
To implement our system, we have engineered two different bacterial strains: *Escherichia coli* and *Shewanella oneidensis*. We worked on two populations of *E. coli* and one population of *S. oneidensis*. First, the cigarette butt is shredded to separate the CA fibers that compose the filter, from each other. These CA fibers are carbohydrates that can serve as a substrate for bacteria. Consequently, the fibers are

degraded by a first population of *Escherichia coli* via secretion of two key enzymes: EG1 and CAE [7]. These enzymes allow producing two other carbohydrates from CA: glucose and acetate. This population of *E. coli* is then modified to metabolize the two carbohydrates to produce lactate[8]. The lactate is excreted by *E. coli* to allow *S. oneidensis* to use it as a substrate. *S. oneidensis* is qualified as electrogenic as it can produce electrons and transport them to a solid conductor, thus producing electricity[9].

With the aim to increase the number of electrons transported to the anode and so the yield of electricity produced, we engineer a second population of *E. coli* to produce flavins. Flavins are defined as cofactors for the transport of electrons[10,11]. Hence, increasing the number of cofactors would increase the number of electrons transported to the electrode and finally increase the amount of electricity produced. These manipulations are performed for us to obtain an interesting yield of electricity[9,12].

Finally, the electricity produced is stored in an accumulator or in a battery containing an electrical output (e.g USB) to plug a smartphone or a lamp for instance.

All the Cinergy project steps are presented in Figure1.



**Figure1: Scheme of the Cinergy project steps**

Degradation of the cellulose acetate into glucose and acetate that will be the substrate for the first *E. coli* population, which will produce lactate. The other population of *E. coli* will produce flavins which will help the transfer of electron from *S. oneidensis* to the anode and give the electricity production.

For this project, 6 different plasmids have been constructed transformed in 3 different bacterial population. The plasmids 1 and 2 have been transformed in the first *E. coli* population, 3 and 4 transformed in the second *E. coli* population and the two last ones in the *S. oneidensis*. The composition of the different plasmids is the following:

- The first plasmid contains three BioBricks: *ldha*, *lldp*, *acs* operon that allows a better intake and secretion of acetate and glucose of the *E. coli* bacteria.
- The second plasmid contains three BioBricks: *cae*, *egl*, *tev* genes that allows the degradation of CA in glucose and acetate
- The third plasmid contains two BioBricks: *ribABDEC* and *ribF* that are co-factors called flavins and help the electron transport of *S. oneidensis* to the anode
- The fourth one contains four BioBricks: *acs*, *zwf*, *gnd*, *pgl* where *acs* help the *E. coli* to boost the Krebs cycle and the three other genes are increasing the flavins synthesis pathway which is not favored in the normal *E. coli* metabolism.

- The fifth one 4 contains BioBricks: *pflb*, *acs*, *mdh*, *fdh* genes where *acs* increases the acetate and lactate of *E. coli* consumption and the three other genes increase the generation of electrons.
- The sixth one 2 contains BioBricks: *lldp* operon and *mtrC* genes where *lldp* operon boosts the acetate and lactate of *E. coli* consumption and *mtrC* is helping to increase the electron transfer and thus, increase the current.

## Material and Methods

### Make competent cells

The preparation of competent cells is done from culture of *E. coli* bacteria from glycerol stock on LB agar plate before incubation for 16 hours at 37°C. The next day, pick up one colony and culture in liquid LB medium (Tryptone 10 g/L, Yeast extract 5 g/L, NaCl 10 g/L) overnight with shaking conditions. On the third day, inoculate LB medium with the saturated culture and incubate until OD600=0,4 is reached. Then, place into an ice bath for 10 minutes before centrifugation at 2,700g for 10 minutes. Resuspend the pellet with ice-cold CaCl<sub>2</sub> 100mM, incubate on ice for 30 minutes and centrifuge again at 2,700g for 10 minutes. Resuspend the pellet with MgCl<sub>2</sub>-CaCl<sub>2</sub> 100mM, incubate on ice for 10 minutes and add 80% glycerol before freezing aliquots in liquid nitrogen and store at -80°C.

### Construction of BioBrick

For *cae*, *egl*, *tev*, *gnd* and *zwf* genes, they have been synthesized by IDT®. The Step one to three are not performed for the construction of these two BioBricks.

#### 1. Production of the insert by Polymerase Chain Reaction

PCR were performed with Q5 MasterMix 2X®, a High-Fidelity DNA Polymerase with 50ng of template DNA according to manufacturer instructions (New England Biolabs). The PCR mix is composed of the enzyme buffer, the four nucleobases (dNTP), forward and reverse primers, the template DNA and the polymerase. The enzyme is added extemporaneously. Samples are incubated into a thermal cycler for 25-35 cycles. Hybridization temperature is set according to the couple of primers used and the elongation time depends on the enzyme capacity and the size of the amplicon. The Q5® enzyme can amplify 2kb per minute. A table of the PCR conditions and the different couple of primers used are given on the Supplementary Figures 1 and 2.

#### 2. Gel Migration

To purify PCR amplicons, gel electrophoresis was performed. The gel used was a 1% agarose gel in TAE buffer (TrisBase, acetic acid, EDTA, H<sub>2</sub>O). PCR samples were prepared by adding 5µL of DNA Gel Loading Dye (6X) (ThermoFisher Scientific) for 30µL sample. Then, the samples were loaded, and migration was conducted between 90V to 110V depending on the gel size. Finally, the gel was



revealed using a G-Box (Syngene) with UV light and a gel photo was taken.

### 3. Gel extraction and purification sample

Sample extraction from the agarose gel was performed on a transilluminator. They were cut out with a scalpel and purified using NucleoSpin Gel and PCR Clean-up® kit (Macherey-Nagel) according to the manufacturer's instructions. Briefly, after resuspension of the gel extracted into 500µL to 800µL of high salt binding buffer (NTI solution) at 50°C, samples were loaded on silica-membrane columns and washed several times with ethanolic solutions (NT3) before the sample elution under low ionic strength conditions.

### 4. Digestion

To insert the GOI sequences into the donor plasmid (pSB1A3 or pSB1K3 the maps are presented Figure 2, generous gift from Glowee) a double enzymatic digestion was performed using EcoRI and SpeI (New England Biolabs) in the 2.1 buffer. The mix was made on ice containing the enzyme buffer, the restriction enzymes, the insert or the vector that must be digested and finally demineralized water QSP 25µL. Samples were then incubated at 37°C for an hour.

After incubation, the vector undergoes an agarose gel electrophoresis before being purified thanks to the NucleoSpin® PCR and gel Clean-up (Macherey-Nagel) as previously described.

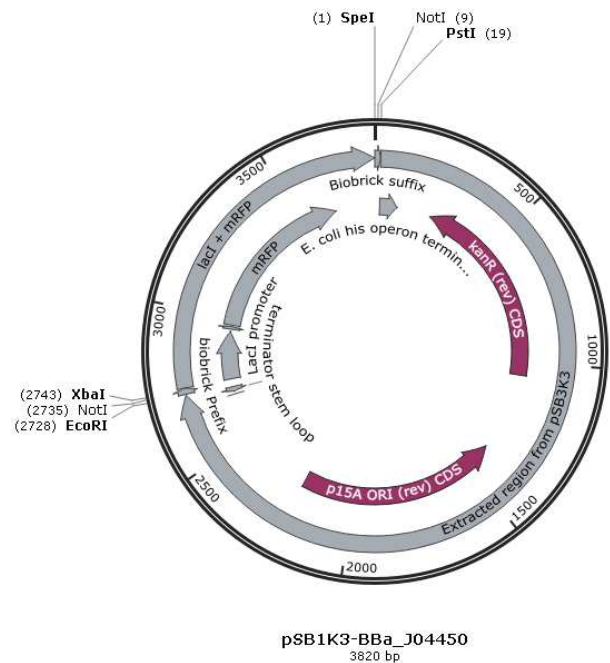
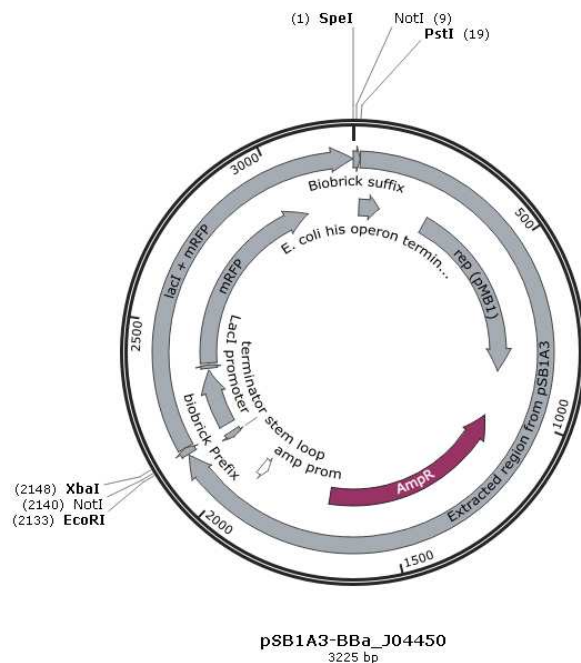


Figure 2: Plasmid map of pSB1K3 and pSB1A3

Both plasmids contain the *LacI* gene that will be removed for the insertion of the gene of interest. *EcoRI*, *NotI*, *XbaI*, *SpeI*, *PstI* are the enzymes required for the construction of the BioBrick for the iGEM competition. Finally, the pSB1A3 has ampicillin resistant gene and pSB1K3 the kanamycin resistant one.

### 5. Golden Gate Assembly (GGA)

Ligation of the different parts of the BioBrick promoter, GOI sequence, terminator is performed, incubated in the ratio of 1:3:1 respectively. GGA required the use of 2 enzymes *Esp3I* and *T4 ligase* (New England Biolabs). A mix was performed and incubated into a thermal cycler for 30 to 50 cycles, table shown in Supplementary Figure 3.

After the double digestion of insert and vector with *EcoRI*/*SpeI* (New England Biolabs), the mixed ligation was made using *T4 Ligase* (New England Biolabs) according to manufacturer's instructions. The mixed ratio is 3:1 for the insert and the vector respectively and was incubated at least 3h at 16°C.

### Transformation

Chemio-competent DH5α bacteria (ThermoFisher Scientific) were thawed on ice and transformed with the donor plasmids harboring the different GOI according to manufacturer's instructions. Depending on the number of samples to be transformed, 35 to 100µL of DH5α were used per plasmid and transformed with 10 to 200 ng of plasmid. After 30 minutes of incubation on ice, a thermal shock was applied to the DNA: bacteria mixture at 42°C for 45 seconds followed by 2 minutes on ice. S.O.C medium (tryptone 2 %, yeast extract 0,5 %, NaCl 10 mM, KCl 2,5 mM, MgCl<sub>2</sub> 10 mM, MgSO<sub>4</sub> 10 mM and glucose 20 mM) was then added to each transformation reaction and bacterial cultures were incubate for one hour at 37°C under shaking for recovery.

Petri dishes were pre-warmed at 37°C. After the recovery incubation, bacteria were pelleted at 3000g for 3 minutes and pellets were resuspended in LB medium. Bacteria samples were spread on petri dish with the corresponding antibiotic and incubated overnight at 37°C.

#### *Bacterial culture*

Clonal amplification was done in liquid LB medium with the adequate antibiotic(s) (Ampicillin 100µg/mL, Kanamycin 50µg/mL or Tetracycline 10µg/mL). The tubes were incubated overnight at 37°C under shaking. These cultures will undergo further characterization to be sure that the vector has correctly inserted the gene of interest.

#### *Bacterial colony screening by PCR*

Bacterial colony screening was performed by PCR on overnight liquid cultures. Two PCR reactions were set up by bacterial culture. In the first reaction, one primer is chosen to hybridize on the insert sequence and the second one a few hundred base pairs downstream. PCR were performed as previously described except that the 50 ng of plasmid DNA were replaced by 2µL of overnight bacterial culture. PCR products were run on an agarose 1% gel as previously described for analysis.

#### *Plasmid DNA purification: Miniprep*

The NucleoSpin® Plasmid kit from Macherey-Nagel was used for DNA purification. The sample is loaded onto the column and DNA is trapped on the silica membrane in the presence of the chaotropic salts. After several washes with high salt or ethanolic solution, DNA is eluted with low-salt concentration solution or water. Finally, the DNA sample is nanodropped to know its concentration.

#### *Characterization of BioBrick*

##### *1. Lactate detection*

An *E. coli* bacteria culture was prepared in LB medium. Once OD600=0,5 is reached, induce the cell with 1M IPTG (final concentration 1mM) for at least 4 hours. Use the lactate detection kit from Libios (Libios K-DATE kit) to measure lactate concentration every 3 hours. For that, prepare a blank, standard and sample solution by mixing the different solution from the kit as described in the Supplementary Figure 4. Incubate the tubes for 3 minutes at 37°C and measure the absorbance at 340nm. Add in each tube the solution 4 (D-LDH) which contains the trigger enzyme. Mix and incubate for 3 minutes at 37°C and measure the absorbance at 340nm (A2). Use calculation table provided by the kit supplier to determine the lactate concentration.

##### *2. pH measurement*

An *E. coli* bacteria culture was prepared in LB medium. Once OD600=0,5 is reached, induce the cell with 1M IPTG (final concentration 1mM) for at least 4 hours. Once induction process is over, either centrifuge and recover the supernatant, or lyse the bacteria to recover intracellular content. Recover the solutions and add a few fibers of clean cellulose acetate

(recovered from cigarette butt filters) in the adequate tube. Incubate at 37°C and measure the pH using pH indicator strips every two hours.

##### *3. Acetate detection*

Prepare an *E. coli* bacteria culture in LB medium. Once OD600=0,5 is reached, induce the cell with 1M IPTG (final concentration 1mM) for at least 4 hours. Use the acetate detection kit from Libios (Libios K-ACTRM kit) to measure sample acetate concentration (see Supplementary Figure 4). For that, prepare a blank, standard and sample solution by mixing the different solution from the kit data not shown. Incubate the tubes for 2 minutes at 37°C and measure the absorbance at 340nm (A1). Add in each tube the solution 4 (D-LDH) which contains the trigger enzyme. Mix and incubate for 4 minutes at 37°C and measure the absorbance at 340nm (A2). Use calculation table provided by the kit supplier to determine the lactate concentration. Incubate the culture for at least 2 hours at 37°C. Once induction process is over, either centrifuge and recover the supernatant or lyse the bacteria to recover intracellular content. Recover the solutions and add a few fibers of clean cellulose acetate (retrieved from cigarette butt filters) in the adequate tube. Incubate at 37°C and measure the acetate concentration every two hours.

#### *Production of electricity*

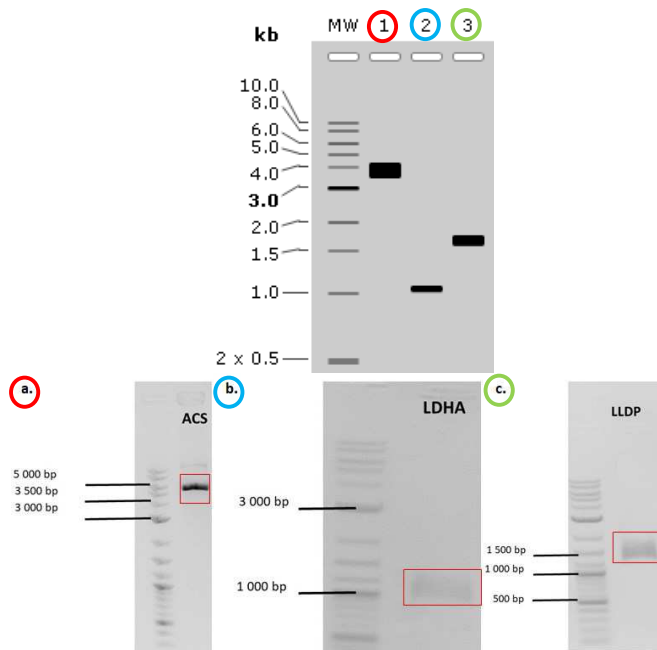
To prove the ability of our bacteria to generate electricity, *S. oneidensis* are inoculated from glycerol stock on LB agar plate and incubated overnight at 30°C. The next day, was prepared from this plate a liquid culture in LB medium supplemented with a lactate solution at 30mM. Cut square-shape of carbon felt and put them in the culture. Finally, connect the anode to a generator with a 1V current applied, incubate and take current measurement every day.

## **Results**

### *1 Molecular Biology*

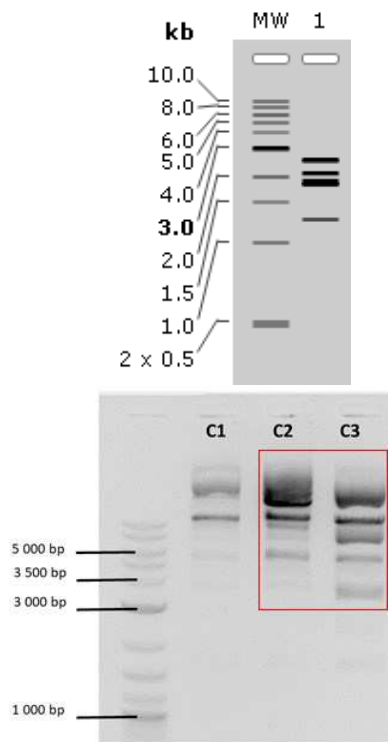
Thanks to the GGA technique being a scarless technique of assembly, all the different BioBricks have been constructed. Another backup method, using two specific restriction enzymes (BspI and PacI), was developed to anticipate any failure of the GGA construction.

Here, we want to showcase the construction of the different BioBricks and their verification by enzymatic digestion and gel migration. Figures for only one BioBrick will be shown here to show the validation methodology. Each step of the construction of the BioBrick containing the *acs*, *ldhA*, and *lldp* genes are presented below. First, all genes were extracted from *E. coli* using colony PCR method. After an overnight bacterial culture, the bacteria were lysed at 95°C for 7min. Then, a PCR was performed with 5µL of the bacterial lysate and the primers corresponding to the gene of interest. The product migrated on agarose gel is shown on Figure 3.



**Figure 3: PCR gel recovery of three genes of interest**  
1% agarose gel, 1kb plus DNA ladder; The upper figure represents the theoretical profile of the band that is wanted, and the second figure corresponds to the obtained results (red circle. *acs* gene expected at 4180 pb; blue circle. *ldhA* gene, 1053 pb, green circle. *lldp* gene, 1723 pb)

Then, using the GGA method, the BioBrick was constructed and transformed into K12 *E. coli* bacteria. Finally, the correct assembly of the BioBrick was verified via enzymatic digestion as shown on Figure 4.



**Figure 4: Verification of the BioBrick Construction**  
Migration perform on a 1% agarose gel. The first part of the figure is the theoretical profile of what is expected with strips at 1 239 bp for LDHA, 1 905 bp for LLDP, 2 114 bp for the plasmid backbone and 4 366 bp for OpACS. The second part of the figure correspond to the obtained results, different cloned were tested and only C2 and C3 had integrated the correct plasmid.

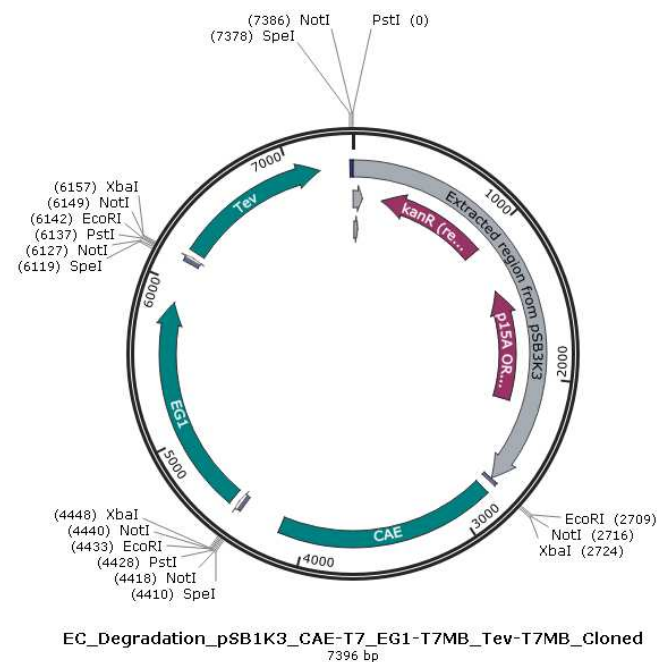
Those steps validate the construction of this BioBrick. All BioBricks were constructed and validated following the same procedure.

## II Characterization

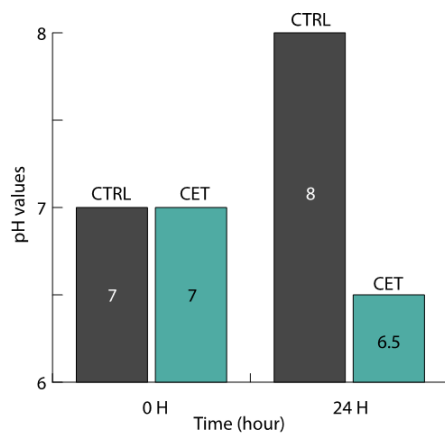
### 1. Cellulose acetate degradation

A critical step of Cinergy is the degradation of CA by a modified *E. coli*. This enzymatic process generates glucose and acetate molecules that are then used as energetic substrates by *E. coli* itself. This degradation is based on the secretion of CAE and EG1. This action is performed by the BioBrick BBa\_K3102042 presented in Figure 5, coding for CAE, EG1, and Tev protease (CET).

Thus, to prove that CA degradation is functional, pH and acetate concentrations are measured. If the degradation of cellulose acetate is functional, acetate molecules accumulate in the media. Thus, pH acidification should be detected. Results are shown on the Figure 6.



**Figure 5: Plasmid Map of BioBrick BBa\_K3102042**  
This plasmid contains the genes: *cae*, *eg1* and *Tev* flanked by a promoter and RBS before the start codon and terminator after the stop codon. There is also a kanamycin resistant gene.

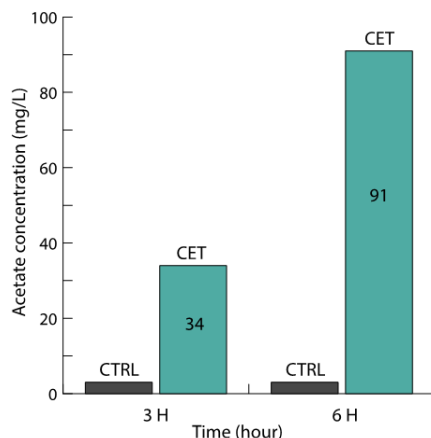


**Figure 6: pH measurement to ensure a good degradation of cellulose**

The pH of the control (CTRL) which is bacteria non modified, and the pH of the modified bacteria with the BioBrick containing *cae*, *egl* and *tev* protease (CET) were compared over time.

For all the samples, the results show an alteration of the pH value: the control (CTRL) presents a slight increase, which may be due to the regular metabolism of the bacteria. However, concerning CET, the pH is becoming more acidic over time. This may indicate the efficient degradation of cellulose acetate fibers. CET seems to be more efficient as the pH is lower.

To ensure that the diminution of the pH value is due to an accumulation of acetate, an Acetate Detection Kit was used on CET transformed bacteria, and non-transformed *E. coli* bacteria, shown in Figure 7.



**Figure 7: Comparison of the bacteria performance in degrading cellulose acetate**

The acetate concentration is measured in the control (CTRL) which is bacteria non modified, and modified bacteria with the BioBrick containing *cae*, *egl* and *tev* protease (CET) were compared over time.

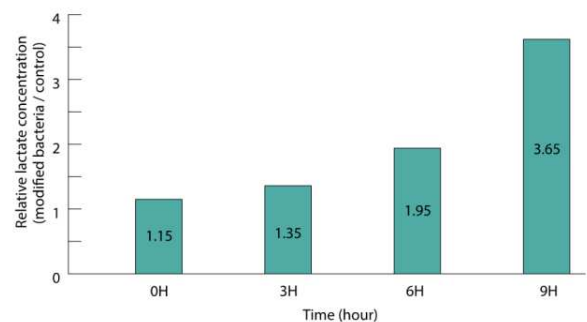
Those data show that the enzymatic process is functional. For CET transformed bacteria, acetate is detected and measured whereas the control does not seem to have any degradation activity. In 3h, the modified bacteria are able to

produce 34mg/L of acetate and even triple this quantity, 91mg/L in 6h. This complements the pH measurement data described above.

Those results demonstrate the efficiency of the CET modified bacteria as the acetate concentration increased.

## 2. Lactate production

An important step of the system is the optimization of lactate secretion by *E. coli* as it constitutes the major energetic substrate of *S. oneidensis*. Lactate thus represents an essential element in the generation of electricity and the results are shown on the Figure 8.



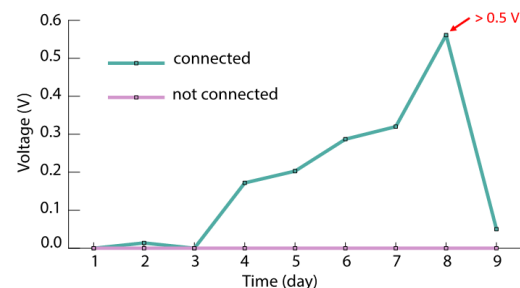
**Figure 8: Measurement of lactate concentration ratio over time between control and modified *E. coli***

The lactate concentration is measured in the control (CTRL) which is bacteria non modified, and modified bacteria with the BioBrick containing *cae*, *egl* and *tev* protease (CET) were compared over time. A ratio of the modified bacteria over the control ones was performed to be able to see if there is an increase of lactate concentration.

This figure suggests that, over time, the suspension containing modified bacteria has a higher lactate concentration than the control. These results thus clearly show the efficiency of the modified organism to secrete more lactate compared to the wild type strain.

## III Electricity production

To make sure that the electrogenic bacteria used in the system is able to produce electricity from lactate molecules, we measured the current generated by *S. oneidensis* using lactate as a substrate. Measurements were taken each day and plotted as shown on Figure 9.



**Figure 9: Measurement of the electricity produced by a non-modified *Shewanella oneidensis***

Measurement of the voltage of *S. oneidensis* connected to a 1V generator (in green) or not connected to a 1V generator (in pink). The highest voltage we were able to measure is 0,56 V.

These results demonstrate the ability of *S. oneidensis* to generate electricity from lactate. We were pleased to observe a voltage up to 0.5 V, corresponding to the average voltage produced by current MFCs. In the future, thanks to the modification that will be performed on *S. oneidensis* to optimize this electron transfer we are hoping to have an even better voltage. However, the presence of a biofilm could not be fully demonstrated as the required material was not available. Those experiments will be performed in the future. Finally, *E. coli* was also present in the medium and the co-culture with *S. oneidensis* does not seem to affect the generation of electricity.

### Discussion

Three main steps had to be proven to demonstrate that the Cinergy process works: the functional degradation of the cellulose acetate fibers into glucose and acetate, the optimized production of lactate and finally the generation of electricity by *S. oneidensis*.

From the results showcased previously, it has been demonstrated that each step was functional in a separate way.

Regarding the enzymatic degradation, two methods were used to prove the efficacy of the reaction: pH measurements and acetate molecules detection using a specific kit. However, as the enzymes were designed with a 10-His tag in N-terminus, purification could also be performed. Test on pure enzymes solutions could be done.

Concerning, the generation of electricity even though voltage was measured up to 0.56V, further characterization experiments must be performed starting with a proper identification of the biofilm formation either by using a color plasmid to be able to follow the formation or used crystal violet or XTT assay. Then, longer tests have to be realized to check the viability of the system through time.

Due to lack of time, each test was only performed once. Thus, replicates of the BioBrick verification, as well as the good degradation of CA and the good lactate overproduction tests have to be performed to ensure the functionality of the system. Furthermore, Cinergy was conceived to be an all-in-one system, meaning that we will have to test it in our prototype to determine if all the steps work well together.

Finally, an optimization process will have to be performed to optimize the electrical yield produced. This includes completing the genetically modification of *S. oneidensis* and *E. coli* (mostly the *plfb* gene knock-out) and the addition of flavins productive engineered *E. coli* to the system.

### Conclusion

These first results seem to be very promising for the future of the Cinergy project. It has been proved that the *E. coli* engineered with cellulose acetate esterase (CEA), endo-1,4- $\beta$ -glucanase I (EG1) and tev protease (Tev) are able to degrade de cellulose acetate constituted the cigarette butts.

The second bacterial strain, *S. oneidensis*, without being optimized, is able to produce electricity up to 0.56V. As this result is the average seen in the literature for the actual MFCs, there is great hope to obtain even better results when the bacteria will be engineered. By stocking the continuous energy produced, it can be considered to charge up a phone without any problem.

Furthermore, in these 9 months of work, all the BioBrick constructions have been obtained. However, due to a lack of time, they have not been all tested and transformed in the final bacterial strain. These tests can be performed soon and if all works as expected a first prototype can be realized in the following months which is the major goal of this project to answer to pollution treat that cigarette butts represent.

Today's world deeply needs a major ecological shift. This can only be performed through the elaboration of green solutions to fight the issues we are facing.

The Cinergy initiative was created in this specific dynamic to treat the pollution produce by cigarette butts. Creating an all-in-one system allowing the generation of free electricity from cellulose acetate filters is the solution developed to prevent smokers to throw their cigarette butts on the ground. Even though the system has been proven to be working, a lot of work, mostly optimization, has to be performed in order to obtain a viable system.

### Acknowledgement

The authors thank Dr. Ferid Nassor from CellTechs (CEA SEPIA/Sup'Biotech) laboratory for the support provided for this project.

The authors thank Dr. Vanessa Proux, Dr. Thierry Picaud, Dr. Frank Yates, Dr. Vanessa Perdiz, Valentina Gligorijevic, Dr. Estelle Mogensen, Dr. Pierre Ougen and Dr. Sophie Mothre for the help provided and the precious advice.

### Conflict of interests

The authors declare no conflict of interest.

### Reference

- [1] Barral Clémence, *Oui, des millions de mégots sont jetés chaque jour dans les rues de Paris*, Le Figaro, consulted on <https://www.lefigaro.fr/actualite-france/oui-des-millions-de-megots-sont-jetes-chaque-jour-dans-paris-20190609>, consulted in June 2019
- [2] NC, *Les mégots de cigarettes*, consulted on <https://www.planetoscope.com/recyclage-dechets/862-megots-de-cigarettes-jetes-dans-la-rue.html>, consulted in February 2019
- [3] NC, *La composition d'un mégot*, consulted on <http://www.gumego.com/la-composition-dun-megot/>, consulted in February 2019
- [4] NC, *14 choses que vous ne saviez pas sur le mégot de cigarette*, <https://consommonssainement.com/2018/04/07/megots/>, consulted in February 2019
- [5] G. Bonanomi, G. Incerti, G. Cesarano, S. A. Gaglione, and V. Lanzotti, "Cigarette butt decomposition and associated chemical changes assessed by  $^{13}\text{C}$  cpmas NMR," *PLoS One*, 2015
- [6] P. Choudhury, U. S. Prasad Uday, T. K. Bandyopadhyay, R. N. Ray, as B. Bhunia, "Performance improvmenet of microbial fuel cell (MFC) using



- suitable electrode and Bioengineered organisms: A review;" Bioengineered. 2017
- [7] O. Haske-Cornelius, A. Pellis, G. Tegl, S. Wur, B. Saake, R. Ludwig, et al., "Emzymatic systems for cellulose acetate degradation," Catalysts. 2017; (10):1-15
- [8] P. K.Bunch, F. Mat-Jan, N. Lee, and D. P. Clark, "The ldha gene encoding the fermentative lactate dehydrogenase of Escherichia coli," Microbiology. 1997; 143(1 997):187-95
- [9] A. Okamoto, R. Nakamura, K. H. Nealson, and K. Hashimoto, "Bound Flavin Model Suggests Similar Electron-Transfer Mechanisms in Shewanella and Geobacter," ChemElectroChem, 2014;1(11), 1808–1812. <https://doi.org/10.1002/celec.201402151>
- [10] Zhenquan Lin *et al.*, "Metabolic engineering of Escherichia coli for the production of riboflavin," *Microbial Cell Factories*, 2014, 13:104.
- [11] M. Mack, A. P. Van Loon, and H. P. Hohmann, "Regulation of riboflavin biosynthesis in Bacillus subtilis is affected by the activity of the flavokinase/flavin adenine dinucleotide synthetase encoded by ribC." Journal Bacteriology, 1998, 180:950–955.
- [12] L. Tao, H. Wang, M. Xie, L. Thia, W. N. Chen, and X. Wang, "Improving mediated electron transport in anodic bioelectrocatalysis," Chemistry Communications, 2015;51(61):12170–3.

## Supplementary figures

*Supplementary Figure 1: Program for PCR cycles incubated between 55°C and 68°C depending on the annealing temperature of the primers*

| Reagents         | MIX for one tube | MIX for 8 tubes |
|------------------|------------------|-----------------|
| Q5 Master Mix 2X | 12,5 µL          | 62.5            |
| Primer Forward   | 1 µL             | 8               |
| Primer Reverse   | 1 µL             | 8               |
| DNA              | Bacteria         |                 |
| H2O ultrapure    | 10,5 µL          | 84              |

*Supplementary Figure 2: Different couple of primers used*

|                    |                                                                                |
|--------------------|--------------------------------------------------------------------------------|
| Primer_ri bA_Fwd   | AAACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCatgCAGCTTAAACG<br>TGTGG         |
| Primer_ri bA_Rev   | AAAGCCATCGAGGCTtaTTTGTTCAGCAA<br>TGGCCC                                        |
| Primer_ri bB_Fwd   | ATAGCCTCGATGGCATTAAAGAGGAGAA<br>AGGATCCatgAATCAGACGCTACTTTCCTC                 |
| Primer_ri bB_Rev   | ATAGCCTCGATGGCATTAAAGAGGAGAA<br>AGGATCCatgAATCAGACGCTACTTTCCTC                 |
| Primer_ri bC_Fwd   | AAAGCTAGCATTAAAGAGGAGAAAGGAT<br>CCatgTTTACGGGGATTGTACAGG                       |
| Primer_ri bC_Rev   | AAACGTCTCTTAATTAACCTcaGGCTTCT<br>GTGCCTGG                                      |
| Primer_ri bD_Fwd   | AATCTTAAGATTAAAGAGGAGAAAGGAT<br>CCgtgCAGGACGAGTATTACATGG                       |
| Primer_ri bD_Rev   | AAGCATATGtcaTGCACCCACTAAATGCA<br>G                                             |
| Primer_ri bE_Fwd   | CGGGCATATGATTAAAGAGGAGAAAGGA<br>TCCatgAACATTATTGAAGCTAACGTTGC                  |
| Primer_ri bE_Rev   | AAAGCTAGCtcaGGCCTTGATGGCTTT                                                    |
| Primer_ri bF_Fwd   | AAACGTCTCGTAACGCTAAGCAAAGAGG<br>AGAAAGGATCCatgAAGCTGATACGCGGC<br>AT            |
| Primer_ri bF_Rev   | AAACGTCTCTTAATTAACCTtaAGCCGGTT<br>TTGTTAGCCC                                   |
| Primer_A CS_Fwd    | AAACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCatgAGCCAAATTCA<br>CAAACACACC    |
| Primer_A CS_Rev    | AAACGTCTCTTAATTAACCTtaCGATGGC<br>ATCGCGATAGC                                   |
| Primer_C 1563T_Fwd | GTGCTGAATGTCTCCGGTCA                                                           |
| Primer_C 1563T_Rev | TGACCGGAGACATTTCAGCAC                                                          |
| Primer_P GL_Fwd    | AAACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCATGAAGCAAACA<br>GTTTATATCGCCAGC |
| Primer_P GL_Rev    | AAACGTCTCTTAATTAACCTTAGTGTGC<br>GTTAACCACCACCC                                 |

|                        |                                                                             |
|------------------------|-----------------------------------------------------------------------------|
| Primer_P FLB_Fwd       | AAACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCatgACCGACAAAAC<br>TGAAGTGT   |
| Primer_P FLB_Rev       | ACCCGTCTCTTAATTAACCTtaCAGACCT<br>TTTGTGAATGTACGCG                           |
| Primer_A CS_SO_Fwd     | AAACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCatgAGCTCGCAGTC<br>TCTCTA     |
| Primer_A CS_SO_Rev     | ACCCGTCTCTTAATTAACCTtaCTCGTTGC<br>GATTTAAGC                                 |
| Primer_C 1609G_Fwd     | ATGTGATTAACGTGTCTGGCCACA                                                    |
| Primer_C 1609G_Rev     | TCTGTGGCCAGACACGTTAATCACAT                                                  |
| Primer_MDH_Fwd         | AAACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCatgAAAGTTGCTGT<br>ACTTGGTG   |
| Primer_MDH_Rev         | CCCCGTCTCTTAATTAACCTtaCTTAACG<br>AAATCAACGCCTAGT                            |
| Primer_T erminator Fwd | ACGCGCGTCTCGATTATTAATTAAGCCC<br>AGGCATCAAATAAAACGAAAGG                      |
| Primer_T erminator Rev | AAACTGCAGCGGCCGCTACTAGTATATA<br>AACGCAGAAAGGCCAC                            |
| Primer_L DHA_Fwd       | AAACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCATGAAACTCGCCG<br>TTTATAGACA  |
| Primer_L DHA_Rev       | AAACGTCTCTTAATTAACCTTAAACCAG<br>TTCGTTCCGGCAG                               |
| Primer_O pACS_Fwd      | AAACGTCTCGTAACGCTAAGCATTAAAG<br>AGGAGAAAGGATCCATGAGCCAAATTCA<br>CAAACACACCA |
| Primer_O pACS_Rev      | AAACGTCTCTTAATTAACCTTAATGCGCG<br>CGGCCTTG                                   |
| P_C1563 T_Fwd          | GTGCTGAATGTCTCCGGTCA                                                        |
| P_C1563 Rev            | TGACCGGAGACATTTCAGCAC                                                       |
| Primer_L LDP_Fwd       | AACGCTAAGCATTAAAGAGGAGAAAGG<br>ATCCATGAATCTCTGGCAACAAACTAC<br>G             |
| Primer_L LDP_Rev       | CCGCGTTAATTAAGGCTTAAGGAATCAT<br>CCACGTTAAGACATAAGC                          |
| Primer_O pLLDP_Fwd     | AAACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCATGACAATACTCC<br>AACTTTTCGC  |
| Primer_O pLSO_Rev      | AAACGTCTCTTAATTAACCTtaGCGGCTG<br>CACTCTTC                                   |
| Primer_MtrC_Fwd        | ACACGTCTCCTAACGGCTAAGCATTAAA<br>GAGGAGAAAGGATCCatgATGAACGCACA<br>AAAATCAAA  |



|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| Primer_<br>MtrC_Re<br>v | CGGCGCCGTCTCTTAATTAAACCTtaCATT<br>TTCACCTTTAGTGTGATCTGC |
|-------------------------|---------------------------------------------------------|

Supplementary Figure 3: Program cycles for GGA

|           |      |       |
|-----------|------|-------|
| 30 cycles | 37°C | 5 min |
|           | 16°C | 5 min |
| 1 cycle   | 55°C | 7 min |
|           | 85°C | 7 min |
|           | 10°C | ∞     |

Supplementary Figure 4: Libios Calculation Sheet

On the **Mega-Calculator** page, fill in the orange boxes and it will provide automatic results in the white boxes.  
To zoom up or down, ensure the Standard tool bar is showing (View > Toolbars) & select a value from the Zoom drop-down list.

1. Enter sample details

Sample details

Blank absorbance values

|   |                |                |
|---|----------------|----------------|
|   | A <sub>1</sub> | A <sub>2</sub> |
| 1 |                |                |
| 2 |                |                |

2. Insert absorbance values for the blank

If duplicate blanks have been run, insert both sets of data and the program will automatically use the average values. If a single set of values are input, these will be used.

3. Insert absorbance values for the samples

Sample absorbance values

| Sample identifier | A <sub>1</sub> | A <sub>2</sub> | A <sub>2</sub><br>corrected | Sample volume (mL) | Dilution (-fold) |
|-------------------|----------------|----------------|-----------------------------|--------------------|------------------|
|                   |                |                |                             | 0.10               | 1                |
|                   |                |                |                             | 0.10               | 1                |
|                   |                |                |                             | 0.10               | 1                |

Results

| Δ Abs (D-Lactic Acid) | D-Lactic Acid (g/L) | Sample (g/L) | D-Lactic Acid (g/100g) |
|-----------------------|---------------------|--------------|------------------------|
|                       |                     |              |                        |
|                       |                     |              |                        |
|                       |                     |              |                        |

4. Extinction coefficient

The calculations are set for readings at 340 nm [extinction coefficient for NADH of 6.3 (l x mol<sup>-1</sup> x cm<sup>-1</sup>)]. For absorbance readings at 365 nm (Hg lamp; ext. coeff. 3.4) multiply the calculated values for D-Lactic Acid by 1.8529. For absorbance readings at 334 nm (Hg lamp; ext. coeff. 6.18) multiply the calculated values for D-Lactic Acid by 1.0194.

5. Sample volume

If a sample volume other than 0.1 mL is used, enter the volume.

6. Sample dilution

If samples are diluted before assay, enter the dilution factor (e.g. 10 for 10-fold).

# DIANE: An aptamer-based device for rapid detection of pathogenic bacteria in biological fluids

Durix L.\*, Fournier T.\*, Cambon E., Charendoff C., De Jode M., Delamare C., Mosagna Q., Sabatou M., Saverat V., Souday R., Vigne L., Mazzuoli M.V., Zidat A., Gopaul D.N.

\*Corresponding authors: [lea.durix@supbiotech.fr](mailto:lea.durix@supbiotech.fr), [tara.fournier@supbiotech.fr](mailto:tara.fournier@supbiotech.fr)

**Abstract - Rapid and precise diagnosis is critical to patient care. Our goal is to develop a point-of-care diagnosis device, DIANE (Diagnosis is Now Easier), to be used for the detection of pathogens in biological fluids. The detection will take only a few minutes with a higher sensitivity than current serological cultures. Our device has been conceived to incorporate key design features based on feedback from physicians and scientists. It is composed by aptamers linked to carbon nanotube electrodes, which will detect bacteria through a voltage change after binding. Aptamers are selectively identified from a library with the Whole-Cell SELEX method and chosen to recognize specific bacteria. Moreover, user-friendliness will be improved by automation with pumps and microfluidics. DIANE will enhance the diagnosis speed allowing doctors to adapt the treatment for each patient. In this study, we successfully selected an aptamer specific to *E. faecium* from the library and we manage to create an aptamer-based biosensor for the detection of *E. faecium* in few minutes. This biosensor will possibly make the diagnosis of critical life-threatening situations quick and easy for medical professionals and NGOs.**

*Index Terms* - Aptamer, Carbon nanotubes, Diagnosis, *Enterococcus faecium*, SELEX

## INTRODUCTION

Sepsis is a major health problem worldwide, with an extremely high morbidity and mortality rate, partly due to a late diagnosis at the onset of the disease. This phenomenon occurs when the immune system develops a very strong response against a bacterial infection, such as against the bacteria *E. faecium*, causing generalized inflammation. Today, one person dies of sepsis every 5 seconds [1]. Treated late, this disease can cause irreparable tissue damage, including organ failure. Infectious diseases account for a total of 40% of the estimated 50 million deaths per year [2] and are thus the cause of more than 6 million deaths [3].

Currently, the diagnosis of sepsis requires a blood culture, a long process taking between 24 and 48 hours. In this context, improving the speed, sensitivity and specificity of bacterial detection is crucial.

Aptamers are *in vitro* selected single-stranded DNA or RNA oligonucleotides that can selectively bind to a wide variety of targets including large protein complexes, ions and whole cells [4]. By folding into tertiary structures, they are able to bind to targets with high affinity and specificity and are thus termed “chemical antibodies” [5]. They also have significant advantages over antibodies. One advantage is that nucleic acid structures can be regenerated several times with little activity loss, whereas protein-based antibodies can only be used once or a few times before the loss of their functionality. Also, DNA aptamers are often highly stable and can be inexpensively synthesized with high reproducibility and purity [6].

Aptamers have the potential to extend the limit of current techniques. They are currently used as aptasensors. These biosensors combine specificity of aptamers for their target with highly sensitive transducers. They could be highly beneficial to clinicians by providing a larger range and more rapid output of diagnostic readings than current technologies and faster implementation of optimal treatments [3]. Different kind of aptasensors are already incorporated in clinical diagnostics or showing a high potential. These aptasensors are mainly based on electrochemical, optical, mechanical, and acoustic approaches. They can be used for real-time biomarker detection (mainly thrombin and IgE), cancer clinical testing (HeLa cells, Leukemia cells), and detection of infectious microorganisms and viruses (HIV-1 Tat protein, prion) [6].

In this article, we developed an aptasensor for the detection of *E. faecium*, one of the most common species in hospitals and human infections included in the list of priority pathogens of the World Health Organization [7,8]. This detection will

occur by combining a specific aptamer to carbon-nanotube electrodes. This aptasensor was integrated into an automated millifluidic device in order to analyze biological samples such as blood, urine and saliva.

## MATERIALS AND METHODS

### Bacterial strains and culture

*Enterococcus faecium* (ATCC 19434) was used to perform the Whole-Cell SELEX. As a negative control, *Escherichia coli* (ATCC 8739) was used. All strains were obtained by the CRBIP of the Pasteur Institute. Bacterial cells were cultured at 37°C in a nutrient LB medium to an OD<sub>600</sub> between 0.5 and 0.6 (equivalent to 10<sup>7</sup> cells/ml), prior to use in experiments.

### Aptamers

The following sequence was used for the SELEX method against *E. faecium*: a random single-stranded 40-nucleotides library. Two primers were used in order to amplify the library by PCR: 5'-AAGCTTGTGAC-3' and 5'-GAATTCGGATCC-3' and the later was phosphorylated in 5' to be digested after the PCR.

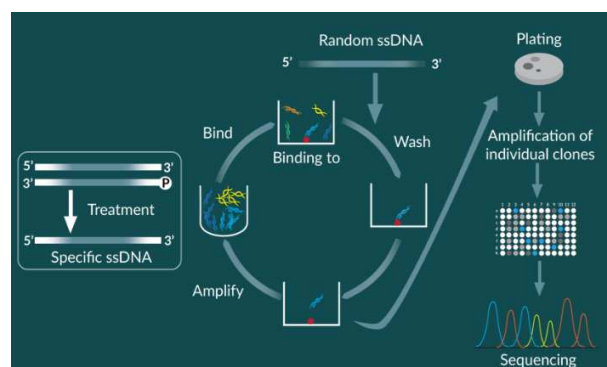
A specific aptamer sequence against *E. coli* published by Kim et al. (2013) was used as a negative control: 5'-CATGAGTGTGTGAAATGTTGGGACACTAGGTGGC ATAGAGCCG-3' [9]. Prior to use, all samples were resuspended with a 10 mM Tris-EDTA pH 8.0 solution. All nucleotides were synthesized by Eurofins Genomics.

### Whole-Cell SELEX

This protocol was inspired by Sefah, et al., 2010 [10], however each steps were optimized by ourselves.

In order to develop the most specific aptamer against *E. faecium*, a Whole-Cell SELEX was performed.

This protocol consisted of several experiments making one round (Fig. 1) and 10 rounds were performed in total. Firstly, the library was amplified by PCR and the mixes were made accordingly to the manufacturer's recommendation (Promega). The following PCR conditions were used in all reactions: 95°C for 5 min, followed by 30 cycles of 95°C for 30 sec, 63°C for 1 min, 68°C for 30 sec, with a final extension step at 68°C for 5 min. Then, the library was digested with the lambda exonuclease in order to pass from double to single-stranded DNA. This step was performed as recommended by the manufacturer's protocol (New England Biolabs). Next, the digested samples were purified by phenol/chloroform extraction followed by ethanol precipitation. Then, the purified library was incubated with 1 mL of *E. faecium* for a time varying at each round (Table 1). Then, the samples were centrifuged at 10,000 x g for 10 minutes and the supernatant was removed. The pellet was resuspended with 100 µL of washing buffer (Phosphate Buffered Saline (PBS) + Tween 20 at 0.05%) and then centrifuged again at 10,000 x g for 10 minutes. These three steps were repeated 3 times in total.



After that, the pellet was resuspended with 100 µL of washing buffer and incubated at 95°C for 5 minutes and then in ice for

FIGURE 1: SCHEMATIC REPRESENTATION OF THE WHOLE-CELL SELEX METHOD

A DNA library was synthesized by Eurofins Genomics containing two specific primers, one of them being phosphorylated in 5' end. After each PCR, a treatment with lambda exonuclease allowed to recover a single-stranded form. After 10 rounds, DH5-α *E. coli* were transformed with the aptamer sequences and bacterial clones were sequenced

15 minutes. Finally, the samples were centrifuged at 13,000 x g for 6 minutes and the supernatant was kept.

TABLE I  
TIME OF INCUBATION FOR EACH ROUND OF THE WHOLE-CELL SELEX

| Round | Time of incubation |
|-------|--------------------|
| 1     | 2 hours            |
| 2     | 2 hours            |
| 3     | 1h45               |
| 4     | 1h30               |
| 5     | 1h15               |
| 6     | 1 hour             |
| 7     | 1 hour             |
| 8     | 45 minutes         |
| 9     | 30 minutes         |
| 10    | 30 minutes         |

WHILE THE NUMBER OF ROUNDS OF SELEX INCREASES, THE TIME OF INCUBATION WAS REDUCED TO ADAPT THE STRINGENCY IN ORDER TO OBTAIN THE MOST SPECIFIC APPTAMERS.

### Sequencing

After transforming *E. coli* DH5-α competent cells with the plasmid pET28:GFP containing the aptamer candidates, the bacteria were spread on LB + Kanamycin plates and grown overnight at 37°C. The next day, each bacterium from the culture were isolated on plates and grown overnight at 37°C. The day after, each isolated bacterium was put into the PlateSeq kit (Eurofins Genomics) by using a pipette tip. The PlateSeq kits were incubated at 37°C overnight and then send to Eurofins Genomics for the sequencing.

### Functionalization of electrodes

The following protocol was based on the one performed by Huang et al. [11]. The single-wall carbon nanotubes (SWCNTs) were oxidized in a silica furnace chamber at 365°C with a synthetic air flowrate of 100 cm<sup>3</sup>/min for 90

minutes. The SWCNTs were refluxed in 2.6 M nitric acid for 4 hours and then filtered and thoroughly rinsed with water in order to completely remove the remaining acid. The filtered SWCNTs were dried overnight at 80°C. A solution of purified SWCNTs at 1 mg/mL in water was prepared and 0.1wt% of sodium dodecylbenzenesulphonate (SDS) was added. A drop of the suspension was deposited on graphite surface electrodes. Then, the nanotubes were settled at room temperature and rinsed with water. After that, the carboxylic groups were activated by letting stand a solution of 100 nmol of N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride (EDC) and 25 nmol of N-Hydroxysuccinimide (NHS) in a 50 mM 2-(N-Morpholino)ethanesulfonic acid (MES) buffer (pH 5) for 30 minutes. Then, a solution of aptamer at 1  $\mu$ M in PBS (1 mM, pH 7.4) and Cetrimonium bromide (CTAB) (0.2 mM) were prepared. The electrode was soaked overnight into 500  $\mu$ L of the aptamer solution.

## RESULTS

After the sequencing of the PlateSeq kits by Eurofins Genomics, the results were analyzed. Out of the 50 clones, only one sequence was composed of 40 nucleotides between the primers (Fig. 2). The length of random sequences is approximately 20–60 bases, we chose 40 nucleotides as an optimized length and a balance between specificity and cost. This aptamer specific to *E. faecium* could be the first revealed, no sequence has been reported in the literature yet.

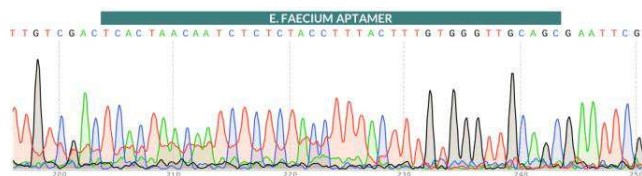


FIGURE 2: SEQUENCING PROFILE OF THE DISCOVERED APTAMER AGAINST *E. FAECIUM*

Sequencing profile was obtained after the bacterial clone sequencing performed by Eurofins genomics.

In order to ensure that the electrodes were properly functionalized by the fixation of the aptamers on their surface, a cyclic voltammetry experiment was performed (Fig. 3). A CV is an electrochemical method to evaluate the reductions (curve from left to right) and oxidations (curve from right to left, the curve returns to its initial potential) of species [12]. The reference electrode used was the electrode Ag/AgCl. The blue curve represents the electrode without SWCNTs and aptamers on it and the other curves are functionalized electrodes with SWCNTs and an aptamer sequence specific to *E. coli* published by Kim et al. [9].

A shift is observed between the non-functionalized electrode and the functionalized one because their response in current differs when electrodes are combined with aptamers. It indicates that layers of aptamers were formed on the electrode

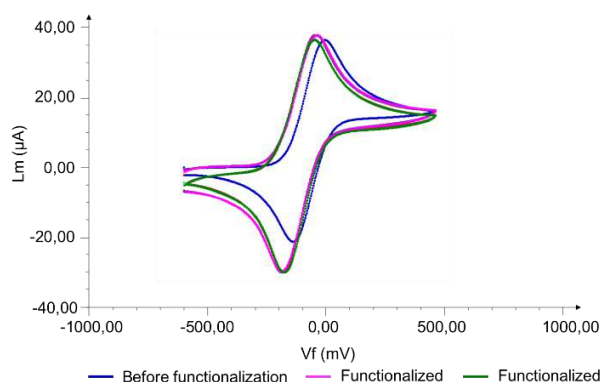


FIGURE 3: CYCLIC VOLTAMMETRY OF THE FUNCTIONALIZED ELECTRODE

Cyclic voltammetry was performed to test the proper fixation of aptamers on electrodes. The blue curve represents the negative control (before the fixation of aptamers). The pink and green curves are the electrodes functionalized with the aptamers. The experiment was performed in triplicate.

which slightly influences the current measured. This experiment was repeated with other aptamers specific to other bacteria and each time the results have the same tendency (data not shown).

To assess the specificity and sensitivity of the aptamer fixed on SWCNTs electrodes, several potential measures were performed. Firstly, the ability of the aptamer to recognize *E. faecium* was tested through the addition of two different concentrations of bacteria (Fig. 4). This figure shows that the aptamer is able to detect *E. faecium* when bounds to functionalized electrodes in few seconds. Moreover, the conformational change of the aptamer induces a higher voltage increase when the concentration of bacteria increases.

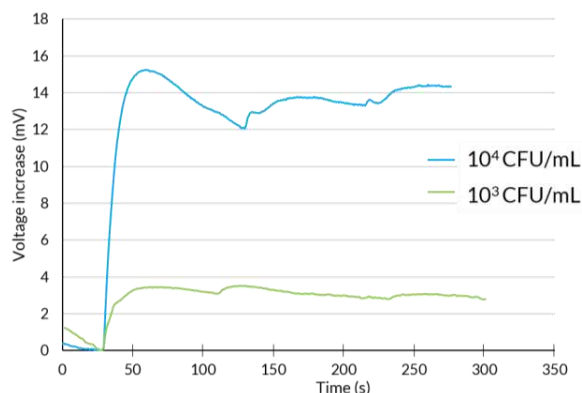


FIGURE 4: MEASUREMENT OF THE POTENTIAL AFTER THE ADDITION OF TWO DIFFERENT CONCENTRATIONS OF *E. FAECIUM* OVER TIME

Upon each addition of *E. faecium*, the voltage was measured over time. The green and blue curves represent samples of *E. faecium* at different concentrations: 10<sup>3</sup> CFU/ml (Green) and 10<sup>4</sup> CFU/mL (Blue). This experiment was performed 10 times and normalized for each concentration.

In order to determine at which concentration of bacteria the electrode is saturated, another potential measurement of different concentrations was performed (Fig. 5). On this figure, the potential increases over time in few seconds and the more concentrated bacteria solution was added, the higher is the potential increase. This dose-dependency could allow us not only to detect pathogens but also follow the evolution of sepsis by monitoring the concentration of bacteria. However, when the concentration was under  $10^2$  CFU/mL, the increase of potential might not have been significant enough to ensure the detection of bacteria. The maximum concentration tested was  $10^8$  CFU/mL but at this concentration no threshold was reached as the voltage was still increasing. Thus, the functionalized electrode can detect at least up to  $10^8$  CFU/mL, but it may detect also more concentrated samples.

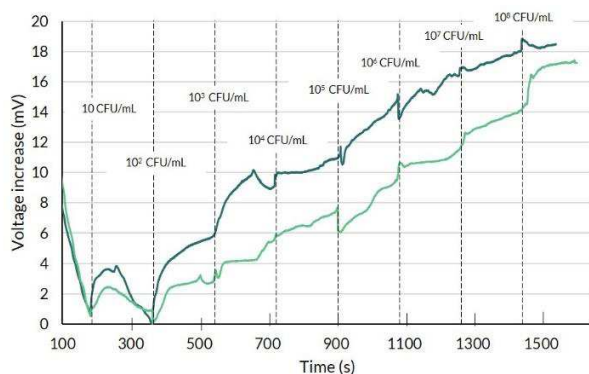


FIGURE 5: MEASUREMENT OF THE POTENTIAL AFTER THE ADDITION OF DIFFERENT CONCENTRATIONS OF *E. FAECIUM* OVER TIME

Upon each addition of *E. faecium*, the voltage was measured over time. The light green and dark green curves are duplicates of the experiment. Each dotted line represents the addition of a sample of *E. faecium* onto the electrode, at different concentrations.

To assess the specificity of the aptamer to only *E. faecium*, potential measures with PBS 1X was performed (Fig. 6). Even if there is an increase of the potential, it decreases right after within less than 100 seconds, contrary to the voltage increase after addition of *E. faecium* which was stable through the time (Fig. 4). Aptamer is not able to recognize PBS 1X but seems to be briefly disturbed. The same experiment was made with *E. coli* (Fig. 7). The curve shows the same profile than with the addition of PBS 1X (Fig. 6).

After the addition of the solution of *E. coli*, the voltage fluctuates and decreases for approximately 60 seconds, but is also not stable. Thus, the mechanical addition of negative controls results in small peaks that decreases immediately, while the addition of *E. faecium* shows a stable change of potential. These results confirmed the specificity of the aptamer for *E. faecium*.

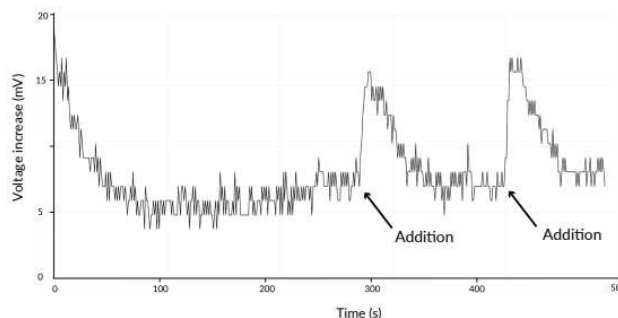


FIGURE 6: MEASUREMENT OF THE POTENTIAL AFTER THE ADDITION OF PBS

Upon each addition of PBS 1X, the voltage was measured over time. Each peak represents the addition of one drop of PBS 1X onto the electrode.

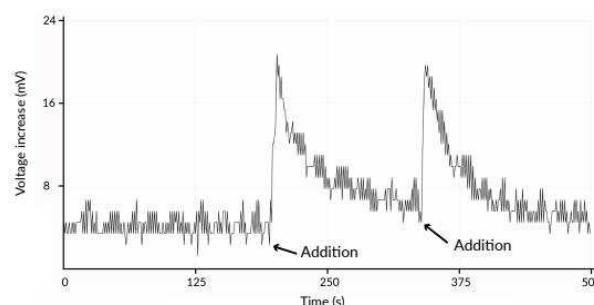


FIGURE 7: MEASUREMENT OF THE POTENTIAL AFTER THE ADDITION OF *E. COLI*

Upon each addition of *E. coli*, the voltage was measured over time. Each peak represents the addition of a concentrated sample of *E. coli* onto the electrode.

## DISCUSSION

Concerning the diagnostic field, aptamers are of great interest because they offer many advantages compared to antibodies. Indeed, aptamers are smaller, easier to produce, cheaper, and their chemical stability allow them to be stored or transported easily [13]. Since decades now, aptamers have been developed as biosensors to detect microorganisms. However, their main drawback is their specificity to one species, being sometimes too much restrictive [14]. In our context, we aim to develop a device for the detection of the pathogen responsible for sepsis. This drawback is more perceived as an advantage as it reduces the possibility to have false positives and enhance the detection rate. However, we still need to be sure that the aptamer will be able to recognize other types of *E. faecium* strains.

Moreover, we demonstrated that our aptasensor is able to detect bacteria within  $10^2$  and  $10^8$  CFU/mL range. This large range will enable to detect bacteria in blood in patients. Indeed, the number of CFU/mL in case of sepsis may vary according to the type of bacteria involved, the advancement of the disease and the age of patients [15]. But, in case of severe sepsis, the blood may contain between 100 and 1,000



CFU/mL. Sometimes, when the number of bacteria is over 1,000 CFU/mL, blood cultures cannot efficiently detect the bacteria involved [15]. Thus, having a more precise aptasensor able to detect more than 1,000 CFU/mL will overcome this issue. However, for the early stages of sepsis, the number of bacteria is lower as it is comprised between 1 and 100 CFU/mL. Because of this low range, blood cultures are not able to accurately determine the involved strain(s) leading to a false negative test [16]. Thus, the lower range of detection of the aptasensor has to be further improved to be able to also detect the firsts stages of sepsis which is already possible for other pathogens [17, 18].

Although aptamers have been selected and tested on our engineered electrodes, many experiments remain to be done in order to characterize them. Flow cytometry could be considered in order to characterize and quantify the affinity constant for its target. Other tests could be done to validate the specificity of our aptamer, especially with Gram-positive bacteria and other *Enterococcus* bacteria. Finally, the selection method via Whole-Cell SELEX process could be improved by adding a negative selection step at the end of all rounds.

### CONCLUSION

Aptamer candidates were successfully selected through the Whole-cell SELEX procedure. We presented the first demonstration of a peptide aptamer-based biosensor for selective and sensitive detection of *Enterococcus faecium* in PBS. This device was realized by the combination of a novel peptide aptamer as a probe molecule and single-wall carbon nanotubes-electrodes. Our results suggest that this detection system is sensitive and specific to *Enterococcus faecium* from a minimal concentration at  $10^2$  CFU/mL and could be adapted to other types of bacteria by selecting novel aptamers.

### ACKNOWLEDGMENT

We would like to thank the Pasteur Institute for providing all the materials and strains necessary. We particularly thank Mathieu De Jode, Maria-Vittoria Mazzuoli, Chloe Charendoff and Deshmukh Gopaul for their help. We also thank Eurofins genomics for the synthesis of the DNA library and the sequencing of our aptamer candidates.

### REFERENCES

- [1] Institut Pasteur, "Sepsis/Septicémie", Institut Pasteur, online the 21/10/19, disponible sur : <https://www.pasteur.fr/fr/centre-medical/fiches-maladies/sepsis-septicemie#symptmes>
- [2] INSERM, "Résistance aux antibiotiques", INSERM, online the 22/03/18, disponible sur : [www.inserm.fr/information-en-sante/dossiers-information/resistance-antibiotiques](http://www.inserm.fr/information-en-sante/dossiers-information/resistance-antibiotiques)
- [3] Fleischmann, C. *et al.*, "Assessment of Global Incidence and Mortality of Hospital-treated Sepsis. Current Estimates and Limitations.", *American Journal of Respiratory and Critical Care Medicine*, 193(3), 259-272 (2016).
- [4] Ali, M. H., Elsherbiny, M. E., & Emara, M., "Updates on Aptamer Research". *International journal of molecular sciences*, 20(10), 2511 (2019).
- [5] Kaur, H., Bruno, J. G., Kumar, A., & Sharma, T. K. "Aptamers in the Therapeutics and Diagnostics Pipelines". *Theranostics*, 8(15), 4016–4032 (2018).
- [6] Ilgu M., *et al.*, "Aptamers for Diagnostics with Applications for Infectious Diseases". *IntechOpen* (2019).
- [7] World Health Organization, "Global priority list of antibiotic-resistant bacteria to guide research, discovery, and development of new antibiotics", World Health Organization, online the 27/02/17, disponible sur : <https://www.who.int/news-room/detail/27-02-2017-who-publishes-list-of-bacteria-for-which-new-antibiotics-are-urgently-needed>
- [8] Furtado I., *et al.*, "Enterococcus faecium and Enterococcus faecalis in blood of newborns with suspected nosocomial infections". *Revista do Instituto de Medicina Tropical de Sao Paulo*, 56(1), 77-80 (2014).
- [9] Kim Y.S., *et al.*, "Isolation and characterization of DNA aptamers against Escherichia coli using a bacterial cell-systematic evolution of ligands by exponential enrichment approach". *Anal. Biochem*, 436, 22–28 (2013).
- [10] Sefah K, *et al.*, "Development of DNA aptamers using Cell-SELEX". *Nature Protocols*, 5(6), 1169-1185 (2010).
- [11] Huang, C., *et al.*, "Layer-by-layer spray deposition and unzipping of single-wall carbon nanotube-based thin film electrodes for electrochemical capacitors". *Carbon*, 61, 525-536 (2013).
- [12] Elgrishi, N., *et al.*, "A Practical Beginner's Guide to Cyclic Voltammetry". *Journal of Chemical Education*, 95, 197-206 (2018).
- [13] Graziani A.C., *et al.*, "High Efficiency Binding Aptamers for a Wide Range of Bacterial Sepsis Agents". *J. Microbiol. Biotechnol.*, 27(4), 838–843 (2017).
- [14] Su L., *et al.*, "Microbial biosensors: a review". *Biosens. Bioelectron*, 26, 1788-1799 (2011).
- [15] Yagupsky P and Nolte F, "Quantitative Aspects of Septicemia". *Clinical microbiology reviews*, 3(3), 269-279 (1990).
- [16] Stranieri I, *et al.*, "Assessment and comparison of bacterial load levels determined by quantitative amplifications in blood culture-positive and negative neonatal sepsis". *Revista do Instituto de Medicina Tropical de Sao Paulo*, 60 (2018)
- [17] Zhao Y.-W., *et al.*, "Application of Aptamer-Based Biosensor for Rapid Detection of Pathogenic Escherichia coli". *Sensors*, 18(8) (2018).
- [18] Zelada-Guillén G. A., *et al.*, "Immediate detection of living bacteria at ultralow concentrations using a carbon nanotube based potentiometric aptasensor". *Angewandte Chemie*, 48(40), 7334 7337 (2009).

### Author information

**Durix Léa**, 5<sup>th</sup> year student at Sup'Biotech, R&D, Health and Pharmaceuticals – Elaboration of protocols/proof of concept, conducting lab experiments

**Fournier Tara**, 5<sup>th</sup> year student at Sup'Biotech, R&D, Health and Pharmaceuticals – Elaboration of protocols/proof of concept, conducting lab experiments

**Cambon Elise**, Jean Monnet University, M2 "Droit de la Création et du Numérique" – Intellectual property, web design

**Delamare Clara**, Jean Monnet University, M2 "Applied Intellectual Property Law" – Intellectual property, entrepreneurship

**Mosagna Quentin**, 2<sup>nd</sup> year student at Chimie ParisTech – Fonctionnalization and characterization of the electrodes

**Sabatou Marie**, double-degree student at Chimie ParisTech and AgroParisTech – Crowdfunding and communication

**Saverat Vincent**, 3<sup>rd</sup> year student at ENS Paris Saclay - Characterization of the electrodes

**Soudy Ronan**, 2<sup>nd</sup> year student at Chimie ParisTech – Design of the millifluidic device, entrepreneurship

**Vigne Laurence**, 2<sup>nd</sup> year student at Chimie ParisTech – Fonctionnalization of the electrodes

**Zidat Alexandre**, 2<sup>nd</sup> year student at ESPCI (Ecole Supérieure de Physique et de Chimie Industrielles) – Programation of the Arduino for the electrodes, web design

**Charendoff Chloe**, PhD student, “Cellular biology of microbial infection” unit, Pasteur Institute

**De Jode Mathieu**, PhD student, “Bacteriophage, bacterium, host” unit, Pasteur Institute

**Mazzuoli Maria-Vittoria**, PhD student, “Biology of gram-positive pathogens” unit, Pasteur Institute

**Gopaul Deshmukh**, Project manager, Pasteur Institute







Presentation by Dr Josquin Courte, who uses microfluidic devices to study the propagation of prion-like proteins in neural networks.



«Deep mutational scanning of essential genes in E. coli using Cas9 recombining » Presentation by Dr Alaksh Choudhury



Opening of the first Fil Rouge Student Symposium : presentation of the program by Biocampus



Paul-Etienne Fontaine, Co-founder of the startup Azuvia presents a history of farming practice



Videoconference with Pr Vincent Laudet, Oceanologic Institute Banyuls-sur-Mer (Occitanie)

On November 13, 2019, the **first Fil Rouge student symposium** was held, organized by the students of Biotech 5 class of 2020, from the Research & Development major. The event took place at the Salle de la Géothermie in Villejuif, near the Sup'Biotech campus. Prior to this day, the students identified and contacted professionals working on themes close to their Fil Rouge project. They invited these professionals **to come and present their work** in front of the students in the class, Sup'Biotech staff and invited guests. Videoconferences were organized for invitees who were unable to make themselves available for travel. During the breaks, ideas were exchanged over coffee and pastries during a poster session around the poster previously prepared by the students. An organic and eco-responsible lunch was prepared by the **student association Biocampus**. This first edition of the Fil Rouge student symposium was rich in meetings exchanges and discussion.

*The Sup'Biotech team and students would like to thank the speakers who took the time to present their work: Paul-Etienne Fontaine, Dr Alaksh Choudhury, Pr Vincent Laudet, the Curcumin B student group, Dr Christelle Langevin, Jules Heraud, Dr Josquin Courte, Dr Isabel Valsecchi, and Dr Jean-Pierre Jourdan. The symposium organizing team also wishes to thank the members of the Biocampus association, especially Maxime Mauviel, Thomas Grossetete and Clémentine Piquemal.*

# Sup'Biotech 1<sup>st</sup> student's Symposium



**9h.15 Introduction**

## DM compost

9h30-10h10, Paul-Etienne Fontaine, Engineer, Azuvia, History of farming practice.

## Endocrine disruptors

11h10-11h50, Vincent Laudet, Pr, Banyuls Oceanographic Observatory, Chlorpyrifos affect the life cycle of coral reef fishes.

## Light sheet and tissue clearing

13h30-14h10, Christelle Langevin, CR INRA, Deep inside host pathogen interactions: Tissue clearing approaches for whole organ imaging.

## Cell culture and microfluidics

14h50-15h30, Josquin Courte, PhD, CNRS UMR 8256, Mechanisms of prion-like propagation (Alzheimer's disease, Parkinson's disease, etc.) in neural networks.

## Curcumin as therapy

16h40-17h20, Jean-Pierre Jourdan, Pharm D, PhD, CHU Caen/UNICAEN, Place of Curcumin in Neuroscience drug design project.

## Quantitative Evolutionary Microbiology

10h10-10h50, Alaksh Choudhury, Post-Doc, IAME, Inserm U1137, Genomic Deep mutational scanning of essential genes in *E. coli* using Cas9 mediated recombineering.

## Curcumin in cancer treatment

11h50-12h30, Antoine Retho, Sup'Biotech 5<sup>th</sup> year, Curcumin and crocin as natural substitutes for current chemotherapeutic agents.

## Organ on a chip

14h10-14h50, Jules Heraud, PhD student, BlackHole Lab, Development of an organ on chip to co-culture human liver slices and immune cells in order to study liver fibrosis.

## Essential oil

16h-16h40, Isabel Valsecchi, PhD student, Pasteur institute, *Aspergillus fumigatus*, a fungal pathogen.

## Poster session and concluding remarks

17h20-18h, Students master projects poster presentation.

**November  
13th, 2019**

**9.15h-18h**

Organized by Sup'Biotech R&D students, class 2020 with the help of the Villejuif Geothermal Power Plant and the association BIOCAMPUS

**Program and more information on :**

<https://www.recherche.supbiotech.fr/symposium2019>



# STUDENT SPACE : A WEBSITE FOR SUP'BIOTECH STUDENTS

## Student Space : Research and Lab Tools

Access reserved to Sup'Biotech students

Useful info:  
Codes for  
Nature journals,  
lab directories...



### Useful info for Sup'Biotech students

- > How to obtain scientific publications :
- > Access your Nature journals
- > Write to a researcher
- > Directories

A list of student  
competitions  
you can join !



### Student competitions

- > Take part in a competition
- > Develop your own project
- > Show off your skills

Agenda of  
selected  
biotech events



### Scientific and Biotech events

- > Selected by Sup'Biotech professors
- > Recommended for Sup'Biotech students

Receive emails  
from the Research  
Department



### Sign up for our Newsletter (for Sup'Biotech students only...)

Nom/Name \*

E-mail \*

Tenez-moi informé car je suis  
passionné par ... / Keep me informed  
because I am fascinated by..... \*

Envoyer

### Interesting news for students



First Sup'Biotech International  
Symposium : 29 Jan 2020

13 Jan 2020



Visiting  
Laboratories

12 Dec 2019



CAR-T cells are the future of  
biomedicine. Read this if you don't  
believe me.

30 Nov 2019



Student Symposium "Fil Rouge  
2019" Nov 13th, 2019

8 Nov 2019

Interesting news  
for students

## Visit now !

<http://www.recherche.supbiotech.fr/espace-etudiants>







## **SUP'BIOTECH PROJECTS JOURNAL**

<http://recherche.supbiotech.fr/sbpj>

Sup'Biotech  
Ecole d'ingénieurs  
66, rue Guy Moquet  
94800 Villejuif



**Service recherche**  
<http://recherche.supbiotech.fr>

**Service des laboratoires d'enseignement**  
[estelle.mogensen@supbiotech.fr](mailto:estelle.mogensen@supbiotech.fr)